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Definitions and conventions

As used in this report, the term “country” does not in all cases refer to a territorial entity that is a state as understood by international law and practice. As used here, the term also covers some territorial entities that are not states but for which statistical data are maintained on a separate and independent basis.

In this report, the following groupings are used:

- “ASEAN-4” comprises Indonesia, Malaysia, the Philippines and Thailand.
- “Emerging Asia” comprises Asian economies such as China, Hong Kong, India, Indonesia, South Korea, Malaysia and Thailand.
- “EMEA” refers to Europe, Middle East and Africa.
- “Euro zone” comprises Austria, Belgium, Cyprus, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Portugal, Slovakia, Slovenia and Spain.
- “LATAM” refers to Latin America.

Abbreviations used for financial data are as follows:

- Currencies: Australian Dollar (AUD), British Pound (GBP), Chinese Yuan (CNY), Euro (EUR), Hong Kong Dollar (HKD), Indian Rupee (INR), Indonesian Rupiah (IDR), Japanese Yen (JPY), Korean Won (KRW), Malaysian Ringgit (MYR), New Taiwan Dollar (TWD), New Zealand Dollar (NZD), Singapore Dollar (SGD), Thai Baht (THB), US Dollar (USD).

Other abbreviations:

ABG	Asian Bond Grant
ABS	Association of Banks in Singapore
ABSD	Additional Buyer’s Stamp Duty
AE	Advanced Economy
ATM	Automated Teller Machine
BdF	Banque de France
BIS	Bank for International Settlements
BoE	Bank of England
BTI	Balance-to-income
CAR	Capital Adequacy Ratio
CCR	Core Central Region
CET1	Common Equity Tier 1
COVID-19	Coronavirus Disease 2019
CRE	Commercial Real Estate
DBU	Domestic Banking Unit
DDRS	Depository Trust & Clearing Corporation Data Repository (Singapore) Pte Ltd
DNB	De Nederlandsche Bank
DOS	Department of Statistics
D-SIB	Domestic Systemically Important Bank

EBIT	Earnings Before Interest and Taxes
ECB	European Central Bank
EDB	Economic Development Board
EME	Emerging Market Economy
ESG	Enterprise Singapore
EU	European Union
F&B	Food and Beverage
FCI	Financial Conditions Index
FI	Financial Institution
FRA	Forward Rate Agreement
FRN	Floating Rate Note
FSB	Financial Stability Board
FSR	Financial Stability Review
FVI	Financial Vulnerability Index
FX	Foreign Exchange
GDP	Gross Domestic Product
GFC	Global Financial Crisis
G-SIB	Global Systemically Important Bank
HDB	Housing Development Board
HQLA	High Quality Liquid Assets
IAM	Integrated Assessment Model
ICR	Interest Coverage Ratio
IIF	Institute of International Finance
IMF	International Monetary Fund
IP	Illiquidity Premium
IR	Interest Rate
ISDA	International Swaps and Derivatives Association
IWST	Industry-Wide Stress Test
JSS	Jobs Support Scheme
LBS	Locational Banking Statistics
LCR	Liquidity Coverage Ratio
LIBOR	London Interbank Offered Rate
LTD	Loan-to-Deposit
LTV	Loan-to-Value
MA	Matching Adjustment
MAS	Monetary Authority of Singapore
MOM	Ministry of Manpower
m-o-m	Month-on-Month
MSD	Macroprudential Surveillance Department
MSR	Mortgage Servicing Ratio
NDF	Non-deliverable Forward
NGFS	Network of Central Banks and Supervisors for Greening the Financial System
NFC	Non-financial Corporate
NIM	Net Interest Margin
NPA	Non-Performing Asset
NPL	Non-Performing Loan

O&G	Oil and Gas
OCR	Outside of Central Region
OTC	Over-the-counter
OTP	Option to Purchase
PR	Permanent Resident
q-o-q	Quarter-on-Quarter
RBC	Risk Based Capital
RCR	Rest of Central Region
REIT	Real Estate Investment Trust
RHS	Right Hand Side
ROA	Return on Assets
SAAR	Seasonally Adjusted Annual Rate
SBF	Singapore Business Federation
SC-STC	Steering Committee for SOR Transition to SORA
SFEMC	Singapore Foreign Exchange Market Committee
SFRP	Special Financial Relief Programme
SGX	Singapore Exchange Limited
SIBOR	Singapore Interbank Offered Rate
SME	Small and Medium-sized Enterprise
SOE	State-Owned Enterprise
SOFR	Secured Overnight Financing Rate
SOR	SGD Swap Offer Rate
SORA	Singapore Overnight Rate Average
S-REIT	Singapore-listed Real Estate Investment Trust
TSC	Transport, Storage and Communications
URA	Urban Redevelopment Authority
VA	Value-Added
VAR	Vector autoregression
VIX	Chicago Board Options Exchange Volatility Index
y-o-y	Year-on-Year
YTD	Year-to-date

Preface

The Monetary Authority of Singapore (MAS) conducts regular assessments of Singapore's financial system, which identify potential risks and vulnerabilities, and review the capacity of the financial system to withstand potential shocks. The analyses and results are published in the annual Financial Stability Review (FSR). The FSR aims to contribute to a better understanding of issues affecting Singapore's financial system among market participants, analysts and the public.

Section 1 of the FSR provides a discussion of the risks in the external environment. This is followed by an analysis of the Singapore corporate and household sectors in Sections 2 and 3 respectively. A review of the financial sector is then provided in Section 4. The final section features in-depth analyses on various topics in financial stability.

The production of the FSR was coordinated by the Macroprudential Surveillance Department (MSD) team that comprises Lily Chan, Evelyn Chen, Fan Jia Rong, Cheryl Ho, Koh Zhi Xing, Khoo Ye-Min, Kwek Kiat Cong, Wendy Lee, Aloysius Lim, Kalena Lim, Rosemary Lim, Vincent Low, Low Yen Ling, Orchi Modhurima, Satish Nagdev, Ng Heng Tiong, Desmond Ong, Alex Phua, Edward Robinson, Tan Aik Khim, Andrew Tan, Daniel Teo, Teoh Shi-Ying, Andrew Wilfred, and Wong Siang Leng. The FSR also incorporates contributions from Irineu de Carvalho Filho, Banking Departments I & II, Corporate Finance & Consumer Department, Capital Markets Intermediaries Department II, Enterprise Knowledge Department, Economic Analysis Department, Economic Surveillance & Forecasting Department, Financial Markets Development Department, Insurance Department, Monetary & Domestic Markets Management Department, Prudential Policy Department, Baskar Chinniah, and Harshitha Gooty. The FSR reflects the views of the staff of MSD and other contributors.

The FSR may be accessed in PDF format on the MAS website:

<https://www.mas.gov.sg/publications/financial-stability-review>

Overview

The COVID-19 pandemic has raised risks to global financial stability...

The COVID-19 pandemic led to an unprecedented contraction in economic activity and heightened risks to global financial stability. Decisive monetary and fiscal policy measures have alleviated near-term risks by easing global financial conditions and maintaining the flow of credit to businesses and households. However, medium-term vulnerabilities have risen as a result of these measures. Corporate leverage has continued to climb from already elevated levels, increasing credit risk. Higher sovereign debt levels amid unprecedented fiscal stimulus have also contributed to debt sustainability concerns. Capital flows to emerging market economies (EMEs) remain volatile, suggesting that market sentiment continues to be fragile with the path of recovery uncertain. A sudden spike in risk aversion could trigger a retrenchment from funds in the non-bank financial sector, causing liquidity stresses in financial markets. EMEs could be susceptible to a resurgence of capital outflows. While recent progress in vaccine development could provide some upside to economic growth, much is still dependent on the effectiveness of the vaccines and their widespread deployment.

...even as domestic financial conditions remain supportive.

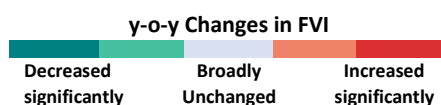
Financial conditions in Singapore have eased since tightening in March this year. Domestic factors (including the exchange rate, interest rate as well as equity and property prices) have played a key role in keeping overall financial conditions relatively supportive.

Corporates, households and the financial sector in Singapore have remained generally resilient so far despite the initial COVID-19 shock.

Corporates, households and banks in Singapore have remained resilient following the onset of the COVID-19 pandemic. The corresponding Financial Vulnerability Indices (FVI) for the respective segments were broadly unchanged on a year-ago basis (see table below).

Sector	y-o-y Change as of	Change
Overall Corporate FVI	Q2 2020 ¹	
Overall Household FVI	Q3 2020	
Overall Banking FVI	Q3 2020	

The FVI is calibrated according to deviations from its historical distribution and divided into five coloured bands, which reflect varying degrees of y-o-y change. An increase/decrease in the FVI indicates that vulnerability has risen/fallen over the previous year.



¹ For the corporate FVI, the latest available data is as of Q2 2020.

Singapore corporate balance sheets were relatively resilient entering 2020, amid elevated but stable debt levels. The COVID-19 pandemic led to a fall in earnings and further rise in debt, worsening leverage risk. However, most firms should be able to withstand short-term pressures on their financial positions. Firms' liquidity has improved with efforts to bolster their cash buffers, as well as support from the government, MAS and the financial industry to ease cashflows and provide continued access to financing. Nonetheless, increased cash holdings have been accompanied by an increase in short-term debt and maturity risk. Foreign currency risk has remained stable as Singapore firms have avoided excessive reliance on foreign currency borrowings.

Household balance sheets in Singapore were relatively healthy at the onset of the COVID-19 pandemic owing to financial buffers built up in the preceding years, and households' debt servicing burden remains manageable under stress. Government transfers and measures extended by MAS and the financial industry mitigated the impact of a sharp fall in employment and incomes in H1 2020. While growth in overall household debt has moderated, leverage risk has edged up, due to a temporary rise in the debt-to-GDP ratio as GDP fell sharply in Q3 2020. Maturity risk has decreased, alongside reduced outstanding unsecured debt.

Singapore's financial sector has also been resilient to the economic and financial impact of COVID-19. The banking sector has experienced some deterioration in credit quality, but strong capital and liquidity positions provide a robust base for banks to continue supporting the economy's demand for credit. Funds in Singapore weathered the market volatility in March this year. Insurers have also remained strong amid the uncertain economic outlook, bolstered by the enhanced capital framework (RBC 2) that came into effect in March 2020. This report also considers the impact of a "low for long" interest rate environment on insurers via a stress test, and assesses the performance of Singapore-listed Real Estate Investment Trusts (S-REITs) during COVID-19.

The economy's overall credit-to-GDP ratio has increased despite the moderation in resident credit growth. The increase largely reflects the decline in GDP as a result of the sharp economic downturn, rather than a significant increase in credit levels. Hence, MAS will maintain the Countercyclical Capital Buffer (CCyB) at 0%.

However, given the uncertain economic outlook, Singapore corporates, households and the financial sector should remain vigilant and prudent.

While the economy is expected to pick up next year, its uneven trajectory will impinge on jobs and corporate profits. The risk of financial stresses remains during this protracted recovery period. Continued vigilance and prudence therefore remain warranted.

Within the corporate sector, smaller firms that tend to be financially weaker, and those in the domestically-oriented and travel-related services impacted by COVID-19 remain vulnerable.

Households should be prudent in taking up new debt and in committing to property purchases as the labour market recovery is expected to be protracted. Whenever possible, they should continue servicing or consolidating their existing obligations to enhance resilience against unexpected shocks.

Banks are likely to face challenging operating conditions in the near term. The prolonged low interest rate environment and asset quality deterioration amid continued uncertainty in the global outlook will exert pressure on banks' profitability, even as they continue to maintain strong underwriting standards and healthy capital buffers. Banks should also continue to actively monitor and manage their foreign currency risks prudently to guard against a renewed tightening of global funding conditions. Similarly, insurers are expected to be impacted by a prolonged low interest rate environment. They should closely monitor their solvency positions and adopt a prudent and forward-looking perspective to capital management.

Macroprudential Surveillance Department, Economic Policy Group
Monetary Authority of Singapore
1 December 2020

1 Global Financial and Economic Environment

The COVID-19 pandemic led to an unprecedented contraction in global economic activity, and heightened stress and risk in the international financial system. In response, authorities took swift and forceful monetary and fiscal measures to support the economy and preserve financial stability. While the decisive policy response has alleviated near-term risks to financial stability, some of these measures may have also raised medium-term vulnerabilities, alongside expectations of an uneven and uncertain economic recovery.

Accommodative financial conditions are necessary to support the economy and stabilise the financial system. However, they have fuelled rising indebtedness against a backdrop of already elevated corporate debt levels and lower levels of activity, increasing credit risk. Fiscal deficits have also widened as a result of unprecedented government support, raising public debt sustainability concerns.

With the path of recovery still uncertain, risk sentiment remains fragile and EMEs could be susceptible to a resurgence of capital outflows. Policymakers need to balance a delicate trade-off between supporting economic recovery and managing financial stability risks.

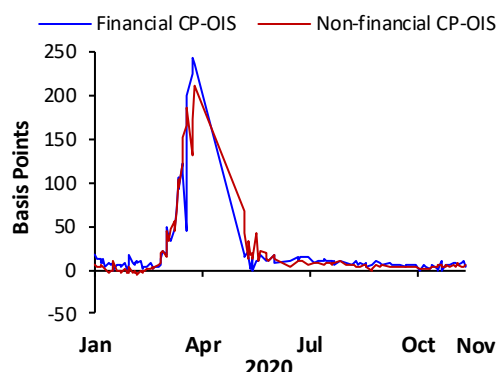
The COVID-19 pandemic has severely impacted global growth and raised risks to global financial stability, prompting authorities to provide swift and extensive support.

The onset of COVID-19 and the measures to contain its spread triggered a sudden fall in global economic activity, presenting significant downside risks to global financial stability. Liquidity stresses abruptly surfaced in the global financial system in March 2020, as worries about the economic impact of the pandemic triggered a global flight to safety and a sharp rise in short-term demand for safe assets (Charts 1.1 and 1.2). Investor flight away from riskier assets manifested in extreme volatility and a sharp repricing in financial markets (Chart 1.3).

Central banks and governments were swift in rolling out policy measures to tackle the market stress. Central banks provided liquidity to the financial system, through direct purchases of corporate bonds and foreign currency swap lines, to restore the smooth functioning of funding markets. Monetary policy was also eased in many major economies, alongside fiscal measures such as government risk-sharing via loan guarantees, which supported the flow of credit to the economy. Credit relief measures such as loan moratoria have also helped to support households and businesses facing cashflow difficulties and tempered the crystallisation of borrower defaults for banks.

Chart 1.1 USD liquidity conditions in the US tightened significantly in March 2020...

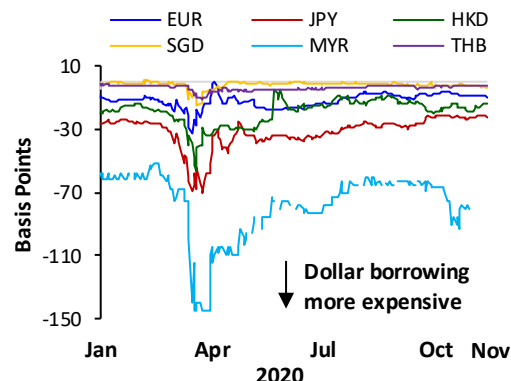
3-month Commercial Paper (CP) vs Overnight Indexed Swap (OIS) Spread



Source: MAS estimates, Board of Governors of the Federal Reserve System, Datastream

Chart 1.2 ...leading to a rise in USD funding costs globally

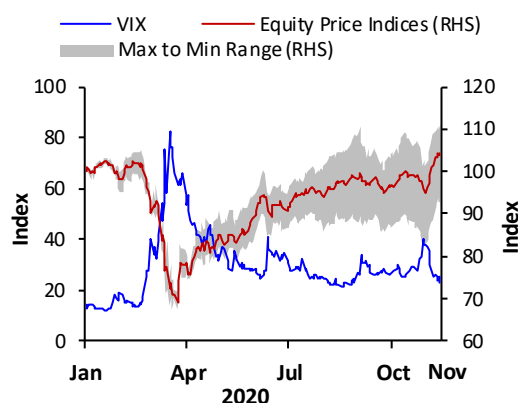
Selected Cross-currency Basis Swaps



Source: Bloomberg

Chart 1.3 Financial markets experienced extreme volatility and a sharp repricing in March 2020

VIX and Equity Price Indices



Source: MAS estimates, Bloomberg

Note: Includes S&P 500, Euro STOXX 600, Nikkei 225 and MSCI Emerging Markets indices.

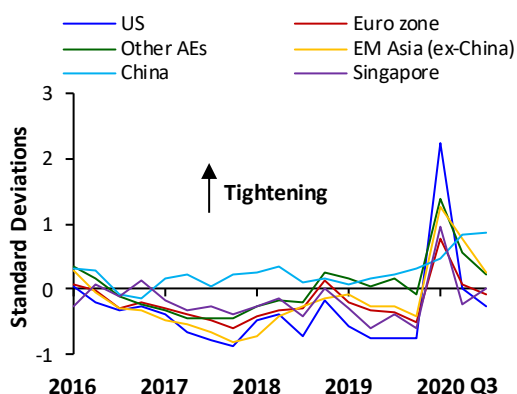
Aggressive monetary policy action has led to an easing of financial conditions, alleviating near-term financial stability risks...

The loosening of monetary policy across major advanced economies (AEs) has contributed to a significant easing of global financial conditions since the market turmoil in March 2020 (Chart 1.4). Financial conditions in Singapore have also eased in line with global developments, and on the back of accommodative domestic conditions. Compared to past crises, domestic factors that include the exchange rate, interest rate, as well as equity and property prices, have played a key role in keeping overall financial conditions relatively supportive. Within ASEAN, funding conditions have likewise improved (Chart 1.5). Meanwhile, financial conditions in China have remained relatively tighter than usual as policymakers have kept policy rates steady since April, scaling back expectations of further monetary policy

loosening and leading to a corresponding rise in interbank rates and bond yields. The policy shift came amid broadening economic recovery, allowing policymakers room to focus on containing rising financial sector risks and continuing with their medium-term objective of restraining the rise in overall banking system leverage.

Chart 1.4 Overall financial conditions have eased significantly since the tightening early in the year

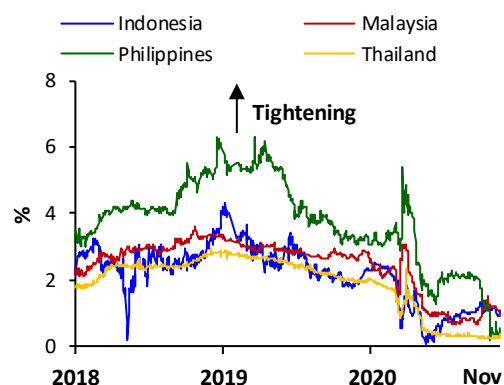
Global and Domestic Financial Conditions Indices



Source: MAS estimates, International Monetary Fund (IMF)

Chart 1.5 Funding conditions in the region have improved amid large-scale policy support

Implied USD Funding Rates



Source: Bloomberg

The global banking system confronted the crisis from a position of resilience. Together with governments and central banks' support measures, these factors have allowed banks to continue lending to households and businesses during the pandemic.

...but fuelling a build-up of vulnerability.

While such accommodative financial conditions have alleviated near-term financial stability risks by allowing credit to be more readily available and have likely forestalled even greater economic damage, they have also contributed to a rise in vulnerability.

Accommodative financial conditions, driven by central bank and government support measures, have helped to boost market confidence and, accordingly, financial asset prices. However, policy intervention has widened a disconnect between financial market valuations and the real economy as risk asset prices have rebounded despite lingering economic uncertainty (Chart 1.6). Any change in market expectations of economic recovery (e.g. premature withdrawal of policy support) could trigger a bout of disruptive adjustment in market pricing.

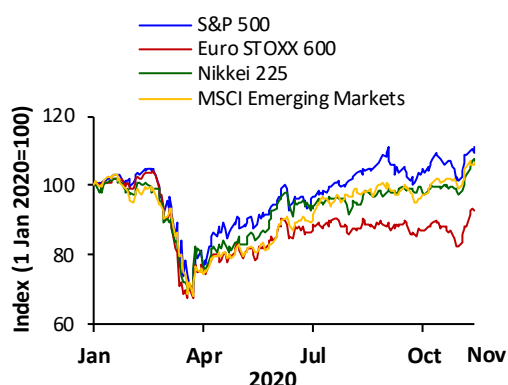
Accommodative financial conditions have also contributed to a rise in medium-term vulnerabilities by fuelling greater indebtedness alongside falling corporate and household incomes. Leverage has increased against a backdrop of already high corporate debt levels noted in the last FSR (Chart 1.7). These developments highlight concerns regarding debt sustainability and corresponding credit risk to the financial sector, especially as the global economy now faces an uncertain recovery path.

Within the region, corporate vulnerabilities have risen sharply over the past year. This is measured by an FVI that MAS constructed based on a principal component analysis for the ASEAN-4² corporate sector (Chart 1.8). The FVI considers four broad categories of financial vulnerability—leverage, liquidity mismatch, maturity mismatch and foreign currency mismatch, using quarterly data for each vulnerability category.

Based on the ASEAN-4 FVI, the rise in vulnerabilities has largely been driven by higher leverage and foreign currency mismatch risks. ASEAN-4 corporates have taken on more debt to meet increased financing needs, leading to growing corporate leverage and a rise in foreign currency-denominated bonds. As emerging market currencies are prone to depreciation pressures in a risk-off environment, corporates in such emerging economies could be more susceptible to foreign currency mismatch risk, especially if their liabilities are largely unhedged.

Chart 1.6 Equity markets have rebounded since the sharp repricing in March

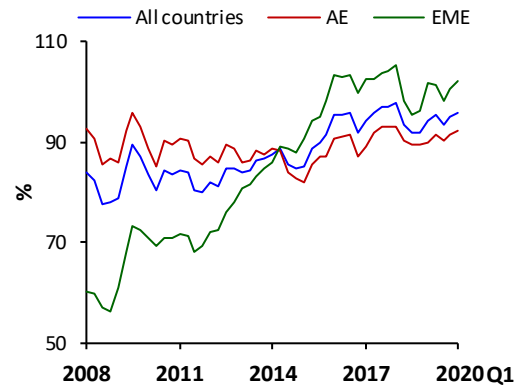
Selected Equity Market Indices



Source: MAS estimates, Datastream

Chart 1.7 Corporate debt has continued to increase from elevated levels

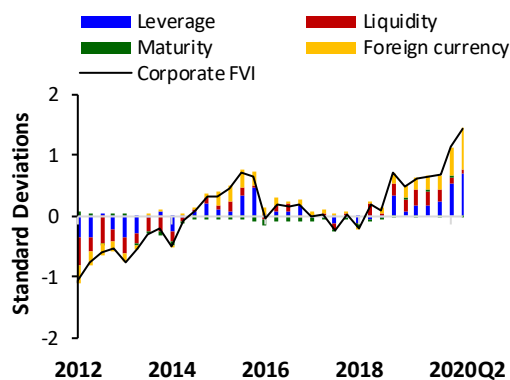
Non-financial Corporate Debt as % of GDP



Source: Bank for International Settlements (BIS)

Chart 1.8 ASEAN-4 corporate vulnerabilities have risen in recent quarters, driven by leverage and foreign currency mismatch risks

ASEAN-4 FVIs



Source: MAS estimates, Asian Development Bank (ADB), Bloomberg, Institute of International Finance (IIF), Haver, Worldscope

² ASEAN-4 comprises Indonesia, Malaysia, the Philippines and Thailand.

Further accumulation of corporate debt could increase credit risk, discourage bank lending, and tighten financial conditions.

Given the uneven and protracted economic recovery in many jurisdictions and the risk of renewed movement controls arising from bouts of epidemic flare-ups, many businesses and households are likely to continue to grapple with weaker revenue and earnings for some time. As more debt is accumulated, business and household incomes may not catch up with their increased servicing costs (for example, in cases of business or job redundancy arising from COVID-19), and borrowers could eventually default.

The materialisation of this credit risk will increase bank credit losses, and may prompt bank capital adequacy and solvency concerns that could weaken their ability and willingness to lend. Financial conditions may subsequently tighten and hinder the economic recovery. This situation could be further exacerbated in regions where banks had already been facing weak profitability amid a prolonged low interest rate environment.

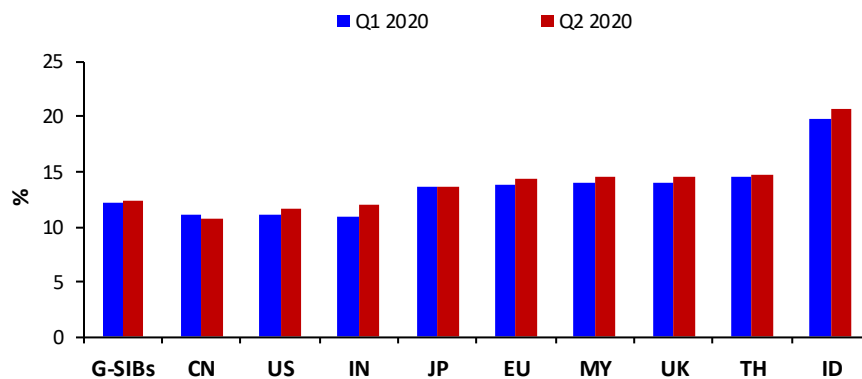
Against these risk factors, large, global systemically important banks (G-SIBs) generally maintain strong capital positions and thus have more buffers to mitigate the deterioration in credit quality (Chart 1.9).³ Banks in general had also been increasing their loan loss provisions to buffer against rising credit losses during the initial months of the pandemic. However, the economic recovery in Q3 across various jurisdictions might have induced banks to reduce loan provisioning and added pressure for them to restart dividend payments. Given the uncertainty over the path of the pandemic, such actions may pose some risk to banks' future loss absorbing capacities.

Conversely, smaller banks may face greater risk due to relatively higher exposures to riskier segments, for instance through loans to small and medium-sized enterprises (SMEs). SMEs typically have less financial buffers to withstand the short-term cashflow shocks, and are more likely to encounter loan repayment difficulties. Smaller banks are also likely to have less geographically diversified portfolios, and therefore higher concentration risk to movement restrictions in individual jurisdictions, compared to larger banks.

³ IMF (October 2020), Global Financial Stability Report.

Chart 1.9 Bank capital positions have remained adequate since the outbreak of COVID-19

Weighted Average of Common Equity Tier 1 to Risk-weighted Assets Ratio, for G-SIBs and selected economies



Source: MAS estimates, SNL Financial

Note: Capital levels weighted by banks' share of total risk-weighted assets within each group.

Managing continued policy support well will mitigate financial stability risks.

Broad-based policy support from governments has helped to stabilise the immediate financial position of firms quickly and prevented liquidity pressures from developing into solvency issues. However, continuing broad-based support including towards unviable corporates could delay the resource reallocation that needs to take place as well as the recognition of unavoidable credit loss, even as the financial stability risk associated with unsustainable debt rises.

Policy support also cannot continue indefinitely, given the increasing strains on fiscal sustainability. Policymakers thus face the challenge of unwinding the support measures in an orderly manner that minimises the risk of a sharp spike in bankruptcies and defaults. Policy missteps from a premature withdrawal of support measures could result in a credit risk "cliff effect" and widespread defaults, particularly by more vulnerable firms and households. As banks recognise a larger amount of loan losses, they may pull back on credit supply. This would forestall the recovery and cause further stress on underlying financial sector vulnerabilities.

Liquidity stresses in the non-bank financial sector are a source of vulnerability.

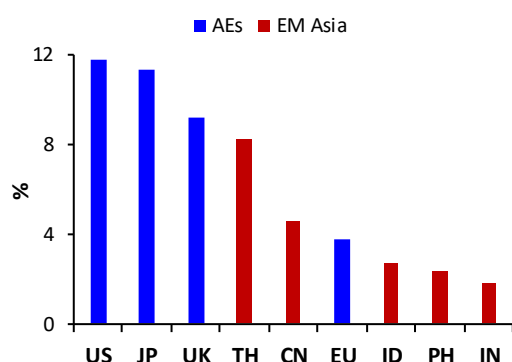
The increased demand for precautionary liquidity placed strains on the non-bank financial sector in March this year. While central bank measures have since provided some reprieve, liquidity stresses in the non-bank financial sector remain a source of vulnerability. Central bank policy support to the non-bank financial sector remains largely indirect, working through banking systems or through asset market intervention. The hoarding of liquidity by banks as well as limits and criteria adopted by asset purchase programmes could reduce the efficacy of these measures. Consequently, adequate support may fail to reach non-bank financial institutions in a timely manner should risk aversion and liquidity stresses return.

Sovereign debt as a percentage of GDP increased, raising public debt sustainability concerns.

The unprecedented fiscal stimulus that governments have rolled out to mitigate the impact of COVID-19 (Chart 1.10) coupled with a reduction in tax revenues has caused fiscal deficits to widen globally in 2020.⁴ Together with the economic contraction caused by COVID-19, sovereign debt as a percentage of GDP rose and could continue to do so should further fiscal support be deemed necessary (Chart 1.11).

Chart 1.10 Fiscal stimuli in response to the COVID-19 shock have been large across AEs and EMEs...

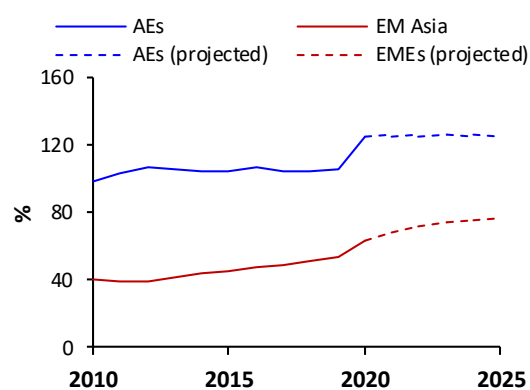
Size of Fiscal Response to COVID-19 as % of GDP



Source: IMF

Chart 1.11 ...leading to higher public debt as a percentage of GDP

Gross Public Debt as % of GDP



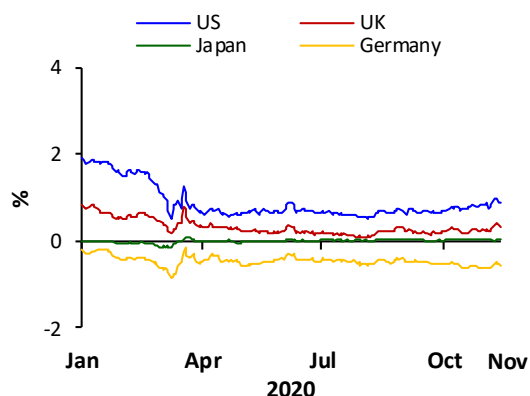
Source: IMF

Accommodative monetary policy and asset purchase programmes by central banks since the onset of the COVID-19 shock have brought down the cost of borrowing and helped to temper sovereign debt sustainability concerns (Charts 1.12 and 1.13). In Europe, spreads of euro zone periphery sovereign debt have been further contained by some fiscal sharing through the inception of the "Next Generation EU" (NGEU) Recovery Fund in July 2020. Sovereign spreads so far appear to be well contained in AEs and ASEAN-4 economies. However, increasing sovereign debt as a percentage of GDP raises the risk that a loss of investor confidence places renewed pressure on debt refinancing.

⁴ The IMF estimates the global aggregate fiscal deficit will widen by 8.8% points to 12.7% in 2020.

Chart 1.12 AE sovereign yields have fallen on the back of central bank actions...

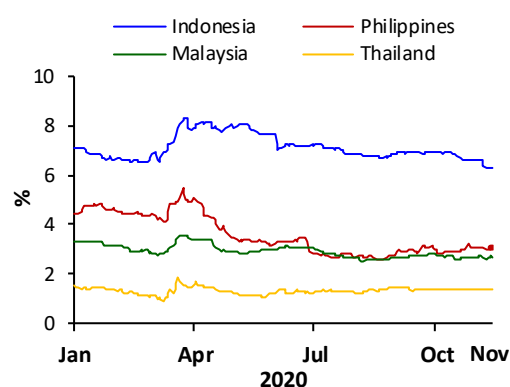
10-year Sovereign Yields for Selected AEs



Source: Datastream

Chart 1.13 ...as have ASEAN-4 sovereign yields

10-year Sovereign Yields for ASEAN-4

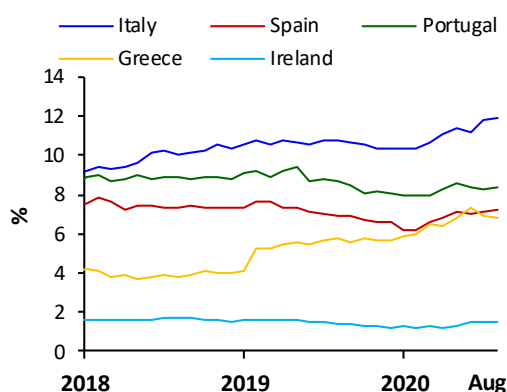


Source: Datastream

The large issuance of government securities to fund fiscal responses to COVID-19 is also likely to have deepened interconnections between sovereigns and banks. For example, banks in euro zone periphery economies have been increasing their holdings of sovereign debt since the pandemic broke out (Chart 1.14). Higher bank exposure raises the likelihood of spillovers to the banking system through two main channels, should sovereign debt sustainability concerns come to the fore. First, a repricing of government securities could lead to significant mark-to-market losses for banks, with knock-on effects to capital positions. Second, a decline in sovereign credit ratings could impact banks' own credit ratings, increasing bank funding costs, which could curtail their willingness and ability to provide credit.

Chart 1.14 Banks' holdings of domestic sovereign debt in the euro zone periphery have increased since the outbreak of COVID-19

Euro Zone Periphery Banks' Holdings of Domestic Sovereign Debt as % of Total Assets



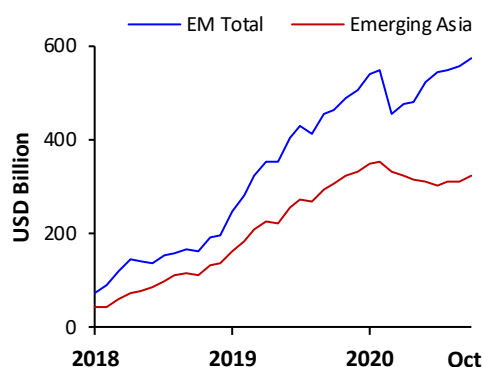
Source: MAS estimates, ECB

Volatile portfolio capital flows could exacerbate EME weakness.

The 2019 FSR flagged that low global interest rate conditions have increased portfolio flows to EMEs, making their financial conditions more sensitive to global shocks. The onset of the COVID-19 shock in March this year led to heightened risk aversion globally, causing EMEs to experience acute, broad-based portfolio capital outflows (Chart 1.15), and currency depreciation (Chart 1.16). Capital flight has moderated since April on the back of supportive policies (in both AEs and EMEs) that shored up confidence and helped improve liquidity conditions. That said, capital flows to EMEs remain volatile (Chart 1.17), and investor confidence fragile.

Chart 1.15 COVID-19 triggered acute capital outflows from EMEs in Q1 2020...

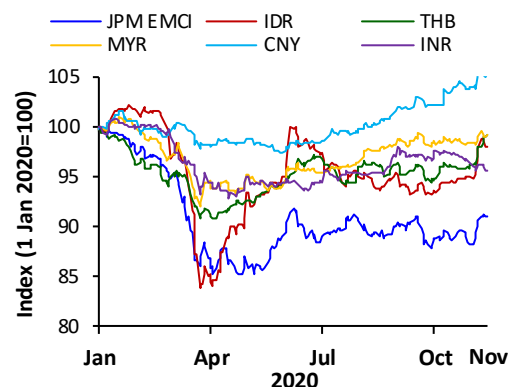
Cumulative Non-resident Portfolio Flows to EMEs



Source: IIF

Chart 1.16 ...which have contributed to the depreciation of EME currencies this year

EME Exchange Rates Against the USD

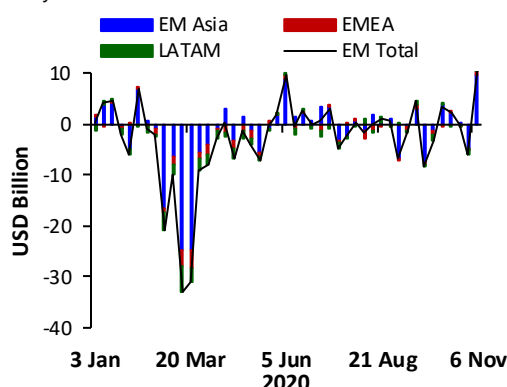


Source: Datastream

Note: JPM EMCI = JP Morgan Emerging Market Currency Index

Chart 1.17 Capital flows to EMEs have moderated, but remain volatile

Weekly Non-resident Portfolio Flows to EMEs



Source: MAS estimates, IIF

Note: EMEA = Europe, Middle East and Africa; LATAM = Latin America

Several developments could trigger risk-off sentiment that would increase EMEs' vulnerability to a resurgence of capital outflows. First, subsequent waves of infection could drive further sudden stops in economic activity, which could be amplified if countries are

perceived to be running out of fiscal or monetary policy space to provide further support. Second, geopolitical risks and corresponding policy uncertainty might also adversely impact investors' confidence.

A recent MAS study on the relationship between capital flows in EMEs and USD movements found that episodes of broad USD appreciation (which may occur, for instance, if EMEs' economic recovery lags AEs, due to difficulties deploying vaccines in EMEs) have been associated with lower net capital flows to EMEs (see Special Feature 1 "Implications of USD Dominance on Capital Flows and Financial Stability in EMEs").

These developments, should they materialise, could exert further downward pressure on currencies and increase interest rates on securities, making it difficult for EMEs to obtain external financing. Financial conditions would tighten, further reducing debt servicing ability of corporates and sovereigns.

While COVID-related downside risks to financial stability must be carefully managed, there is a need to plan ahead for risks arising from longer-term structural shifts in the operating environment.

The global economic outlook will continue to be subject to risks, for as long as the course of the pandemic remains uncertain, with resultant implications for Singapore. Based on MAS' latest Survey of Professional Forecasters in September 2020, respondents cited a tightening in global financial conditions, an escalation of the global COVID-19 situation and heightening US-China tensions as the top three factors⁵ that could potentially weigh on financial market and lending conditions in Singapore. Conversely, recent progress in vaccine development provides some upside to the global economic prospects in the latter half of 2021, although much depends on their effectiveness and deployment. Financial stability risks will thus remain elevated in the medium term. Policymakers will have to tread a careful balance between supporting economic recovery while managing risks to financial stability. These downside risks need to be adequately recognised, especially given the present disconnect between market valuations and the real economy. Policymakers will need to work with industry to strengthen the resilience of the financial system, including by facilitating changes to market structure that address the vulnerabilities revealed by the crisis.

Amid the many challenges posed by the economic effects of the COVID-19 pandemic, there is a need to prepare for other mid- to longer-term shifts which could pose risks to financial stability. With LIBOR (London Interbank Offered Rate) expected to discontinue after 2021, policymakers are working intensively to facilitate the shift in global interest rate benchmarks (see Special Feature 2 "Transitioning to a New Interest Rate Benchmark Landscape in Singapore"). Climate change presents complex risks to financial institutions (FIs) via their exposure to entities that could be significantly impacted by physical or transition risks. FIs would need to build up resources to assess, manage and mitigate these risks (see Special Feature 3 "Climate Change and Financial Stability").

⁵ Of the survey respondents, 76.9% cited a tightening in global financial conditions, followed by 38.5% each for an escalation of the global COVID-19 situation and heightening US-China tensions.

2 Singapore Corporate Sector

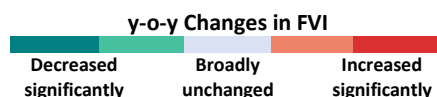
The onset of the pandemic led to a sharp contraction in the global economy, with the Singapore economy also registering its sharpest decline on record in Q2 2020 before rebounding in Q3. As with the experience in most other economies, businesses in Singapore experienced severe earnings shocks.

Corporate balance sheets in Singapore were relatively resilient entering 2020, amid elevated but stable debt levels prior to the COVID-19 pandemic. Swift government support for the corporate sector has further helped to alleviate short-term liquidity pressures, with efforts focused on extending credit to support SMEs. While leverage and maturity risks have increased in the near term, most Singapore firms should be able to withstand short-term pressures on their financial positions. However, corporates with highly leveraged positions and smaller firms with weaker cash buffers remain vulnerable and could come under more severe strain.

Overall liquidity positions improved in Q2 2020, as firms increased cash holdings in light of the global uncertainty. However, higher cash holdings have been accompanied by an increase in short-term debt and maturity risk. Foreign currency mismatch risk has remained stable as Singapore firms continued to avoid over-reliance on foreign currency borrowings.

In the context of still-heightened levels of corporate debt built up since 2011, the sector's leverage risk and overall FVI in Q2 2020 were broadly unchanged, staying at the same elevated band as in Q2 2019.

Corporate Sector FVI (y-o-y Changes)	Q2 2019 ⁶	Q2 2020
Overall Corporate FVI	→	→
Leverage risk	→	→*
Liquidity risk	→	↘
Maturity risk	↘	↗
Foreign currency risk	→	→

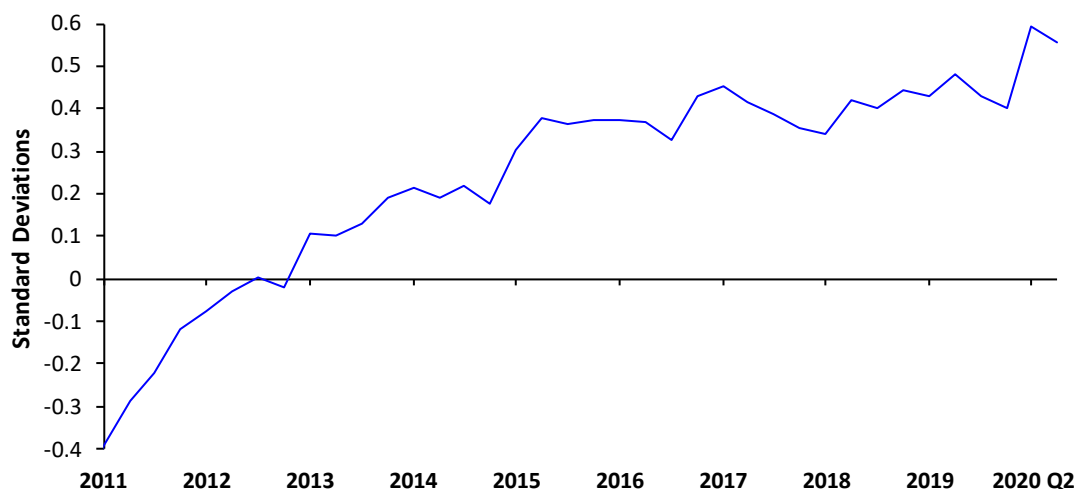


*Leverage risk has risen in 2020 within the same elevated band as Q2 2019 (Chart 2.1).

⁶ The corporate FVI has been revised since the 2019 FSR to improve robustness and accuracy of the indicator by changing how firms with negative earnings were accounted for. The y-o-y changes shown here for 2019 reflect the revised indicator.

Chart 2.1 Corporate sector leverage risk had been elevated since 2015, and rose further in 2020

Corporate Sector Leverage Risk (a Component of the Corporate Sector FVI)



Source: MAS estimates, Thomson Financial, Department of Statistics (DOS)

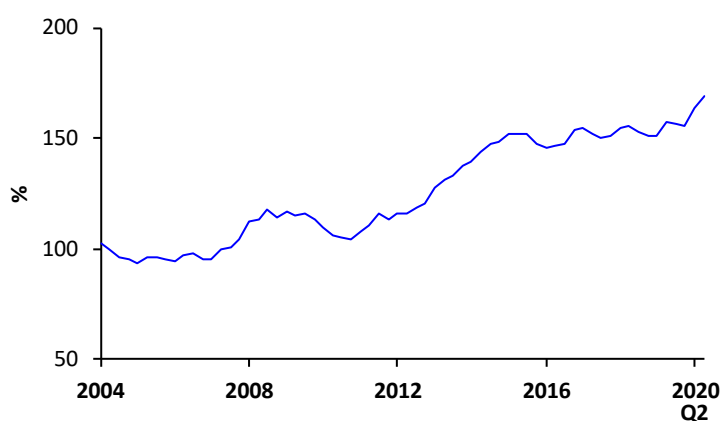
2.1 Leverage risk has worsened as earnings fell and debt rose

Corporate debt rose further in 2020 from already elevated pre-COVID levels.

Driven by the low interest rate environment since the Global Financial Crisis (GFC), corporate debt as a percentage of GDP increased between 2010 and 2014 before stabilising at around 150% of GDP from 2015 to 2019. However, debt ratios have since risen further and reached 163% in Q2 2020 (Chart 2.2), reflecting both a pick-up in corporate debt, as well as a fall in GDP.

Chart 2.2 Corporate debt as a percentage of GDP has risen since 2011

Singapore's Corporate Debt as % of GDP



Source: MAS estimates, BIS, Dealogic

Corporate earnings have fallen significantly across all sectors, led by domestically-oriented firms.

Corporate profitability, measured by return on assets (ROA), has been declining since 2011, and was more recently led by the fall in earnings of firms in externally-oriented sectors on the back of the global slowdown from Q2 2018 to Q2 2019. This trend was exacerbated in 2020, with the median ROA of SGX-listed firms decreasing from 1.9% in Q2 2019 to 0.3% in Q2 2020 (Chart 2.3), as earnings fell across most sectors.

While previous recessions in Singapore were typically driven by externally-oriented⁷ sectors, the COVID-19 shock has disproportionately affected domestically-oriented sectors, reflecting the sharp drop in domestic demand and safe management measures that limited activity (Chart 2.4). The median ROA for the construction sector declined by 3.1 percentage points between Q2 2019 and Q2 2020, and turned negative. The construction sector, already weighed down by a prolonged contraction in demand, suffered further from safe management measures that greatly reduced operating capacity. That said, activity in the construction sector should improve, given the backlog of construction projects that had accumulated due to these safe management measures. The value of contracts awarded—a leading indicator of construction activity—is also expected to rebound in 2021. Restrictions in movement similarly affected travel-related services such as those in the hotels & restaurants sector, and the property sector, which experienced ROA declines of 3.4 percentage points and 2.2 percentage points respectively in the same period.

Firms within the externally-oriented sectors also saw earnings weakness, albeit to a lesser extent. A slowdown in global activity contributed to larger earnings falls in the commerce and manufacturing sectors, which saw ROAs fall by 1.2 percentage points and 0.9 percentage points respectively from Q2 2019 to Q2 2020. Firms in the transport, storage and communications (TSC) sector were affected by disruptions to international travel, and continued weakness in oil-related industries such as shipping and the marine and offshore engineering subsectors. Indeed, the oil trading industry was affected by the collapse of global oil prices, as the COVID-19 pandemic weakened global oil demand faster than oil exporters could credibly commit to cut production.

Debt servicing ability remains manageable for SGX-listed firms despite deterioration in profitability.

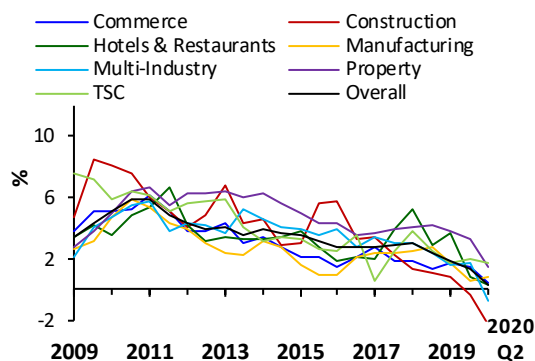
Deteriorating profitability and increasing leverage have weighed on debt servicing ratios, with the median interest coverage ratio (ICR) of SGX-listed firms falling from 2.1 in Q2 2019 to 1.1 in Q2 2020 (Chart 2.5). These trends suggest that firms are at greater risk of not meeting their debt obligations from earnings. Sectors more affected by the pandemic such as construction, hotels & restaurants, multi-industry, commerce, and manufacturing saw median ICRs of below one in Q2 2020. Nevertheless, firms' debt servicing ability remains sufficient in

⁷ Externally-oriented sectors comprise manufacturing, wholesale, transportation & storage, finance & insurance, information and communication technology, and business services. Domestically-oriented sectors include construction, F&B, retail, other services and utilities.

the immediate term—the median ICR of SGX-listed firms remains above one, and cashflows are expected to recover after Q2 2020.

Chart 2.3 Corporate earnings have fallen significantly in 2020

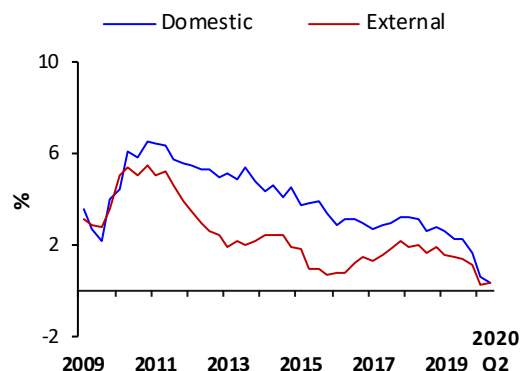
Median ROA of SGX-listed Firms



Source: MAS estimates, Thomson Financial

Chart 2.4 Domestically-oriented firms experienced a larger drop in ROA than externally-oriented ones

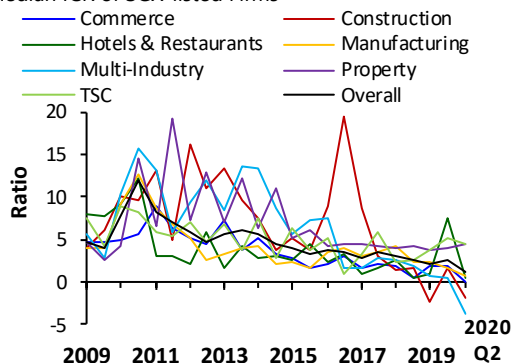
Median ROA of Domestically- and Externally-oriented SGX-listed Firms



Source: MAS estimates, Thomson Financial

Chart 2.5 Deteriorating profitability has weighed on debt servicing ratios

Median ICR of SGX-listed Firms



Source: MAS estimates, Thomson Financial

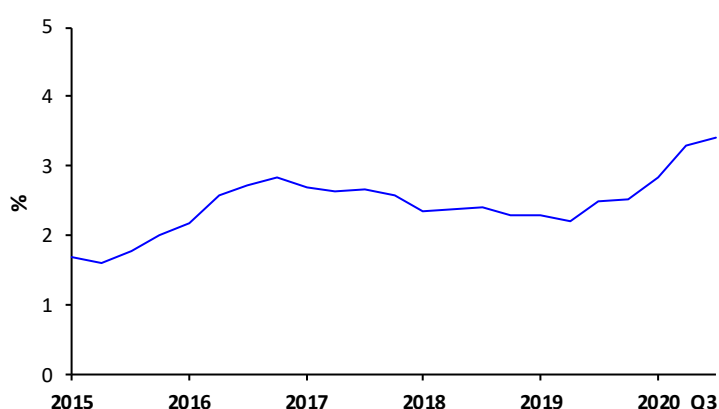
Corporate non-performing loan ratio has risen, and could worsen further if earnings weakness is prolonged.

As earnings declined, the Singapore banking system's corporate non-performing loan (NPL) ratio rose to 3.4% in Q3 2020 compared to 2.5% in Q3 2019 (Chart 2.6). There was a broad-based increase in NPL ratios across sectors, with the exception of building and construction, where NPLs remained relatively stable. The general commerce sector and the TSC sector experienced the largest increases, with NPL ratios increasing from 1.5% and 9.6% in Q3 2019 to 5.9% and 11.7% in Q3 2020 respectively. The higher NPL ratios reflect the prevalence of firms in sectors adversely affected by the COVID-19 pandemic—wholesale trade, retail, food and beverage (F&B), tourism-related industries—and oil-related industries impacted by the collapse of oil prices earlier in the year.

While corporate loan asset quality has deteriorated, banks' provisioning coverage has improved (see Chapter 4 "Singapore Financial Sector"). Corporate NPL ratios have risen only slightly on aggregate, partly due to the extensive support provided by the broad-based Jobs Support Scheme (JSS) and loan relief measures extended by the banking industry to SMEs at the onset of the crisis. Nonetheless, with economic recovery expected to be gradual and uneven, loan asset quality is expected to deteriorate, particularly in sectors with prolonged earnings weakness.

Chart 2.6 Singapore banking system's corporate NPL ratio has risen in 2020

Singapore Banking System's Corporate NPL Ratio



Source: MAS

2.2 Firms have taken proactive steps to retain liquidity

SGX-listed firms' efforts to increase cash holdings have been accompanied by an increase in short-term debt.

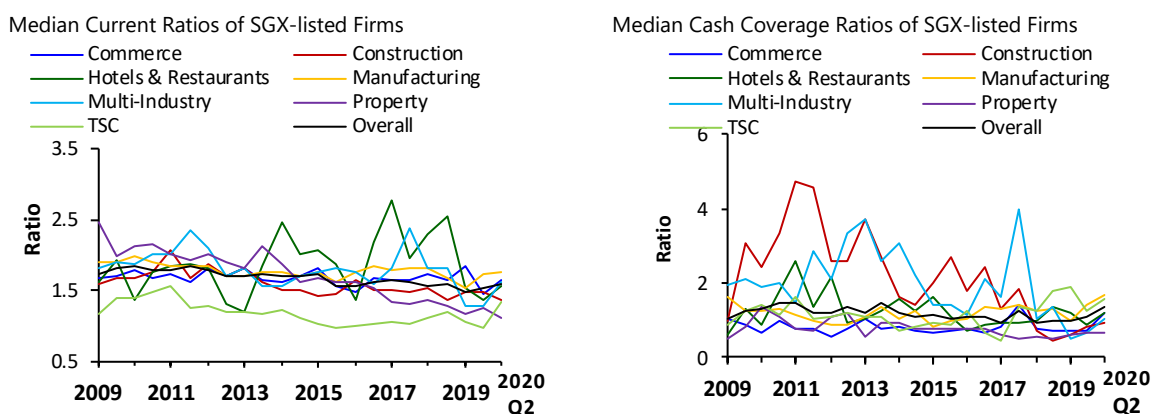
As a response to the severe earnings shock, firms have bolstered cash buffers, resulting in healthier liquidity indicators. The median current ratio of SGX-listed firms remained stable, rising slightly from 1.5 in Q2 2019 to 1.6 in Q2 2020, while median cash coverage ratio (cash to short-term debt) improved from 1.0 in Q2 2019 to 1.3 in Q2 2020 (Chart 2.7).

Higher cash holdings provide firms the necessary buffer for their daily operations, and reflect firms' precautionary demand for cash and their ability to continue accessing the bond market and bank financing. Despite heightened volatility during the onset of the COVID-19 crisis, the Singapore corporate bond market has remained accessible to firms, facilitated by measures⁸ to support financial markets and businesses (see Box A "Singapore's Corporate Bond Market: A Growing Source of Domestic Finance").

⁸ These include the launch of the MAS USD facility, which contributed to the stability of USD funding conditions in Singapore and helped the USD bond market recover and stay relatively resilient.

However, firms' efforts to ease short-term cashflows through increased cash holdings have also increased short-term debt and maturity risks. Short-term debt is likely to be repaid as revenues recover on the back of an economic rebound. However, some firms may face repayment difficulties in a scenario of uneven economy recovery and prolonged earnings weakness, especially if they need to refinance longer-term debt. Maturity risk—measured by the proportion of short-term debt to total debt—rose in 2020 from 40% in Q2 2019 to 42% in Q2 2020, remaining below the historical average (Chart 2.8). The bond maturity profile of Singapore firms also remains well termed out, with bonds due by 2021 making up about 15% of outstanding bonds.

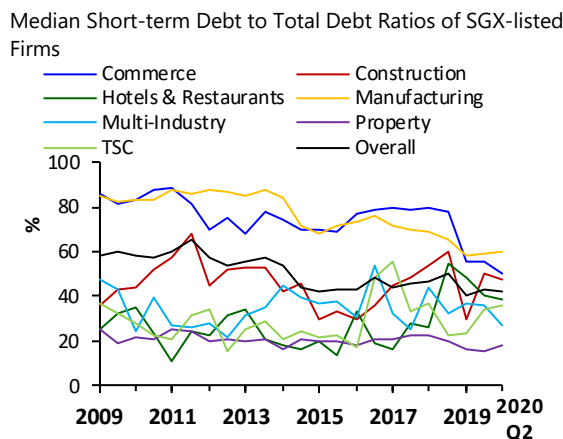
Chart 2.7 Liquidity indicators—current ratio and cash coverage ratio—improved in 2020



Source: MAS estimates, Thomson Financial

Source: MAS estimates, Thomson Financial

Chart 2.8 Short-term debt to total debt ratios have remained low, despite increasing in 2020



Source: MAS estimates, Thomson Financial

Foreign currency mismatch risk remained stable.

Foreign currency mismatch risk has remained stable. Outstanding foreign currency bond value in Singapore as a proportion of total outstanding bond value has remained at around 60% since 2018. Further, foreign currency bond issuance is driven by externally-oriented firms, which derive a sizeable proportion of their revenues from abroad and would be able to offset currency mismatch risk by issuing debt in a matching currency (see also Box A).

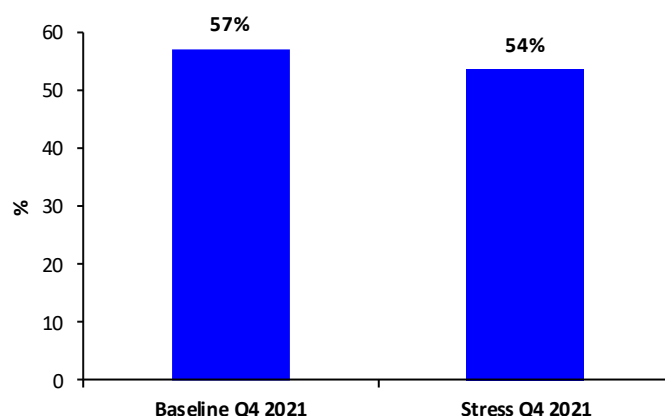
MAS' simulations suggest that Singapore's corporate sector remains resilient to cash flow shocks.

MAS' simulations on SGX-listed firms suggest that corporates would generally have sufficient cash buffers to repay short-term debt, without refinancing. Under a baseline scenario, firms' Q2 2020 cash balances, revenues and expenses were projected to end-2021 using projected sectoral Value-Added (VA) growth figures, to compute cash coverage ratios of cash buffers to short-term debt at end-2021. The simulations showed 57% of firms having cash coverage ratios above 1.5, suggesting they have sufficient cash to repay their short-term debt (Chart 2.9).

Under an adverse scenario where a resurgence of COVID-19 cases and disruptions to economic activity cause 2021 revenues to decline by 20% below the baseline, the majority of SGX-listed firms (of about 54%) would still have resilient cash buffers as of end-2021, which is a small deterioration of 3% of firms relative to the baseline. Most firms with more vulnerable cash buffers at end-2021 are small, with annual revenues of less than SGD100 million.

Chart 2.9 Most Singapore corporates would have sufficient cash buffers to short-term debt as of end-2021, and are resilient to further cash flow shocks

Share of Firms with Cash Coverage Ratio Above 1.5 at end-2021



Source: MAS estimates, Thomson Financial

Smaller firms are expected to have weaker cash buffers and would be more severely impacted by the current crisis.

MAS has also conducted an analysis of the financial position of unlisted firms⁹, based on FY2018 financial data, alongside the more timely and comprehensive data for SGX-listed

⁹ Using financial data from about 65,000 firms which reported financial statements to the Accounting and Corporate Regulatory Authority of Singapore (ACRA), the data was projected to 2020 levels using sectoral VA growth forecasts.

entities. Businesses with low cash buffers relative to liabilities due within a year and on-going operational costs, were identified as relatively more vulnerable.

The study found that SMEs generally have weaker cash positions. The proportion of vulnerable SMEs was estimated to be about 30 percentage points higher than that of vulnerable large firms. After accounting for their projected revenue in 2020, sectors which were most affected by the pandemic—accommodation, F&B, construction, and retail—had a higher proportion of firms with weaker cash positions (including SMEs). These sectors have between 11% and 17% more vulnerable firms compared to the share of vulnerable firms across the rest of the economy. The median cash to current liabilities ratio of the more COVID-impacted sectors was also 12 percentage points lower. These results confirm the expectation that liquidity stresses would hit smaller firms in COVID-impacted sectors more severely. Such firms, with their smaller size and limited access to capital markets, could thus be less able to weather the pandemic's impact on revenue and face more severe strains that could impair their debt servicing ability and business operations.

2.3 Financing conditions remain supportive against an uneven economic recovery

SME financing conditions remain supportive.

Bank lending to SMEs has increased from Q2 2019 to Q2 2020, with outstanding SME loans growing 15% y-o-y as of Q2 2020 (see Chart Panel 2A “Small and Medium-sized Enterprise Financing Conditions”). The expansion was largely a result of government financing schemes, which were enhanced this year with higher government risk share of 90%¹⁰ for qualifying loans extended by FIs. MAS also introduced the MAS SGD Facility, which provides FIs intermediating government financing schemes with low-cost funding. Together, these measures allow FIs to extend more loans at lower rates to SMEs.

Business outlook has declined in 2020, but expectations have improved since Q2 2020.

While the Singapore economy has rebounded strongly in Q3 2020, growth is expected to slow in Q4 and remain modest in 2021. Growth is expected to be uneven, and some pockets of the economy are not expected to recover to pre-COVID levels even by the end of 2021. Against this backdrop, the general outlook for the manufacturing and services sectors for October 2020 to March 2021 remains negative, although it has improved from Q2 2020 (Chart 2.10). In the Q4 2020 Business Expectations surveys by the Economic Development Board (EDB) and the Department of Statistics (DOS), a net weighted balance of 3% of manufacturers and 5% of firms in the services sector predicted a less favourable business situation, although the outlook varies significantly across subsectors.

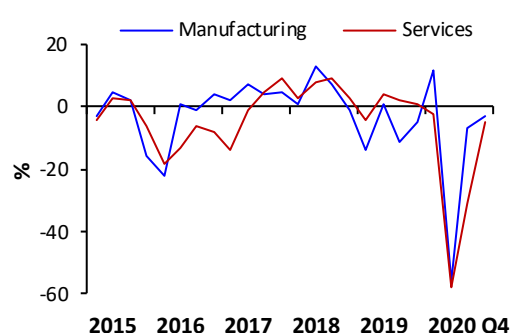
¹⁰ Enterprise Singapore (ESG) had increased its risk share of loans initiated under various loan schemes from 50–70% (pre-COVID) to 90%.

Within the manufacturing sector, most subsectors, apart from the electronics and chemicals subsectors, had an overall positive outlook for business expectations. A net-weighted balance of 16% and 3% of respondents in the respective subsectors had the view that conditions would deteriorate. In comparison, the services sector unsurprisingly had a more bearish outlook, with almost all subsectors having a pessimistic outlook for the next six months. The real estate and accommodation & food services subsectors had the least favourable business sentiment, with 45% and 22% of the subsectors respectively predicting less favourable conditions.

Business sentiment among SMEs has also deteriorated. The Singapore Business Federation (SBF)-Experian SME Index¹¹ contracted from 50.6 in Q3 2019 to 46.3 in Q3 2020 (Chart 2.11), the lowest reading since the inception of the SME index in 2009. According to Experian, business expansion and capital investment expectations remain unchanged from end-2019 to mid-2020, suggesting that SMEs across most sectors remain cautious, scaling back growth and expansion plans amid the ongoing economic uncertainty.

Chart 2.10 Business outlook has expectedly declined in 2020, although it recovered in Q4

General Business Outlook for Manufacturing & Services Sector (Net Weighted Balance¹²)

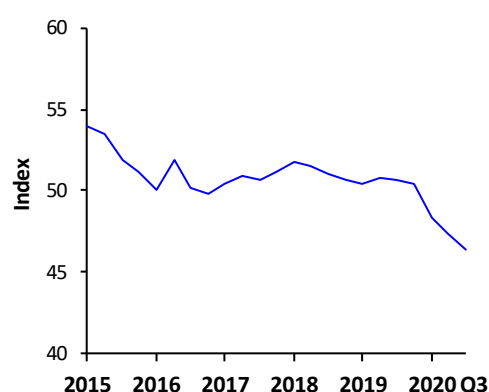


Source: DOS, EDB

Note: The latest data point (surveyed in Q4 2020) refers to the business outlook for October 2020 to March 2021.

Chart 2.11 SMEs expect much lower business activity in the next six months

SBF-Experian SME Index



Source: SBF-Experian

Note: The latest data point (surveyed in Q3 2020) refers to the business outlook for October 2020 to March 2021.

Economic recovery expected in 2021 should bolster corporate financial positions, but pockets of risks remain for smaller and financially weaker firms.

Singapore firms have weathered the initial earnings shock from COVID-19 relatively well, with firms taking proactive steps to retain liquidity, further supported by initiatives from the government, MAS, and the financial industry. These measures eased cashflows and provided

¹¹ The SBF-Experian SME Index is a six-month forward-looking index. It measures the sentiments of SMEs in the areas of turnover, profitability and capacity utilisation. A value above 50 indicates that SMEs expect business to improve in the next 6 months, whereas a value below 50 indicates that they expect lower business activity.

¹² "Net weighted balance" is the difference between the weighted percentages of respondents with a positive outlook and those with a negative outlook. A positive percentage indicates a net positive outlook and a negative percentage indicates a net negative outlook.

continued access to financing. Nonetheless, even as firms eased short-term refinancing concerns through increased cash holdings, debt sustainability in the medium-to-long term poses increased risks.

While the economy is expected to pick up next year, its uneven trajectory will impinge on corporate profits. Within the corporate sector, smaller firms that tend to be financially weaker, and those in the domestically oriented and travel related services impacted by COVID-19 remain vulnerable.

Box A**Singapore's Corporate Bond Market: A Growing Source of Domestic Finance**

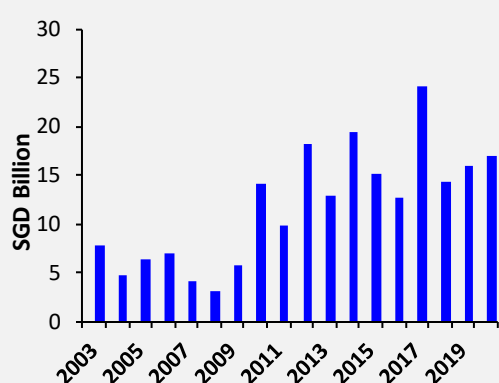
Singapore's corporate bond market¹³ has grown and deepened over the last decade. The bond market plays an important part of the corporate financing landscape, after bank credit.¹⁴ Amid the uncertain global outlook caused by the COVID-19 pandemic, the domestic bond market has held up well and continued to serve corporates' financing needs. However, the pandemic has heightened pre-existing leverage risks among issuers in more affected sectors, and those which face maturing bonds through 2021.

Singapore's corporate bond market has grown, providing SGD and USD funding to a diverse set of corporates.

Corporate bond issuance in Singapore has grown steadily to SGD17 billion in the first three quarters of 2020, compared to an annual average of less than SGD10 billion pre-2010 (Chart A1). Growth has been supported in recent years by the Asian Bond Grant (ABG) Scheme, which was launched on 1 January 2017 and covered up to 50% of the issuance-related expenses for first-time Asian borrowers in the Singapore bond market. The ABG Scheme has catalysed an increase in debutant Asian issuers which rose from an average of SGD14 billion per annum in 2009–2016, to SGD18 billion in 2017–2020 (Chart A1).

Chart A1 Singapore's corporate bond market has grown*

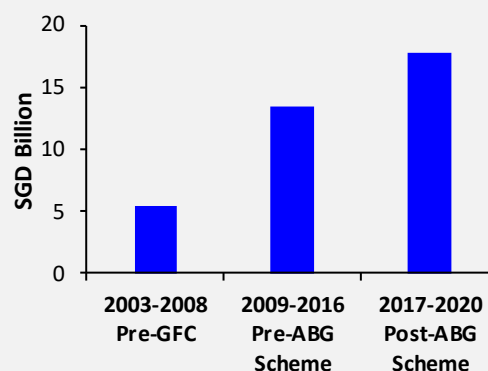
Total Bond Issuances by Non-financial Corporates (NFCs) in Singapore (All Currencies), yearly



Source: Dealogic

*2020 figures were only taken up till Q3 2020

Average Annual Bond Issuances by Non-financial Corporates in Singapore (All Currencies)



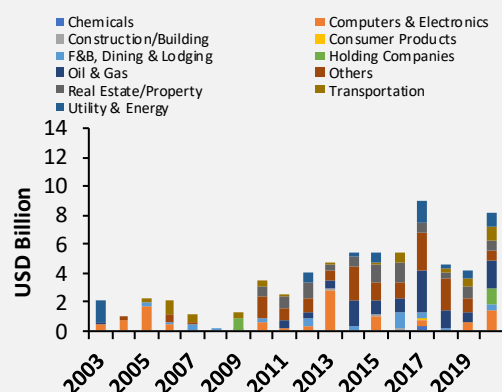
¹³ This includes bonds issued by Singapore-incorporated non-financial corporates (in Singapore or elsewhere), and bonds of foreign-incorporated non-financial corporates issued in Singapore.

¹⁴ Bank financing remains a dominant source of financing for Singapore firms. Based on a 2015 study using SGX-listed firms as a proxy, MAS estimates that 80% of firms derived more than three-quarters of their financing from bank loans.

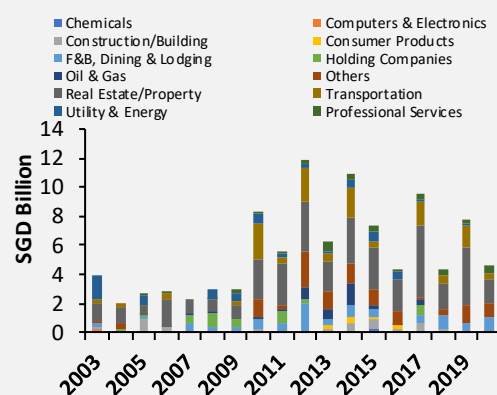
The increase in issuance took place across both SGD- and USD-denominated bonds. The growth in SGD-denominated bonds was attributable to issuance from domestically-oriented firms, such as those from the real estate and transportation sectors (Chart A2). In comparison, the growth in USD-denominated bonds was driven by externally-oriented firms, such as those in the oil and gas (O&G) sector. Many of these domestic based firms derive a sizeable proportion of their revenues from abroad.¹⁵ Firms such as those in the O&G industry with foreign currency revenue streams would be able to offset currency mismatch risks by issuing debt in a matching currency.

Chart A2 USD-denominated bond issuance was driven by externally oriented sectors while SGD denominated bond issuance was driven by domestically oriented sectors*

USD-denominated Bond Issuances by Non-financial Corporates in Singapore, yearly



SGD-denominated Bond Issuances by Non-financial Corporates in Singapore, yearly



Source: Dealogic

*2020 figures were only taken up till Q3 2020

Note: Others includes sectors like Agribusiness, Leisure & Recreation, Telecommunications, etc.

The Singapore bond market has remained accessible in 2020 amid the COVID-19 pandemic.

In late February 2020, the spread of COVID-19 triggered a rapid change in market sentiment as investors factored in a significant slowdown in economic activity. The Singapore SGD and USD bond markets saw a decline in activity between March to April 2020 amid heightened volatility. As market sentiment deteriorated alongside heightened global USD liquidity stress, primary issuance volumes dropped sharply in March 2020 (Chart A3), especially for USD-denominated bonds.¹⁶ Trading volumes and bond prices for SGD-denominated bonds declined in the secondary market, particularly in the high yield segment. Bid-ask spreads also widened, arising from selling pressure.

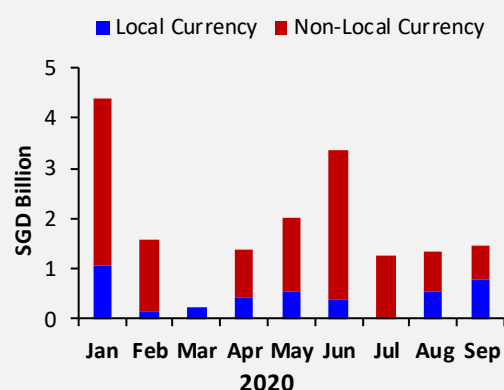
¹⁵ Approximately 80% of firms surveyed in SBF's 2019/2020 National Business Survey, representing Singapore companies with more than SGD0.5 million in share capital, reported having business engagements overseas (up from 70% in 2018/2019). Roughly half of them derived more than 50% of their turnover from abroad.

¹⁶ There was no USD-denominated NFC bond issuance in March 2020, and only two SGD-denominated bonds issued, with an aggregate value of SGD0.25 billion.

Given the weakening economic conditions and increased global uncertainty, MAS introduced several measures to support the recovery and continued accessibility of the bond market in 2020 (Charts A4, A5 and A6). These included the MAS USD facility¹⁷, which contributed to stability of USD funding conditions in Singapore and helped the USD bond market recover and stay relatively resilient. The Global-Asia Bond Grant Scheme (replacing its predecessor, the ABG Scheme), also helped issuers facing slower earnings and rising cost of issuance by providing monetary relief to support their financing needs. In the SGD corporate bond market, sentiment and liquidity improved gradually from May 2020, while corporate bond issuance, particularly from high quality issuers, resumed with a total of SGD5.6 billion raised from investment-grade issuers between May and August 2020.

Chart A3 Bond issuance fell in March 2020 but rebounded quickly

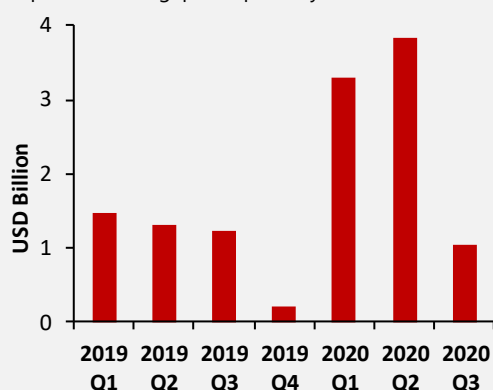
Total Bond Issuances by Non-financial Corporates in Singapore, monthly



Source: Dealogic

Chart A5 USD bond market issuance volume was strong in 2020

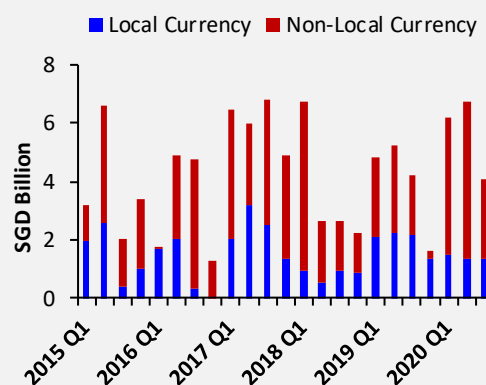
USD-denominated Bond Issuances by Non-financial Corporates in Singapore, quarterly



Source: Dealogic

Chart A4 Bond issuance in 2020 remains relatively resilient

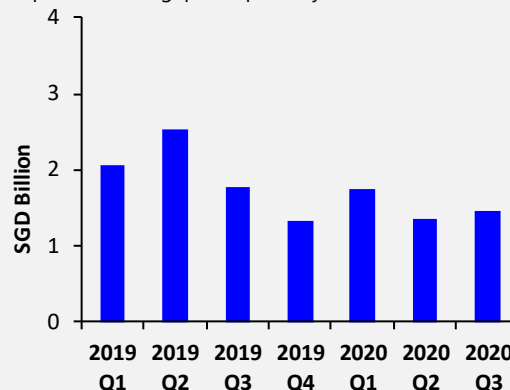
Total Bond Issuances by Non-financial Corporates in Singapore, quarterly



Source: Dealogic

Chart A6 SGD bond market issuance volume remained relatively stable

SGD-denominated Bond Issuances by Non-financial Corporates in Singapore, quarterly



Source: Dealogic

¹⁷ The MAS USD Facility is funded by the USD liquidity swap arrangement between the MAS and the US Federal Reserve. It facilitates USD lending to businesses in Singapore and the region.

Risks facing issuers remain manageable.

The non-financial corporate bond market in Singapore has remained resilient so far with manageable liquidity and refinancing risks. The default rate of SGD- and USD- denominated bond issuers remained low, comprising less than 0.5% of total outstanding SGD- and USD- denominated bonds.

Debt profiles of Singapore firms continue to be sound with firms having a median short-term debt¹⁸ to total debt ratio of 40%, suggesting that liquidity risks of most issuers are contained. The maturity profile of SGD- and USD-denominated bonds is well distributed, with those maturing by 2022 making up less than one quarter of the total outstanding SGD- and USD-denominated bonds (Chart A7). This maturity profile has remained stable since 2018.

Of bonds maturing by 2021, the largest share at 39% (SGD5.9 billion) of outstanding bonds were issued by property-related firms (including S-REITs) (Chart A8). Even as their earnings have weakened and their proportion of short-term debt to total debt increased¹⁹ (for reasons including their efforts in raising cash buffers), they remain well placed to meet their debt obligations. More specifically, S-REITs, which experienced refinancing pressure in the GFC, have also been prudent with leverage, with a median leverage ratio of 35%. This ratio is well within the allowable limit of 50% set out under the Singapore Code of Collective Investment Schemes.²⁰

However, about 20% (SGD3.1 billion) of outstanding bonds maturing by 2021 are from issuers in sectors that are more affected by the COVID-19 pandemic.²¹ These issuers should be monitored closely on their debt repayment and refinancing capacity ahead of the upcoming maturities.

¹⁸ Short-term debt refers to debt coming due within one year.

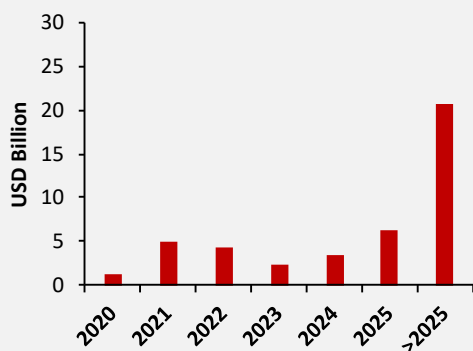
¹⁹ Overall, short-term debt as a proportion of total debt for property related firms increased from 15.3% at end-2019 to 17.7% in Q2 2020. For S-REITs specifically, this proportion increased from 9.5% at end-2019 to 14.6% in Q2 2020.

²⁰ A more detailed analysis on S-REITs can be found in Special Feature 4 "S-REITs' Performance During the COVID-19 Pandemic".

²¹ These are firms in Tier 1 and Tier 2 sectors under the JSS, introduced by the Singapore Government to help companies retain their local employees. Tier 1 sectors include those most directly and severely impacted by the pandemic like aviation and tourism, whereas Tier 2 sectors include those like F&B, retail, and arts and entertainment.

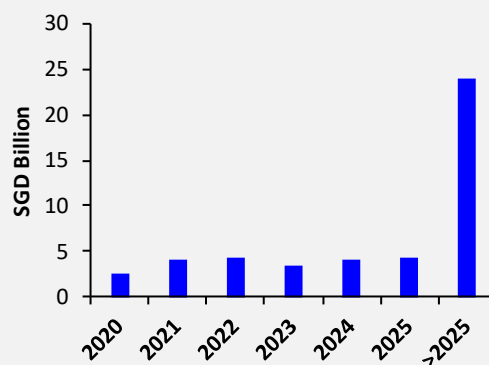
Chart A7 Maturity profile of both SGD- and USD-denominated bonds are well distributed

Maturity Profile of USD-denominated Bonds



Source: Dealogic

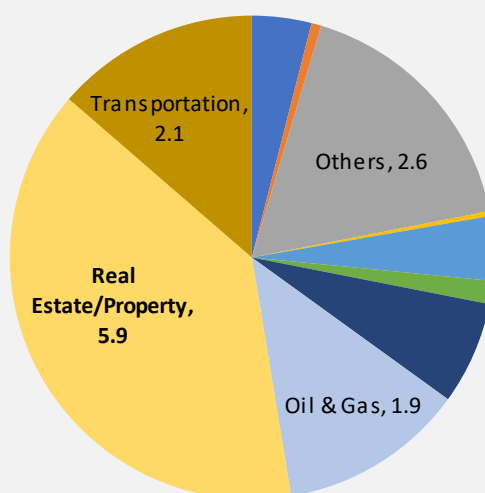
Maturity Profile of SGD-denominated Bonds



Source: Dealogic

Chart A8 Real estate/property firms make up 39% (SGD5.9 billion) of outstanding bonds coming due by 2021

Value of Outstanding Bonds Due by 2021 by Industry (SGD Billion)



Source: Dealogic

Conclusion

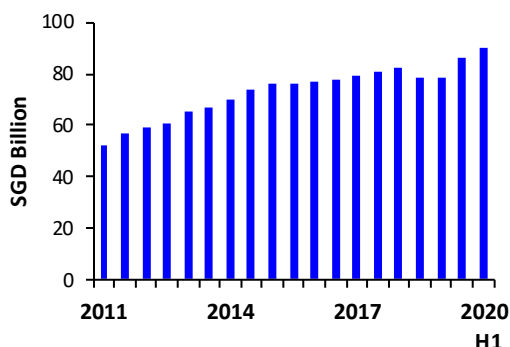
The Singapore bond market has played an increasingly important role in serving the SGD and USD funding needs of a diverse set of corporates over the years. Despite heightened volatility during the onset of the COVID-19 crisis, the market has remained accessible to Singapore firms, supported by the package of policy measures to ease financial and credit conditions.

Non-financial corporate bond issuers in Singapore have remained resilient. Liquidity and refinancing risks are manageable thus far, with a well-distributed maturity profile. However, close monitoring of debt repayment and refinancing capacity is warranted for firms with debt coming due, particularly for those in sectors where revenues may remain depressed.

Chart Panel 2A Small and Medium-sized Enterprise Financing Conditions

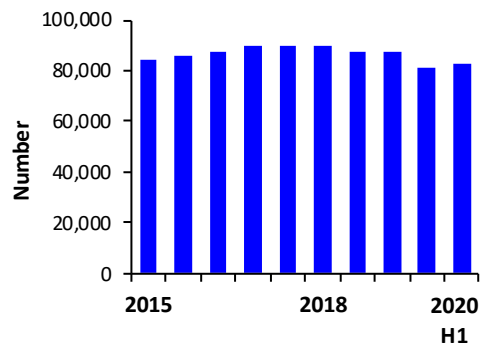
Lending to SMEs has increased over the past year, bolstered by the enhanced government financing schemes introduced in April 2020. The number of SME customers has contracted over the same period, suggesting increased financing needs of each borrower.

Chart 2A1 SME Loans Outstanding



Source: MAS

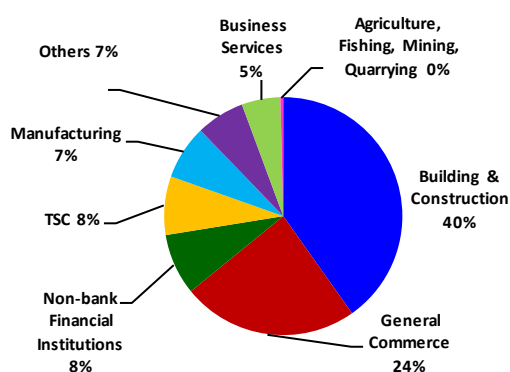
Chart 2A2 SME Customers



Source: MAS

The building and construction and general commerce sectors continued to account for the majority of SME loans.

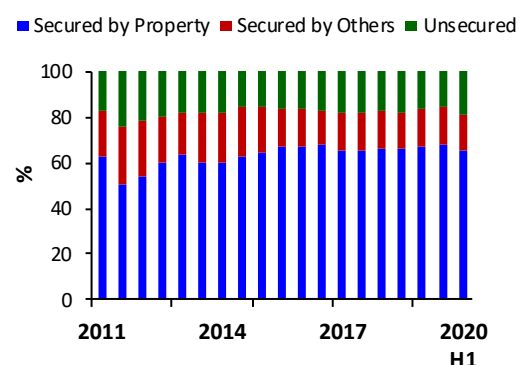
Chart 2A3 SME Loans by Sector (as of H1 2020)



Source: MAS

The share of outstanding SME loans secured by property has decreased.

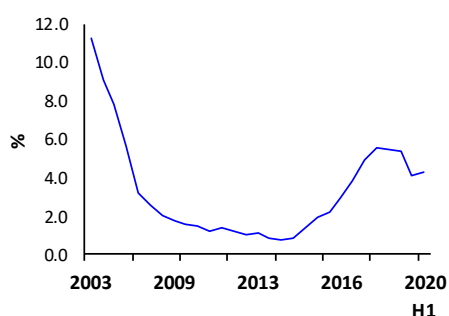
Chart 2A4 Outstanding SME loans by Type of Collateralisation



Source: MAS

The SME NPL ratio improved to 4.3% in H1 2020, as lending increased to the segment.

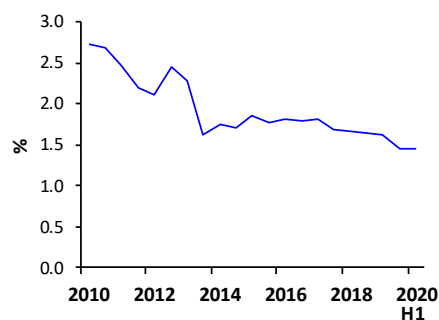
Chart 2A5 SME NPL Ratio



Source: MAS

Banks' Net Interest Margins (NIMs) on SME loans remained stable at 1.5% in H1 2020.

Chart 2A6 NIM on SME Loans

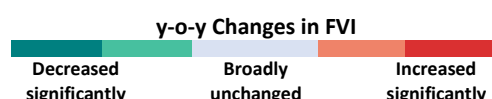


Source: MAS

3 Singapore Household Sector

Household balance sheets in Singapore were relatively healthy at the onset of the COVID-19 pandemic, reflecting the financial buffers that were built-up in the preceding years. The household sector FVI had been on an improving trend as borrowers turned more cautious in taking on leverage, alongside the property cooling measures and strengthening of unsecured credit rules since 2010. However, COVID-19 led to a sharp fall in employment and wage incomes in H1 2020, reflecting the near cessation in economic activity in labour-intensive sectors. Government transfers and measures extended by MAS and the financial industry have provided important support. The unprecedented extent of the fall in domestic activity could ultimately impact a subset of households with significant cashflow constraints.

Household sector FVI (y-o-y Changes)	Q3 2019	Q3 2020
Overall Household FVI	↘	↗
Leverage risk	↘	↗
Maturity risk	→	↘



The overall household sector FVI increased in Q3 2020, as sharply-lower GDP resulted in a temporary rise in the household debt-to-GDP ratio, even though outstanding overall household debt fell. Maturity risk has decreased, alongside reduced outstanding unsecured debt. However, the asset quality of both secured and unsecured debt weakened slightly, although it remains manageable at this juncture.

The labour market recovery is likely to be protracted as resident unemployment is expected to remain elevated in 2021.²² While household balance sheets remain resilient in aggregate, households who are highly leveraged or employed in sectors that are severely impacted by the pandemic would be more vulnerable. MAS recognises that some homeowners could face difficulties servicing their mortgages in the period ahead, and has worked with the financial industry to extend industry-wide relief measures for individuals facing financial difficulties due to COVID-19.²³ Given the uncertain economic outlook, households should avail themselves of these support measures if needed and factor in possible volatility in future income streams when considering large purchases and loans.

²² For more details, see Section 3.1 of the October 2020 Issue of MAS' *Macroeconomic Review*.

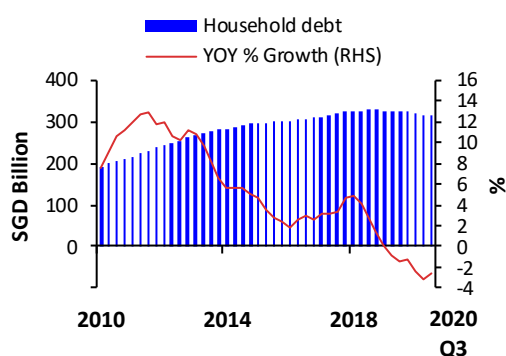
²³ For details on these support measures, see <https://mas.gov.sg/regulation/covid-19/supporting-individuals-through-covid-19>

3.1 Leverage risk has edged up despite a fall in outstanding housing loans

Aggregate household debt continued its downward trend since the introduction of the last round of property market cooling measures in July 2018, coming in at –2.5% y-o-y in Q3 2020 (Chart 3.1). As housing loans account for three quarters of total household debt (Chart 3.2), falling debt levels reflect in part the effectiveness of the cooling measures in moderating private residential property market transactions and in bringing prices closer to economic fundamentals.

Chart 3.1 Household debt has continued to fall in recent quarters

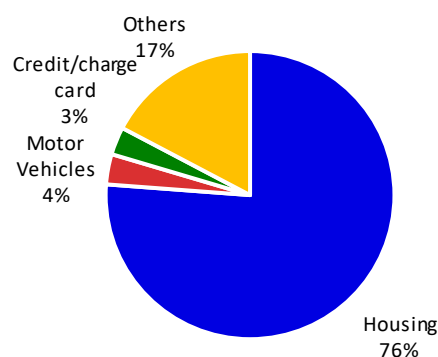
Household Debt in Singapore



Source: MAS estimates, DOS

Chart 3.2 Housing debt is the most significant form of household debt

Breakdown of Outstanding Household Debt (as of Q3 2020)



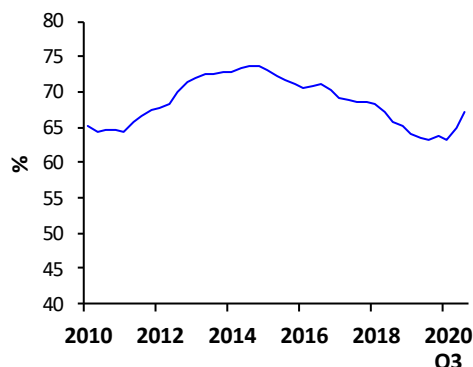
Source: MAS estimates, DOS

However, as a result of the severe economic impact of COVID-19, nominal GDP fell by a larger –10.4% y-o-y over the same period. Hence, household debt as a percentage of GDP rose slightly, reversing its pre-COVID decline (Chart 3.3). Since the Circuit Breaker²⁴ in Q2 2020, household debt-to-GDP increased 1.9 percentage points from 63.1% in Q1 2020 to 65.0% in Q2 2020, and a further 2.1 percentage points to 67.1% in Q3 2020. At the same time, household debt as a percentage of income, which has remained stable at two times since 2015 (Chart 3.4), is expected to increase slightly in the near term as accumulated labour market slack weighs on wages.

²⁴ The Circuit Breaker refers to the set of movement restrictions and closure of non-essential businesses in Singapore from 7 April 2020 to 1 June 2020.

Chart 3.3 Household debt rose slightly as a percentage of GDP, as GDP fell more than household debt

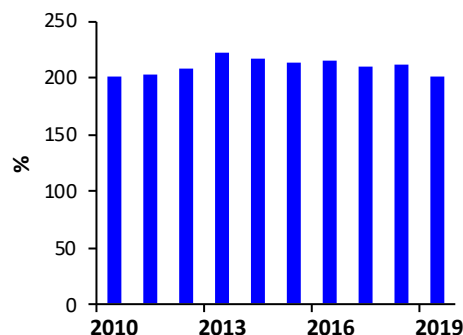
Household Debt as % of GDP in Singapore



Source: MAS estimates, DOS

Chart 3.4 Household debt as a percentage of income has stabilised

Household Debt as % of Income in Singapore



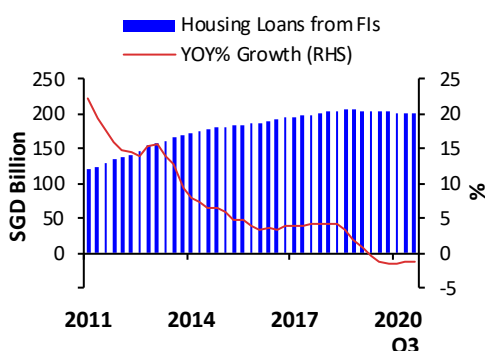
Source: MAS estimates, DOS

Outstanding housing loans have declined gradually while new housing loans remained stable.

Housing loans are a key determinant of overall household sector leverage and hence financial vulnerability. The growth of outstanding housing loans has been gradually declining, in line with slower market activity following the July 2018 cooling measures (Chart 3.5). Despite the suspension of showflat viewings during the Circuit Breaker period and subsequent safe management measures that limited footfall after they reopened, new housing loans remained stable in Q3 2020 (Chart 3.6). These trends reflect the relatively stable transaction volumes and strong refinancing activity as borrowers took advantage of lower interest rates on their mortgages.

Chart 3.5 Outstanding housing loans declined gradually in 2020

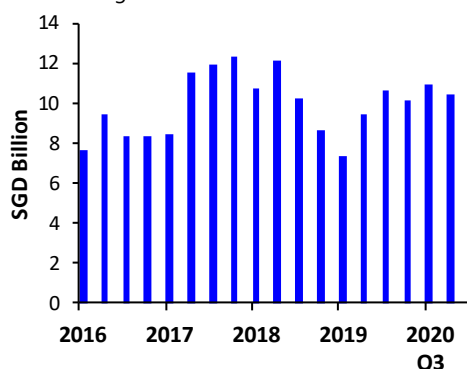
Outstanding Housing Loans



Source: MAS estimates, DOS

Chart 3.6 New housing loans have remained stable despite onset of COVID-19

New Housing Loans



Source: MAS estimates

Note: Housing loans granted by FIs for private and public housing (excludes loans granted by HDB).

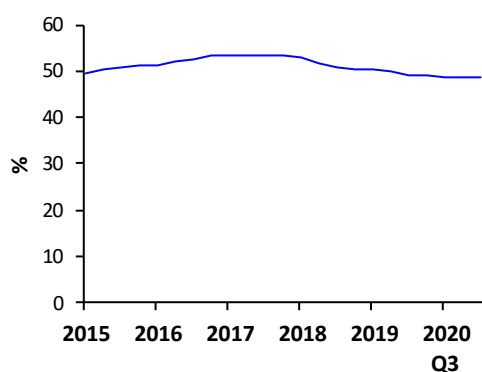
Mortgage credit profile is supported by stable collateral value and loan moratoriums, but credit risk could increase in the period ahead.

The credit risk profile of housing loans remains sound, with macroprudential measures introduced in preceding years encouraging prudent borrowing and improving equity buffer. The average loan-to-value (LTV) ratio of outstanding housing loans has fallen from about 53.5% in 2017 to 48.5% in Q3 2020 (Chart 3.7).

Property market prices and collateral values of housing loans have weathered the impact of COVID-19 well thus far. While private residential property prices declined by 1% q-o-q in Q1 2020 at the onset of COVID-19, they have since recovered, rising by 0.3% and 0.8% q-o-q in Q2 and Q3 2020 respectively.

Chart 3.7 Average LTV has been on a declining trend

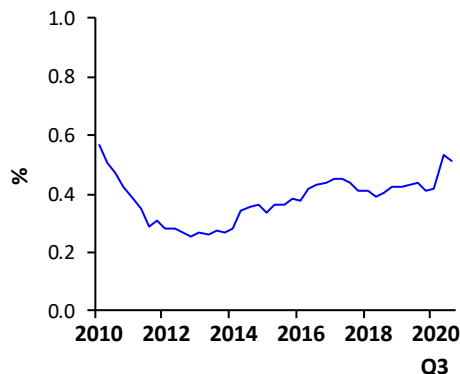
Average LTV Ratio



Source: MAS

Chart 3.8 While housing NPL ratio remains low, it has increased slightly in the last two quarters

Housing NPL Ratio



Source: MAS

The first half of 2020 saw a sharp fall in employment, and many workers who retained their jobs were put on a shorter work-week. MAS and FIs in Singapore introduced measures in March 2020 to support individuals facing financial difficulties through deferred repayment for residential property loans (see Section 3.4 for further details), alongside other Budget support measures such as the JSS. These measures limited the extent of job losses and contributed to low housing NPL ratios of less than 1%, in spite of a slight increase, in Q3 2020 (Chart 3.8).

As household resilience is ultimately tied to employment and income, credit risk for housing loans could increase further in the period ahead if the economic downturn persists. The unsecured credit charge-off rate, a leading indicator for credit quality of housing loans, has crept up in Q3 2020²⁵, suggesting that more households could face difficulties in their housing payments. Close monitoring of housing loans from more vulnerable households is

²⁵ Borrowers in initial periods of distress are likely to miss payments for their credit card bills but continue servicing their secured housing loans. See Section 3.3 for more details of unsecured credit developments.

necessary in the upcoming months given the expectation that the labour market recovery will be protracted.

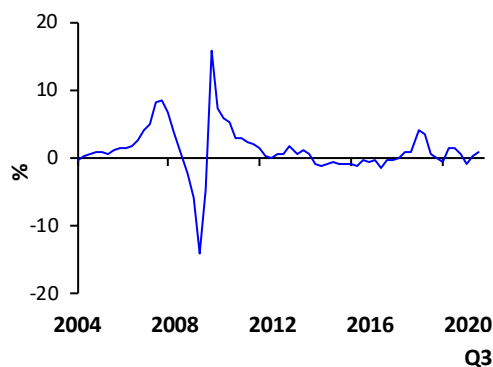
3.2 The private residential property market has remained resilient in 2020

Prices returned to pre-COVID-19 levels in Q3 2020 following an initial decline in Q1 2020.

Private residential property prices increased by 0.8% q-o-q in Q3 2020, slower than the 1.3% experienced in the same period last year, in part reflecting some continued cautiousness in buyer sentiment (Chart 3.9). However, the cumulative price increase over the first three quarters of the year of 0.1%, against the fall in real GDP (average y-o-y) of -6.4%, as well as the less pronounced volatility in overall prices compared to the GFC period, points to some degree of underlying support and resilience in the market. Unlike previous periods where regional price trends were aligned, the Rest of Central Region (RCR) and Outside of Central Region (OCR) saw price gains of 0.3% and 1.4% in 2020 to date respectively, compared to price declines of 3.4% in the Core Central Region (CCR) (Chart 3.10).

Chart 3.9 Property prices saw slight increases following an initial decline in Q1 2020

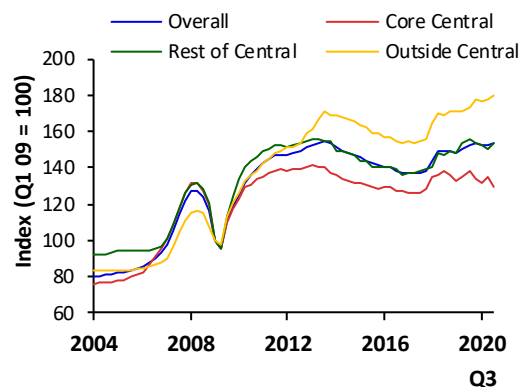
Private Property Price Index (q-o-q Changes)



Source: Urban Redevelopment Authority (URA)

Chart 3.10 RCR and OCR market segments performed better than CCR in recent quarters

Private Property Price Index by Region



Source: URA

Transaction activity fell in Q2 2020, but recovered in Q3 following the lifting of Circuit Breaker measures.

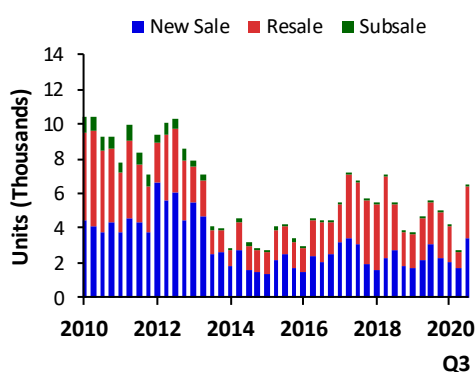
Transaction activity in both new and secondary sales plummeted in Q2 2020 following the implementation of Circuit Breaker measures, which led to the suspension of showflat and property viewings. Transaction volumes fell by more than 50% m-o-m from March to April, and continued to be depressed through May. Activity subsequently picked up in June, following the reopening of sales galleries and home viewings. The improvement in transaction activity continued into Q3 2020, supported by the accommodative interest rate environment

and as some sales may have been delayed from the Circuit Breaker period in Q2 into Q3 (Chart 3.11). Property agents adopted digital sales methods, such as online-only presentations and virtual viewings, to facilitate sales in view of on-going safe distancing measures. More recently, new sales by developers declined m-o-m in October, with fewer units launched. While this decline in sales came after the implementation of restrictions on the re-issuance of options to purchase (OTPs) in September²⁶, it remains to be seen whether the lower sale numbers were due to the new restrictions or the fewer launches. In aggregate, average quarterly transaction volumes in 2020 were broadly unchanged from 2019.

Transaction activity was driven by resident purchases, with foreign demand remaining low, as travel restrictions remained largely in place. As of Q3 2020, residents (citizens and permanent residents (PRs)) accounted for around 95% of all transactions, and Singapore citizens' share of transactions inched up to a historic high of 81.9% (Chart 3.12).

Chart 3.11 Transaction activity rebounded in Q3

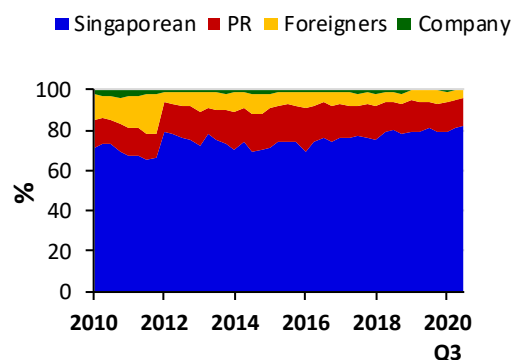
Number of Private Residential Property Transactions



Source: URA

Chart 3.12 Recent transaction activity has been driven by residents

Private Residential Property Transactions by Nationality



Source: URA

Rentals have moderated alongside the increase in vacancy rates.

Vacancy rates for private residential properties increased from 5.4% to 6.2% between Q2 and Q3 2020, reversing the downward trend since 2016 (Chart 3.13). Rentals declined for the second consecutive quarter in Q3, as the private residential property rental price index fell by 0.5% (Chart 3.14). The weakness in rentals was observed for both landed (−0.1%) and non-landed (−0.6%) properties.

Rental trends across regions mirrored price trends. Amongst non-landed properties, rentals fell by 2.1% in the CCR but rose by 0.3% and 1.0% in the RCR and OCR respectively.

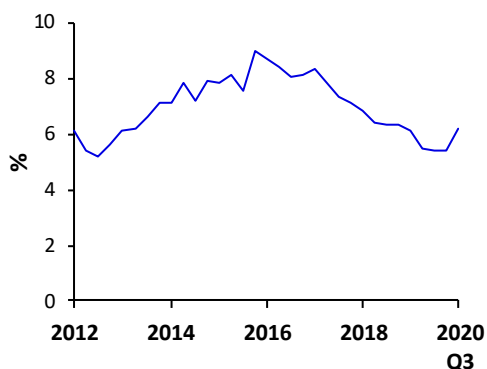
Should demand for rental properties continue to fall, borrowers relying on rental income to meet their mortgage instalments on investment properties could face difficulties in

²⁶ Under URA's revised licensing conditions, developers are restricted from (i) providing an upfront agreement to purchaser(s) to re-issue the OTP; and (ii) re-issuing the OTP to the same purchaser(s) for the same unit within 12 months after the expiry of the earlier OTP, and to inform purchaser(s) of this condition upfront.

repayment. Prospective buyers should accordingly factor in the possibility of further weakness in rental income when committing to purchases of investment properties.

Chart 3.13 Vacancies increased as the labour market contracted...

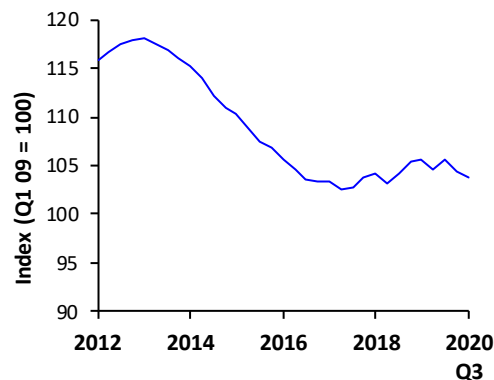
Vacancy Rate for Private Residential Property



Source: URA

Chart 3.14 ...at the same time as a decline in rental prices

Private Property Rental Price Index



Source: URA

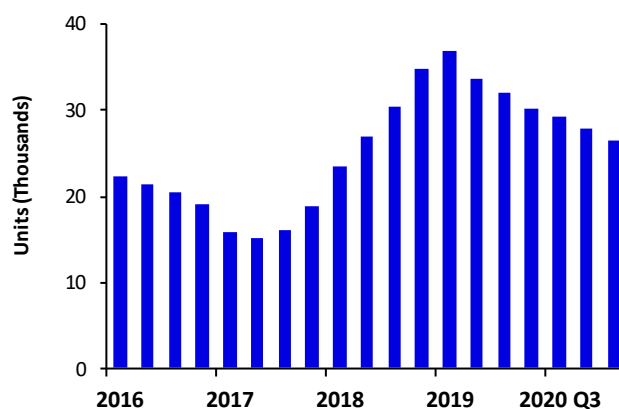
Developers' declining unsold inventories provide them more flexibility to meet Additional Buyer's Stamp Duty remission deadlines.

The number of unsold units in the supply pipeline declined for the sixth consecutive quarter as the recovery in developers' sales outpaced additions to unsold supply inventory (Chart 3.15). Given the progressive decline in unsold inventory, some developers have been keen to replenish their landbank as reflected by the healthy participation for the Government Land Sales (GLS) tenders which closed in October 2020.

The government announced temporary relief measures in 2020, extending the deadlines for Additional Buyer's Stamp Duty (ABSD) remission and Project Completion Period for property developers whose construction timelines and sales of housing units were disrupted by the COVID-19 pandemic. These measures provided developers some flexibility to manage their sales timelines. Alongside the continued decline in unsold inventory, developers are under less immediate pressure to move units to meet ABSD remission deadlines.

Chart 3.15 Developers continued to sell down their inventory of private residential units

Total Launched but Unsold Private Residential Units



Source: URA

3.3 Household maturity risk remains stable

Tighter unsecured credit rules in preceding years have mitigated risks during this COVID-19 period.

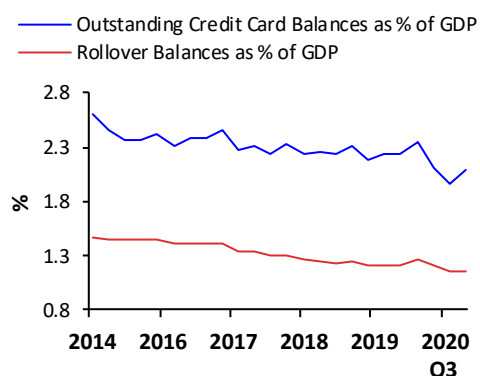
The strengthening of unsecured credit rules by MAS since June 2015 has mitigated maturity risk, by encouraging households to better manage their shorter-term liabilities.²⁷

In 2020, as households pulled back on spending and consumption during the Circuit Breaker period and the safe management measures thereafter, the ratio of outstanding credit card balances to GDP and rollover balances to GDP moderated further (Chart 3.16). The number of revolvers also fell from 27% to 22% between Q3 2019 to Q3 2020 (Chart 3.17), in part due to the slowdown in household consumption. The lower numbers could also be reflective of the relief measures offered by banks for individuals with difficulties making repayment for their unsecured revolving credit facilities, by allowing borrowers to convert such debt to term loans at lower interest rates.

²⁷ The industry-wide borrowing limit requires FIs to suspend unsecured credit to borrowers with outstanding unsecured balances of more than 12 times their monthly income. The requirement was phased in from June 2015 with a balance-to-income ratio (BTI) of 24, to the current BTI of 12 times from June 2019.

Chart 3.16 Credit card debt moderated in Q2 and Q3 2020

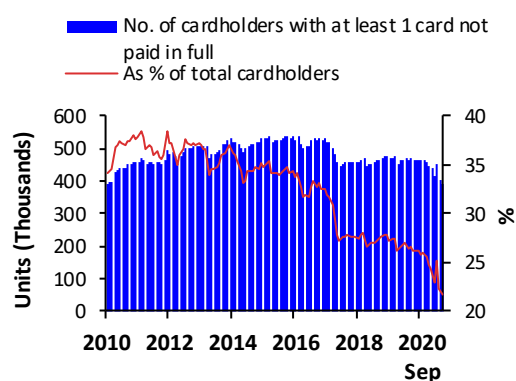
Credit Card Balances as % of GDP



Source: MAS estimates

Chart 3.17 Number of revolvers has fallen

Revolvers as Share of Total Cardholders



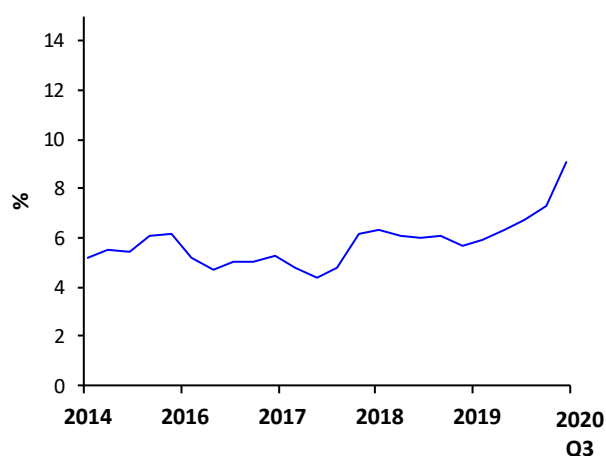
Source: MAS estimates, Credit Bureau Singapore (CBS)

Worsening credit quality of short-term debt could increase household maturity risk.

While outstanding unsecured balances have fallen as a result of reduced reliance on revolving unsecured debt, there has been a slight worsening of credit quality for more vulnerable households. The credit card charge-off rate²⁸ rose from 5.9% to 9.1% between Q3 2019 and Q3 2020 (Chart 3.18). The increase was largely due to a reduction in rollover balances which forms the denominator of the charge-off rate, although the amount of balances charged off also increased slightly. The current charge-off rate is higher than the recent peak of 6.1% in Q3 2015.²⁹

Chart 3.18 The credit card charge-off rate rose

Credit Card Charge-off Rate



Source: MAS estimates

²⁸ Charge-off rates refer to the bad debts written off during the year divided by the average rollover balance.

²⁹ In 2015, Singapore experienced a sharp slowdown in business investment and almost slipped into a technical recession, which raised the amount of irrecoverable unsecured debts.

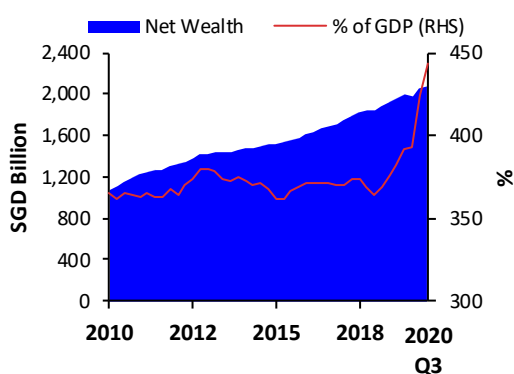
3.4 Household sector balance sheets remain resilient to ongoing risks

Household net wealth has remained stable.

Household net wealth (defined as household assets less household debt) increased from 3.8 to 4.4 times of GDP from Q3 2019 to Q3 2020 (Chart 3.19). While the increase is partly due to the fall in GDP, asset values continued to hold up despite the economic slowdown. Further, liquid assets such as cash and deposits continued to exceed total liabilities (Chart 3.20), providing households a financial buffer against income shocks.³⁰

Chart 3.19 Value of wealth remains stable and increasing against falling GDP

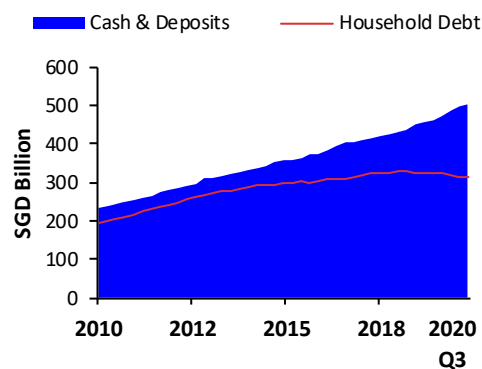
Household Net Wealth



Source: Ministry of Manpower (MOM), DOS

Chart 3.20 On average, households have sufficient liquidity even after paying debt obligations

Household Cash and Debt



Source: MOM, DOS

MAS' simulations suggest that Singapore households' debt servicing burden remains manageable under stress.

MAS' simulations indicate that in general, households' mortgage servicing ratios (MSR) remain manageable. Median MSR is less than 60% in the current baseline scenario, which takes into account income losses in 2020 but assumes no further deterioration. The median MSR remains less than 60% when further income shocks of up to 25% are applied, in an adverse scenario arising from disruptions to economic activity due to an outbreak resurgence and a delay in vaccine deployment.

³⁰ October 2020 Issue of MAS' *Macroeconomic Review* (Section 4.1, page 75) considers the rise in household deposits as an indication of a heightened desire to hold more precautionary balances.

3.5 Ongoing uncertainties in the economic outlook could weigh on household asset quality

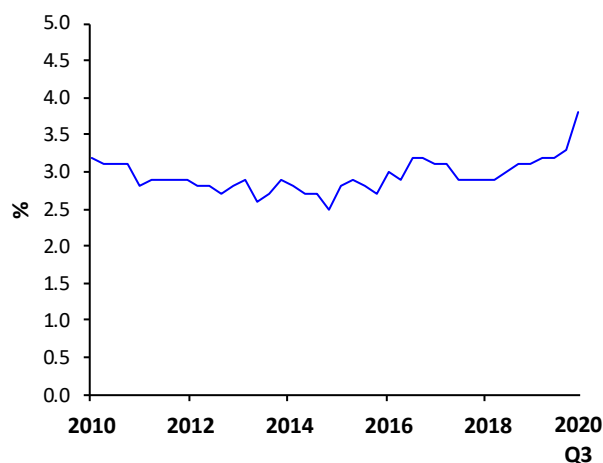
Recovery in the labour market is expected to be protracted.

Labour market slack rose during the COVID-19 pandemic and domestic labour market conditions are likely to improve only gradually next year. The resident unemployment rate has increased from 3.2% in Q3 2019 to 4.7% in Q3 2020 (Chart 3.21) and is expected to weigh on resident wage growth, which will worsen household debt servicing ability into 2021.

Extended support provided by the government, including the JSS to help firms retain local employees, and the newly-introduced Jobs Growth Incentive to encourage the creation of new jobs in sectors with brighter prospects, will provide some support to households.

Chart 3.21 Resident unemployment rate rose sharply

Resident Unemployment Rate (Seasonally-adjusted)



Source: MOM, DOS

Some households will continue to face challenges into 2021, although credit relief measures will ease cashflow pressures.

MAS and the financial industry introduced a package of measures in March 2020 to help ease the financial strain on individuals caused by the COVID-19 pandemic. The measures included deferring repayment of residential property loans and allowing conversion of revolving unsecured debt to term loans at a lower interest rate. As of Q3 2020, about 36,000 borrowers have been granted mortgage relief, and 8,700 borrowers have been granted revolving unsecured debt relief.

Given that some individuals will continue to face cashflow pressures into 2021 and may need more time to transition to full loan repayments, further relief measures were announced on 5 October 2020. FIs will allow eligible individuals to apply to make reduced instalment payments on their property loans, pegged at 60% of their monthly instalment, for a period of

up to nine months. The 60% reduced monthly instalment is intended to cover interest and partial principal payments. These measures will ease individuals' cashflow, while allowing borrowers to resume paying down the principal loan amount. As for the unsecured revolving credit facilities, eligible borrowers would also be able to apply to convert existing unsecured credit balances to a lower-cost term loan until 30 June 2021.

Households should assess their financial positions and seek early assistance if they face repayment difficulties.

MAS has been monitoring vulnerabilities in household borrowings and keeping close tabs on the number and profile of applicants for mortgage relief. Borrowers who have difficulties making payments should approach their banks early to discuss alternative repayment options if they are unable to resume servicing payments. As reduced instalment plans and loan tenure extensions will result in higher overall interest costs, individuals should carefully consider the accumulated interest costs they will eventually have to bear, and balance these against their need for temporary cashflow relief.

Given that an uncertain economic outlook could have dampening effects on income streams, households should remain prudent in taking up new debt and in committing to property purchases. Whenever possible, they should continue servicing or consolidating their existing obligations, which would be useful to enhance resilience against any unexpected shocks.

4 Singapore Financial Sector

Singapore's financial sector has thus far been resilient to the economic and financial impact of COVID-19. The banking sector has experienced some deterioration in credit quality, but strong capital and liquidity positions provide a robust base for banks to continue supporting the economy's demand for credit. That said, banks are likely to face pressures on profitability going forward. They should continue to maintain credit underwriting standards and actively monitor and manage foreign currency risk. Volatility in markets seen globally in March 2020 resulted in significant downward pressure on financial assets. Funds in Singapore, including REITs, have weathered the shocks well thus far, but stresses on the non-bank sector bear close monitoring. Insurers are expected to be impacted by lower investment income due to the low interest rate environment and valuation losses from higher yields on corporate bonds.

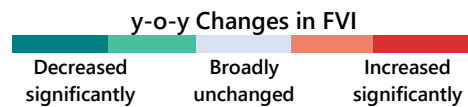
4.1 Banking Sector

The banking sector in Singapore entered the COVID-19 downturn from a position of strength, and has played an important role in supporting the economy's requirements for credit in the first three quarters of 2020. Banks continued to maintain strong capital and liquidity positions. Overall credit growth has remained firm, supported mainly by interbank lending, although subdued economic activity and weak external demand have weighed on non-bank lending.

Despite some deterioration in asset quality, the banking system's NPL ratio continues to be low. Banks pre-emptively increased their loan provisions, improving coverage ratios. Various policy measures introduced by the authorities to ensure stable liquidity conditions in funding markets and continued credit provision to businesses have also helped to mitigate the potential impact of the downturn. MAS' desktop stress test results showed that key banks in Singapore have the capacity to withstand further negative shocks.

Nonetheless, banks are likely to face challenging operating conditions in the near term. The prolonged low interest rate environment, deterioration of asset quality and the possible need to further increase provisions in view of the potential credit losses will pose downside pressure on banks' profitability, even as they continue to ensure strong underwriting standards and healthy capital buffers. Banks should also actively monitor and manage their foreign currency mismatch risk prudently against risks of renewed tightening of global funding conditions.

Banking sector FVI (y-o-y Changes)	Q3 2019	Q3 2020
Overall Banking FVI	→	→
Resident leverage risk	→	→
Non-resident leverage risk	→	→
Liquidity risk	→	→
Maturity risk	→	→



The banking FVI has remained broadly unchanged over the past year. Leverage, liquidity and maturity vulnerabilities have been stable, as credit growth moderated alongside weakening economic activity. Vulnerabilities are expected to ease as banks increase credit provisioning against the backdrop of challenging business conditions and gradual tapering of government support into 2021.

Banks have continued to facilitate credit supply amid weakening economic conditions.

Overall credit growth accelerated sharply in March 2020 before tapering down to a steady pace of 4.9% y-o-y in October 2020. Growth was bolstered by a strong expansion in interbank loans (see Chart Panel 4A “Banking Sector: Credit Growth Trends”), which occurred alongside liquidity measures to ensure stable USD and SGD funding conditions in Singapore.³¹

Meanwhile, non-bank loan growth weakened significantly since March 2020, as a result of the subdued domestic economy and weak external demand. The decline in lending to non-residents was largely driven by a reduction in loans to Emerging Asia (see Chart Panel 4B “Banking Sector: Cross-border Lending Trends”).

Resident non-bank loan growth has more recently registered a contraction of –0.8% y-o-y in October 2020, in line with persistently weak levels of activity in the most affected sectors (see Chart Panel 4A “Banking Sector: Credit Growth Trends”). Lending to residents declined at a slower pace, supported by various measures introduced by authorities to facilitate credit supply amid the temporary disruption in economic activity (see Box B “Credit Conditions amid the COVID-19 Pandemic”).

Based on a survey³² conducted by MAS in September 2020, close to 40% of banks expect demand for corporate credit to increase in the next six-month period, bolstered by working capital needs as well as the gradual resumption of economic activities. Nevertheless, most banks expected the challenging operating environment and weak commodity prices to adversely affect corporates’ financial performance. As a result, the majority of the respondents

³¹ In March 2020, MAS announced a USD60 billion facility to support more stable funding conditions in Singapore and facilitate USD lending to businesses in Singapore and the region, backed by a USD swap facility with the US Federal Reserve. MAS has also ensured sufficient SGD liquidity in the banking system via its money market operations.

³² MAS conducted a survey of banks in September 2020, to seek industry participants’ views on the prevailing credit conditions and expectations over Q4 2020 and Q1 2021.

have underscored the importance of prudent underwriting standards, in anticipation of an increase in credit risk.

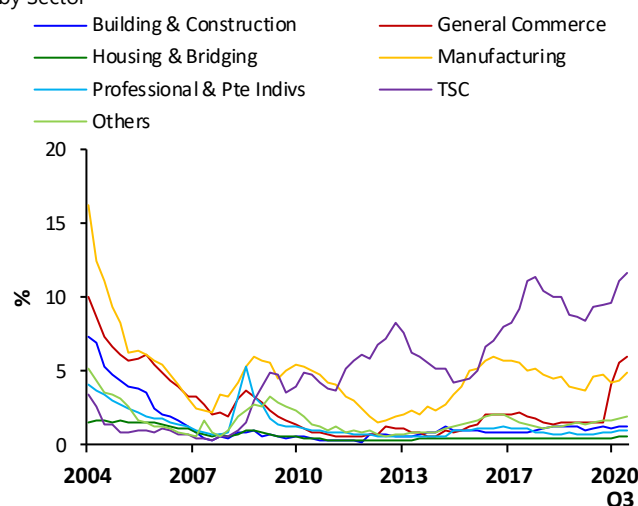
Banks have ample capital buffers and have pre-emptively increased provisions in anticipation of credit losses.

The banking system's overall NPL ratio rose to 2.7% in Q3 2020 from 2.0% in the same period last year (see Chart Panel 4C "Banking Sector: Asset Quality and Liquidity Indicators").

The decline in asset quality was broad-based across all sectors (Chart 4.1). The NPL ratio for the general commerce sector increased to 5.9% in Q3 2020. The higher NPL ratio reflected a preponderance of firms in sectors adversely impacted by the COVID-19 pandemic—wholesale trade, retail, F&B, tourism-related activities and O&G firms—affected by the collapse in oil prices earlier in the year.

Chart 4.1 Increase in NPL ratios was broad-based across all sectors

Banking System NPL Ratios by Sector



Source: MAS

If the pandemic-induced downturn persists, liquidity pressures faced by vulnerable firms may develop into insolvencies, and consequent job losses may lead to a weakening in consumption and possible delinquencies on other portfolios. The potential for further weakening in asset quality is partially reflected in the special mention ratio, which had risen from 3.0% in Q3 2019 to 3.8% in Q3 2020.

Banks pre-emptively increased their provisions, improving their provisioning coverage ratios, in anticipation of a deeper and more prolonged economic downturn. Total provisioning coverage³³ in the banking system rose to 103.1% in Q3 2020, while specific provisioning coverage improved to 65.0%. These buffers are further augmented by general provisions held at the head offices of foreign bank branches.

³³ The banking system's overall provisioning coverage is computed as the sum of general and specific provisions as a share of unsecured NPLs.

The banking system in Singapore has thus entered this crisis from a position of strength. The strong capital positions and higher provisioning buffers have facilitated banks' role during the crisis in accommodating short-term credit demand from firms and households facing liquidity constraints and revenue and income shortfalls. MAS has worked with the financial industry to extend loan moratoriums for certain loans to borrowers facing temporary cashflow pressures as a result of the pandemic. In October 2020, the relief measures were extended to assist borrowers requiring more time to repay their loans because of further revenue and income disruptions to defer part of their loan repayments temporarily (see also Box B). The extended industry measures aim to strike a balance between extending credit relief to those who continue to need it during the current downturn, and to help businesses and individuals transit gradually to full loan repayments, to ensure their debt sustainability over the medium term.

Banks themselves have built up sizable capital buffers over the past decade and are well-placed to weather further downside risk. To further assess the resilience of our key banks, MAS conducted a desktop stress test incorporating COVID-19 impacts for the domestic systemically important banks (D-SIBs) (see Box C "MAS' Desktop Stress Test of D-SIBs"). MAS simulated an adverse scenario of continued disruptions to economic activity in 2021 due to a resurgence in virus outbreaks, delays in vaccine development and renewed lockdown measures globally. Under such an adverse scenario, results showed that all D-SIBs remained solvent, with their aggregate common equity tier 1 capital adequacy ratios (CET1 CAR) well above MAS' minimum regulatory requirements.

Banks' overall liquidity positions continue to be strong.

Banking vulnerabilities arising from overall liquidity and maturity risks have remained unchanged from a year ago.

Over the past year, there has been a surge in deposits in Singapore's banking system due to the increased liquidity in financial markets arising from global central bank monetary policy actions. The disruption to economic activities that led to constrained consumption opportunities and greater uncertainty had also resulted in a higher level of precautionary savings among both corporates and households.

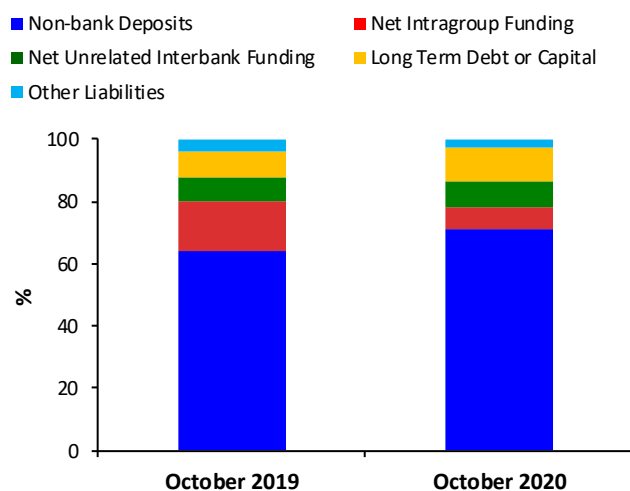
High deposit growth has contributed to the strong liquidity positions of banks. D-SIBs continue to maintain healthy buffers over the minimum regulatory all-currency and SGD liquidity coverage ratio (LCR) requirements. SGD loan-to-deposit (LTD) ratio declined to 72.9% in October 2020. Similarly, foreign currency LTD ratio moderated to 108.1% in October 2020, due to a contraction in loans and a rise in deposits.

On an aggregated basis, USD-denominated liabilities in the banking system remain more than sufficient to fund USD-denominated assets. Local banking groups' USD LTD ratios have consistently stayed well below 100%. The funding structure of foreign banks has also remained healthy, with long-term debt and stable non-bank deposits constituting 81% of total funding (Chart 4.2). Additionally, foreign banks rely on relatively stable net intragroup funding from

their head offices, while net unrelated interbank funding remains low at less than 10% of their total funding.

Chart 4.2 Stable non-bank deposits and long-term debt or capital account for 80% of foreign banks' total funding

Funding Structure of Foreign Banks in Singapore



Source: MAS

However, USD funding risk bears monitoring. The sharp tightening in global financial conditions in March 2020, due to pressures in short-term funding markets, showed that banks remain vulnerable to fluctuations in short-term funding sources. Banks could also amplify the impact of turbulence in international financial markets, by constraining credit provision to the economy as they face liquidity shortfalls.

To alleviate liquidity strains, MAS would continue to provide USD via the MAS USD Facility, following the extension of MAS' USD60 billion swap arrangement with the US Federal Reserve. In the meantime, Singapore banks have been diversifying their USD funding sources across a wide range of geographies, industries, and funding instruments. Nevertheless, there remains a downside risk of renewed tightening in financial conditions. MAS therefore encourages banks to continue efforts to strengthen their USD liquidity profile. Banks should also manage their foreign currency risk prudently under a variety of business conditions by developing liquidity contingency plans.

Local banking groups continue to support the domestic economy while maintaining healthy asset quality and provisioning coverage.

Local banking groups continue to support the domestic economy, participating in the government risk-sharing loan schemes and extending loans to businesses and individuals. Local banking groups' resident loan books expanded through Q3 2020 (on a q-o-q basis), while lending to non-residents declined slightly in Q3 2020 (Chart 4.3).

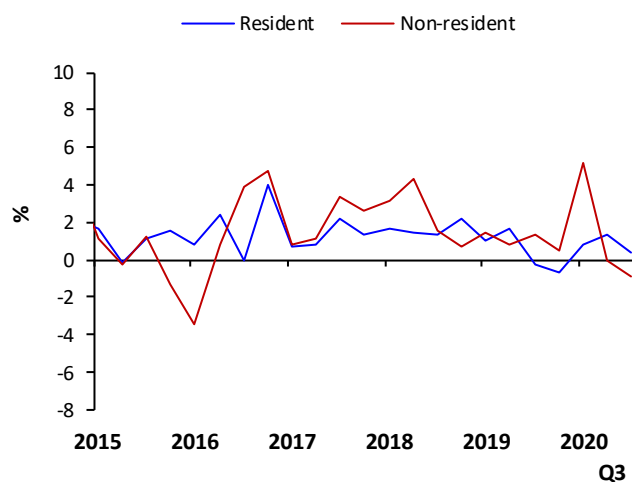
Local banking groups' net profit remains healthy, averaging SGD2.8 billion per quarter over the first three quarters of 2020 (see Chart Panel 4D "Banking Sector: Local Banking

Groups"). While net profit has fallen from the previous year, due to lower net interest margins (NIMs) and substantially higher provisions in anticipation of credit losses from the pandemic, non-interest income has held steady.

Asset quality has remained stable for the local banking groups, due in part to the credit relief measures in place. The Q3 2020 NPL ratio was unchanged at 1.5%, while provisioning coverage rose to 234%. Strong capital buffers, supplemented by pre-emptive provisions, would help banks withstand further increases in credit costs.

Chart 4.3 Local banking groups continued to support resident credit demand through 2020

Local Banking Groups' q-o-q Non-bank Loan Growth by Residency



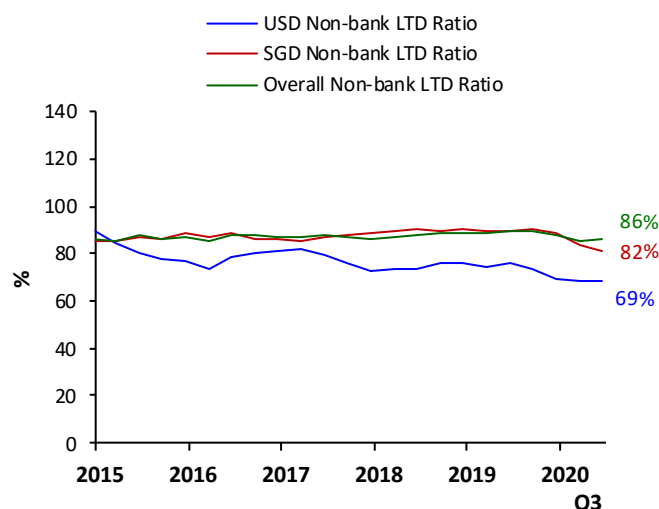
Source: MAS, Local Banking Groups' Financial Statements

Local banking groups' capital and liquidity positions remain sound.

Local banking groups have strong capital and liquidity positions (see Chart Panel 4D "Banking Sector: Local Banking Groups"). Average CARs and all-currency LCRs remain well above regulatory requirements. The overall LTD ratio of local banking groups remains below 100%, with the USD LTD ratio at a healthy 69% (Chart 4.4).

Chart 4.4 Overall, SGD and USD LTD ratios remain low

Local Banking Groups' Non-bank LTD Ratios



Source: MAS, Local Banking Groups' Financial Statements

4.2 Non-Bank Sector

Funds in Singapore have weathered volatility in markets well.

Market volatility in both global and local markets in March 2020 resulted in significant downward price adjustments in financial assets, but funds in Singapore have remained stable despite the market turmoil. The Straits Times Index fell 30% in March before rebounding, but remains some 12% lower from the beginning of 2020 at the time of publication. Amid global market volatility, Singapore markets saw some risk aversion and flight to safety that led to some strains in the high yield corporate bond market in Q1 2020, although the corporate bond market has recovered and remains accessible with the introduction of support measures (see also Box A).

Fund managers took proactive steps to mitigate liquidity risk, including re-sizing their exposures, increasing cash holdings and reducing leverage. 27 funds (11.2%) out of a total of 242 funds authorised for offer to retail investors reported single-day outflows exceeding 5% of their respective assets under management between April and October 2020. Nevertheless, these fund managers were able to meet all redemption requests in an orderly manner, without having to gate or suspend the funds.

The decline in economic activity due to the implementation of the Circuit Breaker in Q2 also had an impact on S-REITs, which have weathered the earnings shock relatively well even as revenue declines exerted pressure on their cash holdings (see Special Feature 4 "S-REITs' Performance During the COVID-19 Pandemic").

Insurers remained resilient in the midst of an uncertain global outlook.

Corporate credit spreads widened following the onset of the COVID-19 pandemic, even as government bond yields remained low. Wider credit spreads accentuate asset-liability mismatches³⁴ for insurers and lowers valuations on their corporate bond portfolios.

The enhanced valuation and capital framework for insurers (also known as the revised Risk Based Capital framework or RBC 2), which came into effect on 31 March 2020, introduced features such as Matching Adjustment (MA) and Illiquidity Premium (IP), that have helped to mitigate the impact of short-term volatility from credit spread movements.³⁵ RBC 2 also incorporated a transitional measure in the calculation of insurers' financial resources³⁶, which has allowed insurers more time to rebalance their investment portfolios and to mitigate the impact of the low interest rate environment on capital buffers.

In addition, stress tests conducted on selected life and composite insurers on the impact of low to negative interest rates showed that insurers were able to mitigate the impact of a "low for long" interest rate environment by taking a series of management actions, including regular reviews of investment portfolios, to strengthen their solvency positions. Overall, insurers remain resilient amid the uncertain global outlook (see Chart Panel 4E "Insurance Sector").

Insurers should closely monitor their solvency positions and adopt a prudent and forward-looking view on capital management. They should continue to maintain strong capital buffers and pre-emptively consider the need to obtain or raise fresh capital to bolster their capital position where necessary. Insurers should also be prudent when making discretionary payments, including the distribution of dividends or profits to their shareholders or head office.

³⁴ Low government bond yields affected life insurers through lower risk-free discount rates used to value liabilities. The increase in liabilities outweighed any increase in the asset values of bonds held as life insurers typically have liabilities of much longer duration than assets.

³⁵ The MA and IP allow for an adjustment to be added to the risk-free discount rates in determining liabilities, so that resulting liabilities will move more in tandem with assets in response to credit spread movements. The use of MA, which is subject to an insurer meeting MAS' requirements and obtaining MAS' approval, provides capital relief and reflects the fact that insurers, particularly life insurers, hold assets for the long term to back their long-term liabilities and are not exposed to short-term spread movements unlike investors with shorter investment horizons.

³⁶ The transitional measure accounts for the differences in which the risk-free discount rate used to value SGD denominated liabilities is derived under both the previous and current capital framework. This has the effect of increasing the solvency position. The transitional measure will grade down linearly and will be phased out by end of 2021.

Box B

Credit Conditions amid the COVID-19 Pandemic

This Box describes the credit conditions for Singapore amid the COVID-19 pandemic and highlights some of the measures that have been undertaken to facilitate credit supply to the real economy.

The COVID-19 pandemic has placed considerable strain on corporate revenues and household incomes.

Corporates, particularly SMEs, have faced significant cashflow constraints due to business activity disruptions and delayed customer payments. While weak underlying economic prospects have led to a moderation in overall credit demand for business expansion and capital investment purposes, working capital requirements have increased to meet near-term cash flow needs. Weak labour market conditions have also negatively affected household incomes.

Authorities in Singapore have responded with various relief measures to facilitate credit supply and mitigate the economic fallout from the crisis.

Singapore introduced various credit relief measures to facilitate credit supply to borrowers. The government enhanced its risk share proportion and loan quantum for loans extended to corporates by FIs under the Enterprise Singapore (ESG) temporary bridging loan programme and working capital loan scheme (ESG schemes). A new MAS SGD facility was introduced to complement the ESG schemes by providing a low-cost source of funding to FIs to facilitate lending to SMEs. For existing loans, SMEs and retail borrowers facing temporary cashflow pressures can apply to their FIs for assistance under the Special Financial Relief Programme (SFRP) for selected loan types. Together, these measures ease restrictions in the flow of credit to the economy, and help otherwise financially healthy corporates and households address temporary liquidity stresses.

There was firm take-up for the various credit relief measures, from both corporates and households. As of Q3 2020, about SGD14.5 billion of loans disbursed to some 18,000 firms, were approved under the ESG schemes. The take-up rate of loans under the ESG schemes was strong especially during the “Circuit Breaker” period from April to June but has since moderated from July. Overall, total SME loan amounts including those approved under the ESG schemes between March and September 2020, were comparable to total SME credit flows over a similar seven-month period a year ago.³⁷

Similarly, there was strong take-up for debt moratoriums. Corporate and individual borrowers applied for moratoriums of approximately SGD11.6 billion of SME loans and

³⁷ Data on historical SME loan approval amounts are from the 2019 and 2020 annual SME financing surveys.

SGD29 billion of property (commercial and mortgage) loans as of Q3 2020. The SME and property loan amounts subject to moratorium constitute about 13.7% and 14.1% of total outstanding SME and property loans respectively. The take-up rates were similar to those of the ESG loan schemes, with a peak in April when the moratoriums were first introduced, and which subsequently moderated from May onwards.

Banks are in a strong financial position to support the real economy's demand for credit.

Government efforts to facilitate credit supply were supported by the robust financial health of banks in Singapore. As of Q3 2020, key Singapore banks continued to maintain a high aggregate CET1 CAR of about 14.1%, representing a strong base for lending under current challenging cyclical conditions.

In July 2020, MAS had called on Singapore-headquartered locally incorporated banks to cap FY2020 dividends to 60% of their FY2019 dividends as a pre-emptive measure to bolster their resilience and capacity to support lending amid the uncertainties ahead. This cap on dividends is estimated to increase the CET1 CAR across the three local D-SIBs by about 0.3–0.4 percentage points on average. The improvement in CARs strengthens banks' capacity to accommodate further increases in credit supply to the economy as needed, while ensuring that CET1 CARs remain well above MAS regulatory requirements.

The projected increase in resident credit-to-GDP ratio by end-2020 largely reflects the collapse in economic activity, alongside a modest rise in resident debt levels.

The resident credit-to-GDP ratio rose from 170.8% in end-2019 to 186.5% in Q3 2020. The (nominal) GDP decline, as a result of the sharp economic downturn, contributed to the bulk (13.9 percentage points) of the rise in the ratio, while credit growth contributed a much lesser extent (1.8 percentage points).

Lending to the domestic economy has been sustained over the first three quarters of 2020. The increased demand for working capital to address near-term cash flow needs has partially substituted for the fall in credit demand for business expansion activities and capital expenditures. At the same time, credit relief measures in the form of debt moratoriums and increased risk share for ESG loans have facilitated credit supply despite increased credit risk that was prevalent during the COVID-19 crisis.

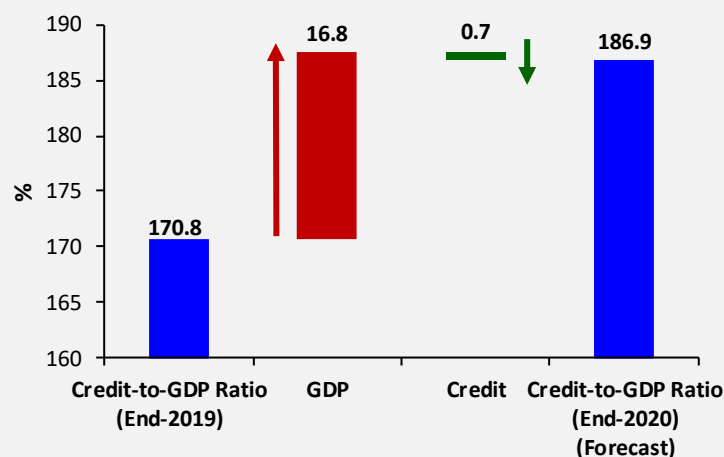
In sum, the various credit relief measures have helped address the temporary cashflow constraints of corporates and households. Nevertheless, the credit relief measures are not meant or expected to reverse the overall trend of moderating credit growth against the significantly lower level of economic activity and spending in the broader economy in 2020. Indeed, as of Q3 2020, total resident credit growth has moderated further on a y-o-y basis from 1.6% in Q4 2019 to –0.2% in Q3 2020.

Factoring in the impact of the credit relief measures, resident credit growth is forecast to moderate to about –0.4% y-o-y by end-2020. Meanwhile, real GDP growth is projected at –6.5% to –6.0% for the year. The resident credit-to-GDP ratio is expected to rise from 170.8% in end-2019 to 186.9% by end-2020, with the contraction in GDP levels contributing to more than 16 percentage points of the increase in the ratio (Chart B1).

The resident credit-to-GDP ratio is projected to stabilise and may decline modestly in 2021. Economic activity is expected to rebound in 2021 as revenues and incomes recover alongside the resumption of business activity. However, despite the anticipated above-trend growth in 2021, the growth momentum of the economy is expected to remain modest and uneven across sectors. Hence, continued uncertainty over the economy's recovery trajectory and lower risk appetite could imply that credit demand would remain relatively subdued if firms and households hold back on investment and discretionary spending. In addition, the increased leverage that some borrowers undertook during the crisis would limit their ability to take on additional credit. Banks are also likely to put in place enhanced measures to ensure the smooth transition of their customers out of the various credit relief programmes. Hence, lending to resident corporates and households is expected to increase only at a moderate pace in 2021. Taken together, the slight pick-up in resident credit and rebound in economic activity would likely result in a fall in the credit-to-GDP ratio in 2021 by about 5–6 percentage points, after its temporary spike this year.

Chart B1 Resident credit-to-GDP ratio is expected to increase by end-2020 as the full economic impact of COVID-19 is felt

Resident Credit-to-GDP Ratio



Source: MAS estimates

MAS has worked with the financial industry to extend credit relief measures to support borrowers who need more time to resume loan repayments.

While the credit relief measures have helped ease cashflow constraints and prevented a sharp contraction in credit supply, there was a need to consider the appropriate next steps after the expiration of the initial round of moratorium measures in December 2020. Prolonged extension of these measures could increase the overall debt burden for some borrowers from the interest accrued on deferred repayments. Relief extension might also run the risk of temporarily supporting some borrowers whose long-term viability has been significantly impaired by the pandemic. Such an outcome would hinder productive reallocation of resources and generate additional credit losses for banks. However, an abrupt withdrawal of the relief measures, under conditions of elevated uncertainty, might also mean that many viable debtors would not be able to service their debt given the significant temporary decline in corporate revenues and household incomes.

MAS and the financial industry have carefully weighed these considerations and announced in October 2020 a series of extended measures to help individuals and SMEs address continued cashflow difficulties. The extended measures would allow some borrowers to defer part of their loan repayments temporarily. For instance, individual borrowers who are able to furnish proof of income impact would be able to make reduced monthly instalments pegged at 60% of the usual monthly instalments (from 100% under the previous measures) on their property loans for a period of up to nine months from take-up, though not beyond December 2021. SME borrowers can also apply to defer 80% (from 100% previously) of principal repayment for either three or six months from 1 January 2021 onwards under the Extended Support Scheme—Standardised.

Conclusion

In sum, the credit relief measures, along with banks' strong financial positions, have helped to maintain credit supply and address corporates' and households' temporary cashflow constraints amid the current crisis. While resident credit growth has already moderated since the onset of the pandemic, concerns remain that the debt moratoriums and government loan schemes might cause vulnerable corporates and households to incur excessive debt, which could in turn generate credit losses for banks. The extended industry measures introduced in October are thus aimed at assisting debtors to repay their loans and mitigate credit risk by gradually transiting borrowers to full loan repayments. The measures therefore aim to attain a balance between safeguarding the resilience of the banking system, and maintaining credit supply to assist borrowers with their debt repayment.

Table: Summary of Key Credit Relief Measures

Credit Relief Measures	Details
SFRP Relief Measures (In effect from April to December 2020)	➤ Individual borrowers can choose to defer repayment of principal or both principal and interest for residential property, renovation and education loans. ➤ Individual borrowers can choose to defer payment of principal for commercial and industrial property loans. ➤ Eligible individual borrowers can choose to convert their outstanding balances from revolving unsecured credit facilities to term loans with effective interest rate capped at 8%. ➤ SME borrowers can choose to defer payment of principal for secured term loans.
Extended Support Measures for Individuals (In effect from January 2021)	➤ Eligible individual borrowers can apply to make reduced instalments pegged at 60% of monthly instalment for property loans for up to nine months, but not beyond December 2021. ➤ Loan tenure extension of property loans for up to three years granted on a case-by-case basis. ➤ Loan tenure extension of up to three years on renovation and student loans. ➤ Eligible individual borrowers can choose to convert their outstanding balances from revolving unsecured credit facilities to term loans with effective interest rate capped at 8%. This is an existing relief measure that is already in effect prior to January 2021, and the application period has been extended to 30 June 2021.
Extended Support Scheme for SMEs (In effect from January 2021)	Extended Support Scheme — Standardised ➤ SME borrowers in more affected sectors (Tier 1 and 2) can apply to defer 80% of principal repayment for six months from 1 January 2021 to 30 June 2021. ➤ SMEs in all other sectors can apply to defer 80% of principal repayment for three months from 1 January 2021 to 31 March 2021. ➤ <i>Tier 1 and Tier 2 sectors include aviation and aerospace, tourism, hospitality, conventions and exhibitions, built environment, licensed food shops and food stalls (including hawker stalls), qualifying retail outlets, arts and entertainment, land transport, and marine and offshore.</i> Extended Support Scheme — Customised ➤ The scheme facilitates multi-lender restructuring for multi-banked SME borrowers requiring bespoke assistance.
Enhanced Government Loan Schemes for Corporates (In effect from April 2020 to March 2021)	Enterprise Financing Scheme — SME Working Capital Loan ➤ Government risk share was increased to 90% (from 50% and 70% for young companies), and the maximum loan quantum was increased from SGD600,000 to SGD1 million. Temporary Bridging Loan Programme ➤ New loan scheme initiated in March 2020 ➤ Government risk share was increased to 90% (from 80%) in April 2020, and the maximum loan quantum was also raised from SGD1 million to SGD5 million.

Extended Government Loan Schemes for Corporates (In effect from April 2021 to September 2021)	Temporary Bridging Loan Programme ➤ Programme extended by six months (from 1 April to 30 September 2021). ➤ 70% government risk share (down from 90% previously). ➤ Maximum loan quantum of SGD3 million (down from SGD5 million previously).
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More information can be found at the following links:

1. <https://www.mas.gov.sg/news/media-releases/2020/mas-and-financial-industry-to-support-individuals-and-smes-affected-by-the-covid-19-pandemic>
2. <https://www.mas.gov.sg/news/media-releases/2020/mas-and-financial-industry-provide-additional-support-for-individuals>
3. <https://www.mas.gov.sg/news/media-releases/2020/mas-and-financial-industry-extend-support-for-individuals-and-smes>
4. <https://www.enterprisesg.gov.sg/campaigns/budget-2020/access-financing>

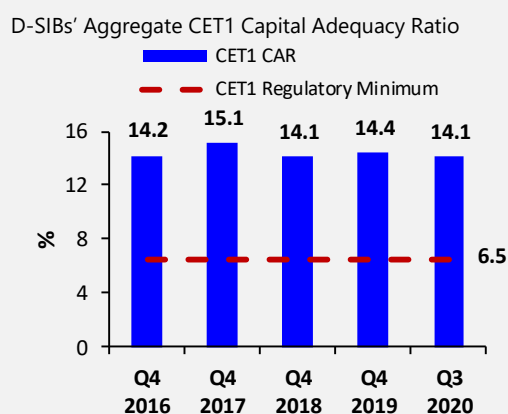
Box C

MAS' Desktop Stress Test of D-SIBs

D-SIBs in Singapore entered the COVID-19 crisis with strong capital buffers.

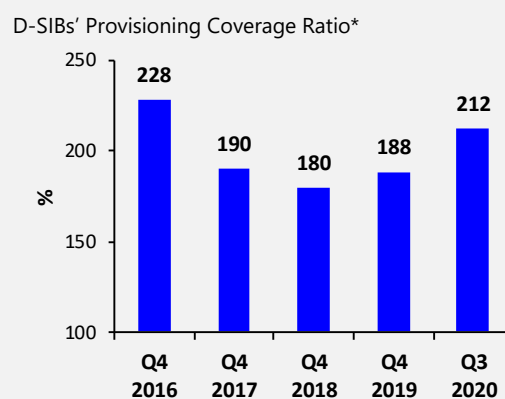
D-SIBs confronted the COVID-19 crisis with strong CET1 CARs at about 14.4% on aggregate as of Q4 2019 (Chart C1). Despite the economic fallout from COVID-19, D-SIBs' capital buffers have remained broadly stable through Q3 2020, dipping by only about 0.3 percentage points since end-2019. At the same time, D-SIBs' resilient pre-provisioning earnings have bolstered their capital buffers while they set aside adequate provisions in anticipation of further credit losses (Chart C2). D-SIBs' balance sheets could be more significantly impacted in 2021 as credit relief measures are gradually phased out amid an uneven economic recovery.

Chart C1 D-SIBs entered the COVID-19 crisis with healthy CET1 CARs



Source: MAS

Chart C2 D-SIBs have ample provisions to buffer against credit losses



Source: MAS

*Total Provisions/Total Unsecured NPAs

MAS conducted a desktop stress test to assess D-SIBs' resilience to economic disruptions.

The desktop stress test employed two scenarios incorporating sharp declines in economic activity over a two-year horizon.³⁸

The first scenario ("Central scenario") simulates the current macroeconomic profile, characterised by the sharp global recession experienced in 2020 (Chart C3). Singapore's real GDP levels dip sharply in the first half of 2020 before recovering in Q3 2020 (Chart C4), with a projected recovery in economic activity in 2021 as the pandemic is brought under control.

³⁸ The desktop stress test is not meant to be a prediction of the losses that banks are expected to incur, but is rather intended to provide an assessment of the capital impacts that might arise in the current conjuncture and how they compare to the capital buffers that banks have in place to absorb the credit losses.

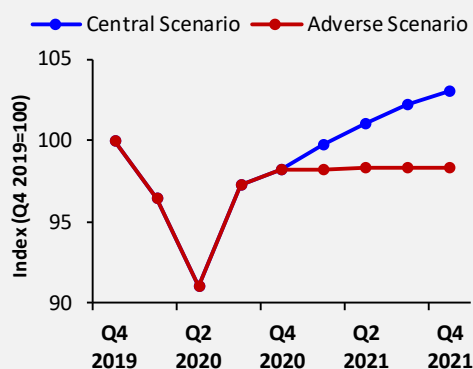
However, despite the anticipated above-trend growth in 2021, the growth momentum of the domestic economy is expected to remain modest and uneven across sectors. Singapore's real GDP levels would remain depressed, recovering to pre-crisis Q4 2019 levels only in the second half of 2021. The overall domestic unemployment rate would increase, peaking in Q3 2020 while remaining elevated into 2021 relative to Q4 2019 levels.

The second scenario ("Adverse scenario") is similar to the Central scenario across 2020, but features an alternative downside risk outcome in 2021, with continued disruptions to economic activity due to delays in vaccine development and deployment alongside a global resurgence in virus outbreaks. Global GDP would follow a below-trend trajectory in 2021, reflecting the economic challenges faced as countries are forced to implement renewed lockdown measures (Chart C3). Domestic economic activity would continue to recover from its trough in 2020, but at a significantly slower pace relative to the Central scenario (Chart C4). Singapore would experience larger declines in output in 2020 and grow at a sluggish pace in 2021, with economic activity only recovering back to pre-crisis Q4 2019 levels beyond 2021. The overall domestic unemployment rate would also increase significantly to peak in mid-2021.

While both the Central and Adverse scenarios feature unprecedented steep declines in economic activity in a single year (2020), these stress scenarios are broadly similar in overall severity to those adopted for past stress test exercises, including the 2019 Industry-Wide Stress Test (IWST) exercise conducted in conjunction with the 2019 IMF Financial Sector Assessment Programme.³⁹

Chart C3 Both scenarios feature a global recession, with a slower recovery under the Adverse scenario

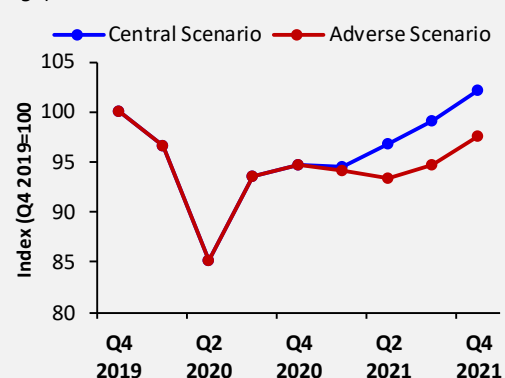
Global Real GDP Levels



Source: MAS

Chart C4 Both scenarios feature a sharp decline in domestic economic activity

Singapore Real GDP Levels



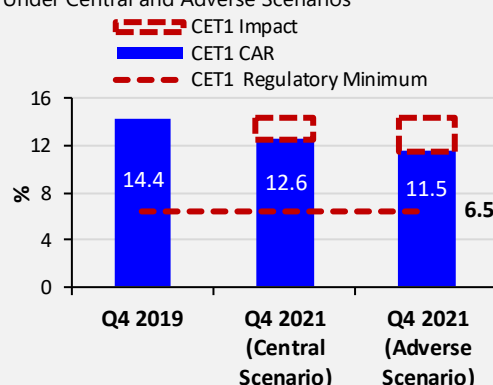
Source: MAS

D-SIBs' aggregate CET1 CAR is projected to keep well above their minimum regulatory requirements under the Central and Adverse scenarios.

Even under the Adverse scenario, the lowest point for D-SIBs' CET1 CAR reached in 2021 would still be well above MAS' minimum regulatory requirement of 6.5% (Chart C5). Overall, the desktop stress test results suggest that D-SIBs would have the capacity to absorb severe shocks under both scenarios while continuing to provide credit to businesses and households to support economic activity.

Chart C5 D-SIBs' aggregate CET1 CAR remains well above regulatory minimum under stress

D-SIBs' Aggregate CET1 CAR Under Central and Adverse Scenarios



Source: MAS estimates

Corporate credit losses could lead to a fall in D-SIBs' aggregate CET1 CAR by 0.8 and 1.4 percentage points at end-2021 under the Central and Adverse scenarios...

Output losses under both scenarios would place significant pressures on profitability and cashflows in the corporate sector, particularly for SMEs, as mobility restrictions and lockdown measures impede business activity. Under the Central scenario, the stronger economic recovery in 2021 should help to bolster corporates' financial positions and debt servicing ability, with corporate defaults expected to ease as business activity resumes and revenues recover. Under the Adverse scenario, corporate loan asset quality would deteriorate significantly, with higher credit losses arising from exposures to an increasing number of firms whose prospects may have been permanently impaired by the slower-than-anticipated resumption of economic activity in 2021.

...while D-SIBs' retail credit losses could cause their aggregate CET1 CAR to decline by around 0.1 and 0.2 percentage points under the Central and Adverse scenario.

Under both scenarios, weak labour market conditions are expected to negatively affect household incomes due to rising unemployment and a decline in wages. Retail credit losses

from the mortgage portfolio would be heavily dependent on residential property prices, as these prices would affect the amounts banks can recover following a default. Residential property prices are assumed to remain stable under the Central scenario, largely reflecting the likely actual outcomes in the domestic residential property market this year. Conversely, the Adverse scenario prescribes a cumulative fall of about 10% in residential property prices over 2020–2021. Nevertheless, D-SIBs remain resilient to losses incurred from defaulted property loans under the Adverse scenario given the healthy LTV ratios of their mortgage books.⁴⁰

Credit relief measures and management actions are expected to cushion the capital impact.

The desktop stress test also incorporated the effects of the relief measures introduced to facilitate credit supply to corporates and households. Enhanced government risk sharing for loans granted and debt moratoriums introduced by FIs for existing borrowers earlier in 2020 have helped to temporarily address cashflow challenges faced by corporates and households, reducing their near-term risk of default. An extension of credit relief measures introduced in October 2020 involving further partial deferments next year for eligible borrowers will also facilitate a more gradual transition to full loan repayments.

The credit relief measures for corporates and households are estimated to improve D-SIBs' aggregate CET1 CAR by an average of 0.2 percentage points by end-2021 under both scenarios (Chart C6). Under the Central scenario, the credit relief measures would help otherwise financially healthy corporates and households to tide over a temporary period of revenue and income disruptions. Borrowers would thus be expected to resume loan repayments when economic activity recovers in 2021. Under the Adverse scenario, the credit relief measures would initially forestall defaults from a larger number of distressed borrowers. Weak economic prospects in 2021 are expected to impair the viability of more borrowers, and banks would incur higher credit losses in second half of 2021 and into 2022 after the expiration of the extended credit relief measures. The CET1 impact of these credit losses would however be partly absorbed by the conservative provisioning already undertaken by D-SIBs.

Prudent management of dividend distributions could also potentially help absorb the impact of the economic shocks on banks' capital positions. In July 2020, MAS called on Singapore-headquartered locally-incorporated banks to cap their total dividends per share for FY2020 at 60% of FY2019 levels to pre-emptively bolster their resilience and capacity to lend amid the uncertainties ahead. The dividend savings from this management action in 2020 were incorporated into the desktop stress test results for the relevant D-SIBs, and are estimated to improve their aggregate CET1 CAR by about 0.3 percentage points on average by end-2021 under both stress scenarios (Chart C6).

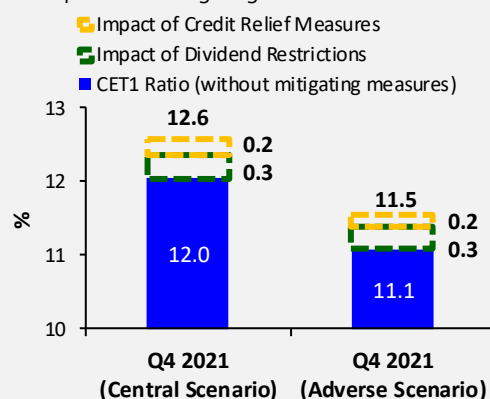
⁴⁰

Average LTV ratios of banks' mortgage books stood at 48.5% as of Q3 2020.

Overall, the projected CET1 CARs of 12.6% and 11.5% would have been about 0.5 percentage points lower under both scenarios in the absence of the credit relief measures and dividend restrictions.

Chart C6 Credit relief measures and dividend restrictions are expected to further bolster D-SIBs' capital positions

D-SIBs' Aggregate CET1 Capital Impact from Mitigating Measures*



Source: MAS estimates

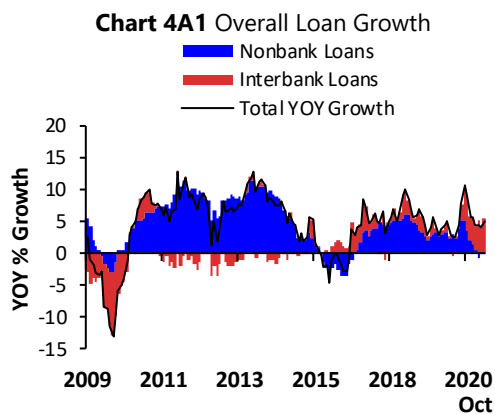
*Figures in the chart may not add up due to rounding.

Conclusion

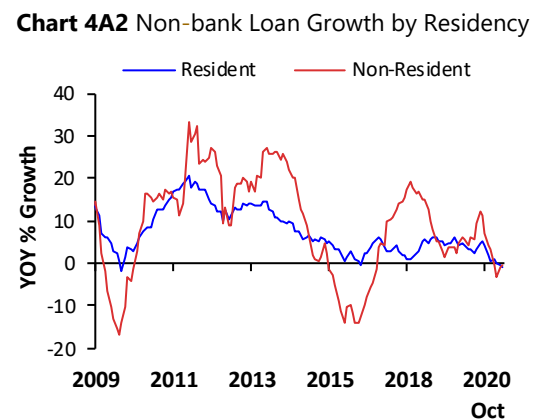
MAS' desktop stress test results suggest that D-SIBs are strongly positioned to weather the uncertainties arising from the COVID-19 pandemic, with CET1 CARs remaining well above MAS' regulatory requirements under both the Central and Adverse scenarios. D-SIBs have also undertaken pre-emptive loan provisioning that would be important in buffering their capital positions from the impact of potential credit losses. The credit relief measures and the guidance on dividend caps have further bolstered D-SIBs' capital positions, preserving their capacity to continue supplying credit to the economy while strengthening their resilience to further shocks should economic prospects turn out to be weaker than expected in 2021.

Chart Panel 4A Banking Sector: Credit Growth Trends

Overall loan growth increased in March 2020 before tapering to 4.9% y-o-y in October 2020, supported by a strong expansion in interbank loans. Non-bank loan growth has slowed significantly, as external demand weakened.



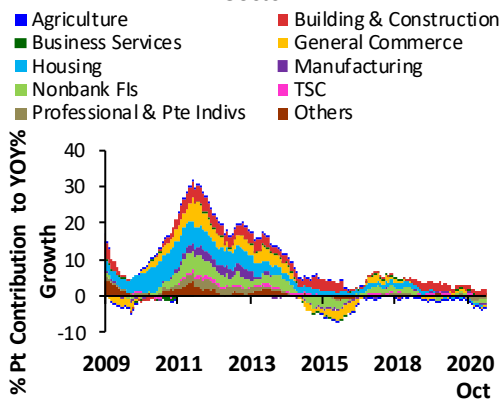
Source: MAS



Source: MAS

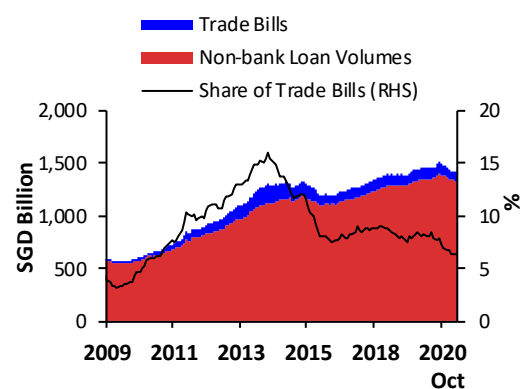
Resident non-bank credit growth has most recently registered a contraction, while trade financing volumes have moderated.

Chart 4A3 Domestic Banking Unit Non-bank Loans by Sector



Source: MAS

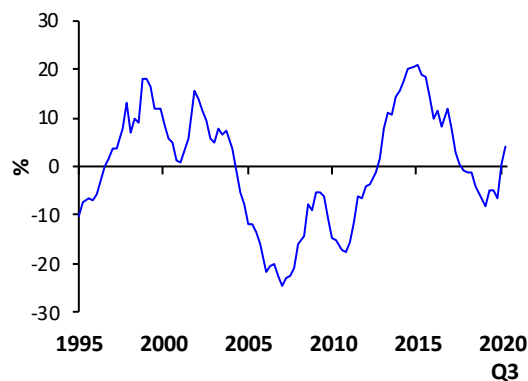
Chart 4A4 Share of Trade Bills in Non-bank Loans



Source: MAS

The credit-to-GDP gap for Singapore has increased to 7.3% as of Q3 2020.

Chart 4A5 Credit-to-GDP Gap

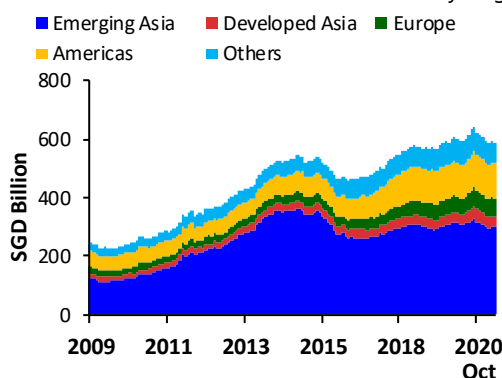


Source: MAS

Chart Panel 4B Banking Sector: Cross-border Lending Trends

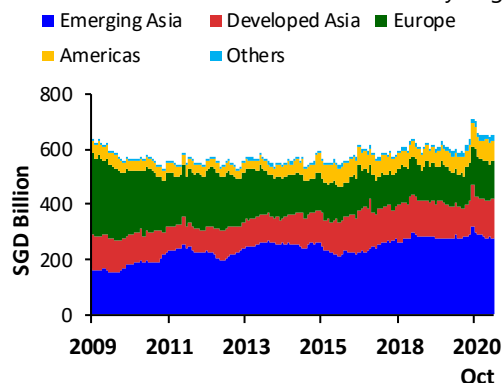
Non-resident non-bank loan growth has declined since the peak in March 2020. Meanwhile, the expansion in interbank loans was broad-based across all regions.

Chart 4B1 Cross-border Non-bank Loans by Region



Source: MAS

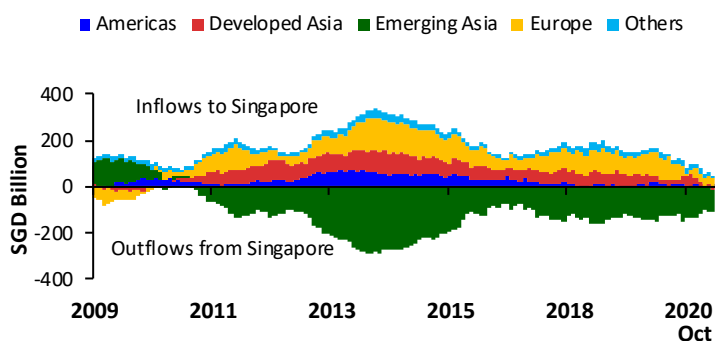
Chart 4B2 Cross-border Interbank Loans by Region



Source: MAS

Net lending to Emerging Asia has declined.

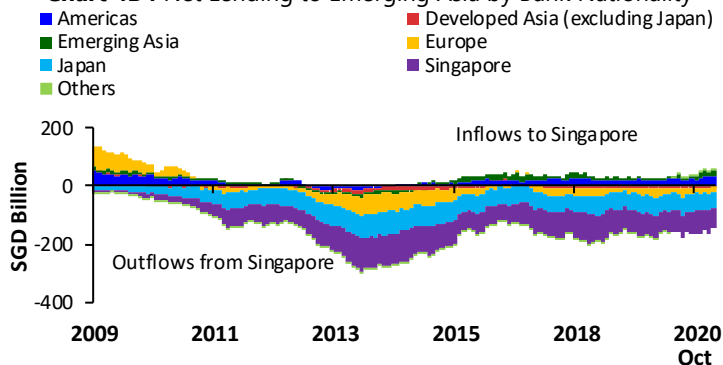
Chart 4B3 Net Lending by Region



Source: MAS

Local and Japanese banks continue to be the main net lenders to Emerging Asia.

Chart 4B4 Net Lending to Emerging Asia by Bank Nationality

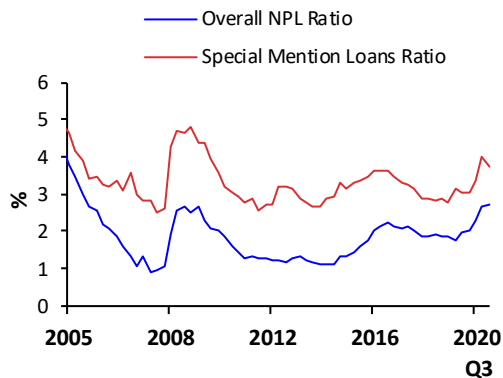


Source: MAS

Chart Panel 4C Banking Sector: Asset Quality and Liquidity Indicators

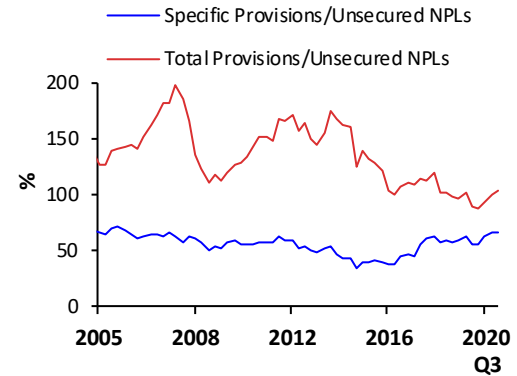
Overall NPL ratio increased to 2.7% in Q3 2020. Provisioning coverage improved to 103.1% as banks pre-emptively increased provisions in anticipation of further credit losses.

Chart 4C1 Banking System's Asset Quality



Source: MAS

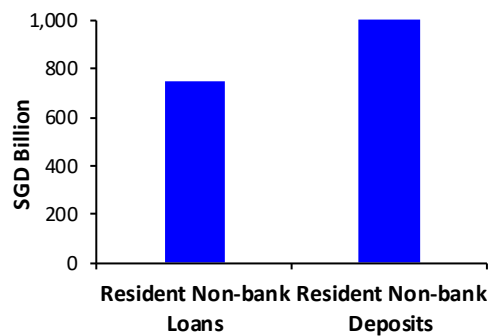
Chart 4C2 Banking System's Provisioning Coverage



Source: MAS

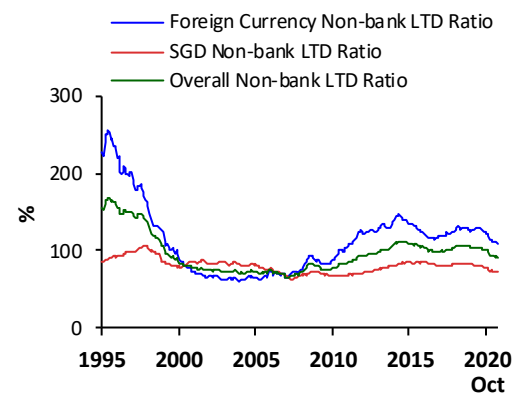
Banks' liquidity positions remain strong. Resident deposits are more than sufficient to fund resident loans. Foreign currency LTD ratio has also improved to 108.1% as of October 2020.

Chart 4C3 Banking System's Domestic Non-bank Loans and Deposits (as of October 2020)



Source: MAS

Chart 4C4 Banking System's Non-bank LTD Ratios

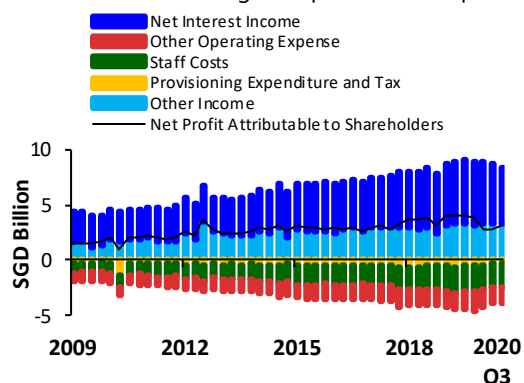


Source: MAS

Chart Panel 4D Banking Sector: Local Banking Groups

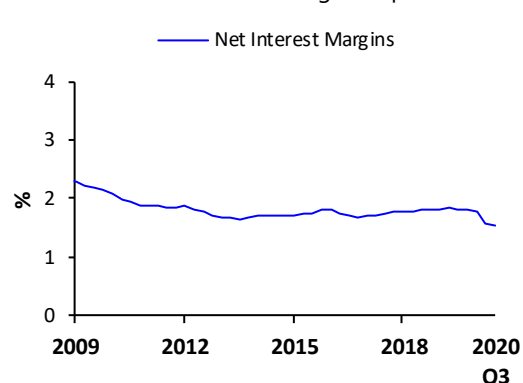
Local banking groups' net profit continues to be healthy, although it has declined due to lower NIMs and higher provisions.

Chart 4D1 Local Banking Groups' Profit Components



Source: MAS, Local banking groups' financial statements

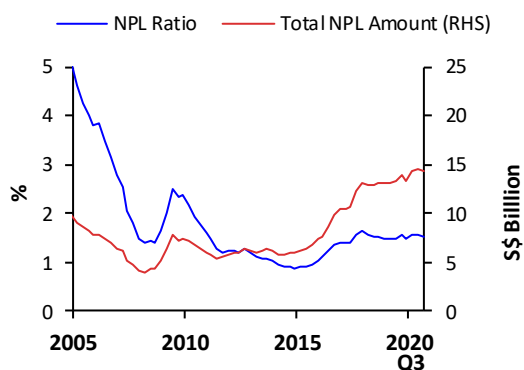
Chart 4D2 Local Banking Groups' NIM



Source: MAS, Local banking groups' financial statements

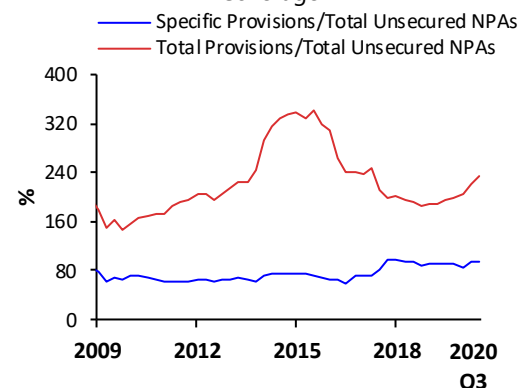
Asset quality has remained stable with the NPL ratio unchanged at 1.5%, due in part to various credit relief measures in place, while provisioning coverage rose to 234% in Q3 2020.

Chart 4D3 Local Banking Groups' NPLs



Source: MAS, Local banking groups' financial statements

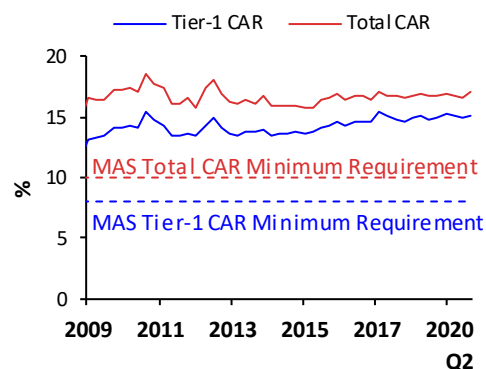
Chart 4D4 Local Banking Groups' Provisioning Coverage



Source: MAS, Local banking groups' financial statements

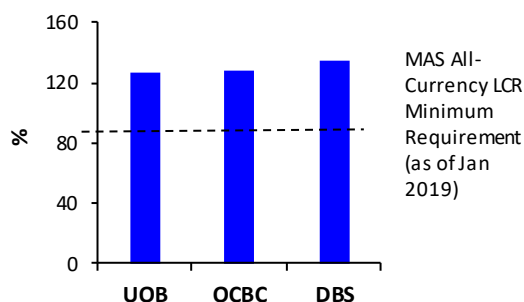
Local banking groups' capital and liquidity positions remain strong, with average CARs and all-currency LCRs well above regulatory requirements.

Chart 4D5 Local Banking Groups' CARs



Source: MAS, Local banking groups' financial statements

Chart 4D6 Local Banking Groups' All-Currency LCR (Q3 2020)

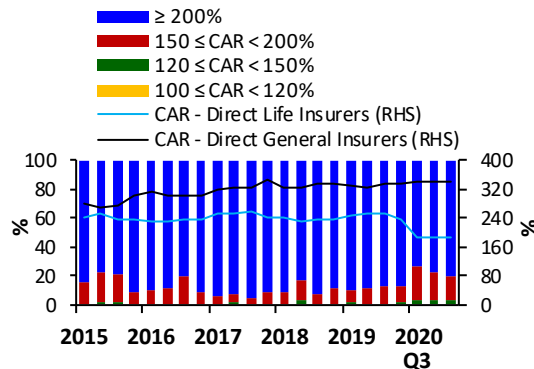


Source: MAS, Local banking groups' financial statements

Chart Panel 4E Insurance Sector

The average CARs for the direct life and direct general insurance industry remain well above regulatory requirements.⁴¹

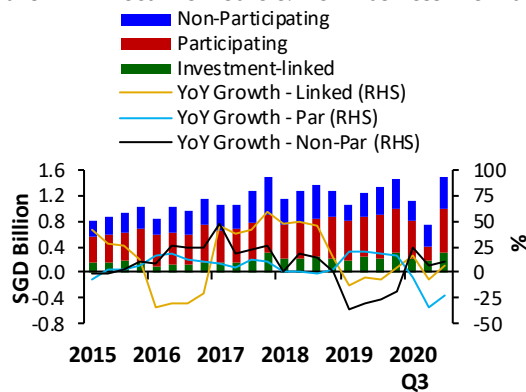
Chart 4E1 CARs of Direct Life and Direct General Insurers



Source: MAS

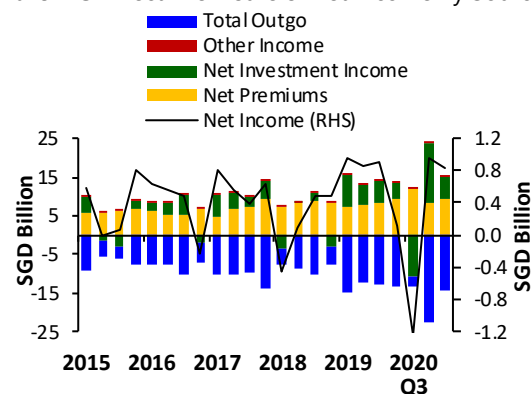
New business premiums of the direct life insurance industry fell in 2020, largely due to reduced sales of participating products. Net income decreased largely due to poorer investment performance as compared to 2019.

Chart 4E2 Direct Life Insurers: New Business Premiums



Source: MAS

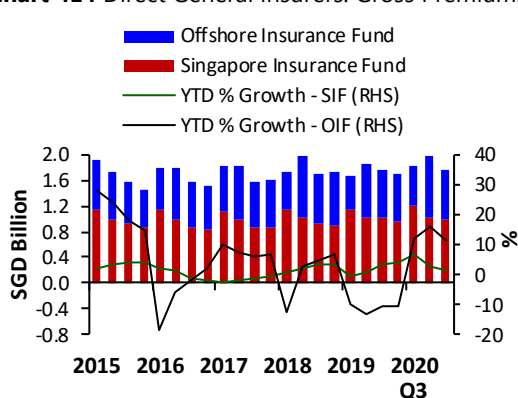
Chart 4E3 Direct Life Insurers' Net Income By Source



Source: MAS

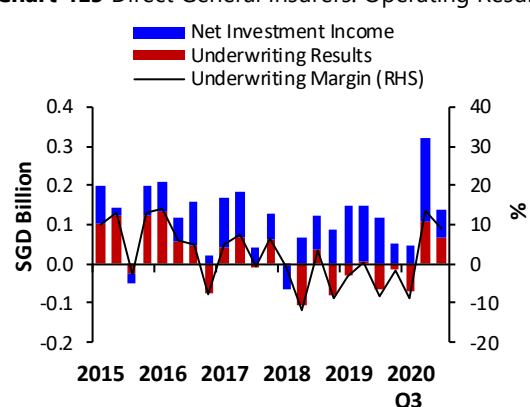
Gross premiums of the direct general insurance industry increased in 2020, mainly due to the increase in Offshore Insurance Fund (OIF) business. The industry reported positive underwriting results and investment profits in 2020.

Chart 4E4 Direct General Insurers: Gross Premiums



Source: MAS

Chart 4E5 Direct General Insurers: Operating Results



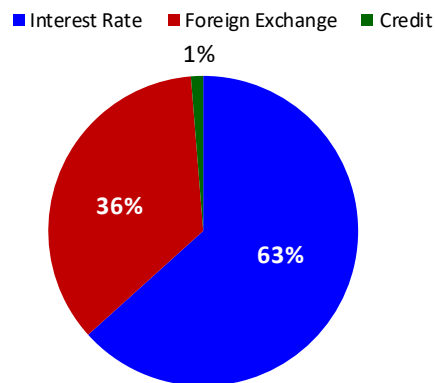
Source: MAS

⁴¹ With effect from Q1 2020, the valuation and capital framework for insurers in Singapore has been enhanced (revised Risk Based Capital framework, RBC 2). While the RBC 2 is not a major overhaul of the RBC, there are nonetheless some fundamental differences which make the CAR of the two regimes less directly comparable.

Chart Panel 4F Over-the-counter Derivatives

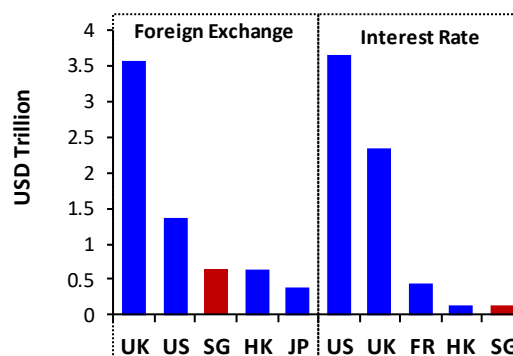
Based on trade reporting data, interest rate (IR) is the largest over-the-counter (OTC) derivative asset class booked and traded in Singapore, followed by foreign exchange (FX).

Chart 4F1 OTC Derivatives Market in Singapore by Asset Class and Notional Outstanding (End-Aug 2020)



Source: MAS estimates, Depository Trust & Clearing Corporation Data Repository (Singapore) Pte Ltd (DDRS)

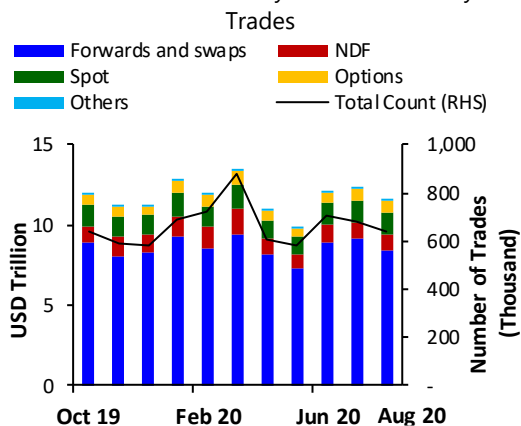
Chart 4F2 Breakdown of Top Five Economies in the Global OTC Derivatives Market by Annual Turnover



Source: 2019 BIS Triennial Survey

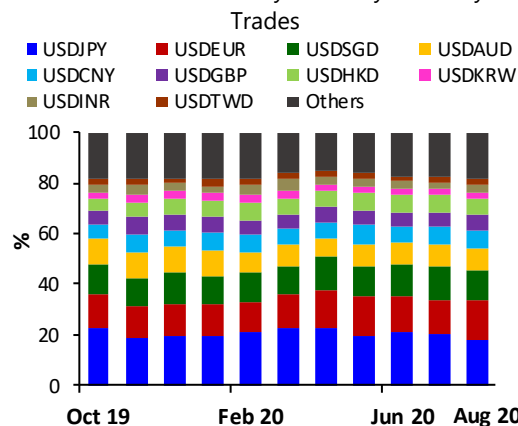
The FX derivative market in Singapore remains dominated mainly by forwards and swaps, with the majority of trades conducted in developed market currencies.

Chart 4F3 FX Derivatives by Product: Monthly New Trades



Source: MAS estimates, DDRS

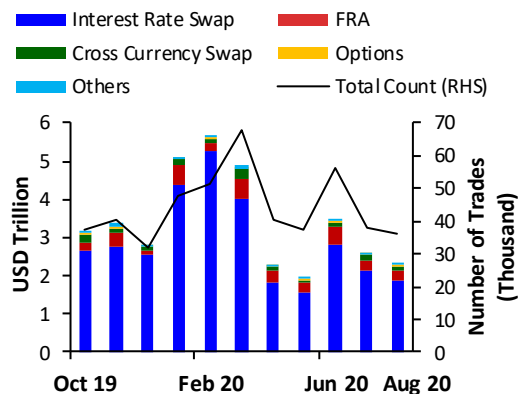
Chart 4F4 FX Derivatives by Currency: Monthly New Trades



Source: MAS estimates, DDRS

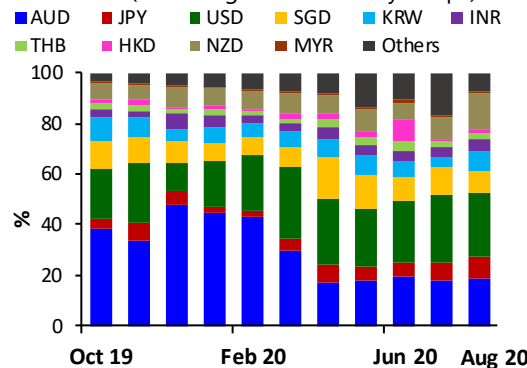
IR derivatives booked and traded in Singapore are mostly denominated in Asia-Pacific currencies, with the majority of trades being IR swaps.

Chart 4F5 IR Derivatives by Product: Monthly New Trades

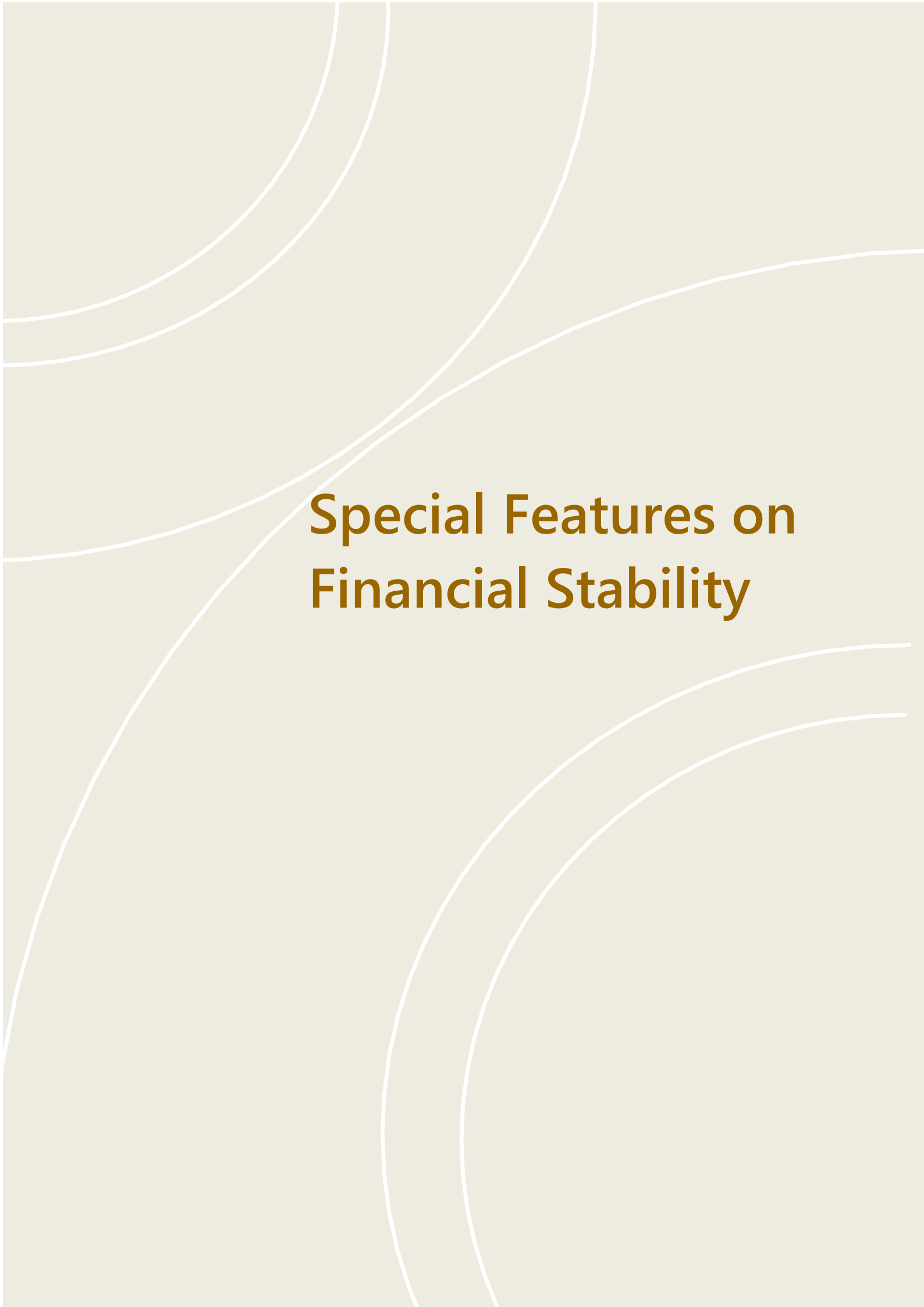


Source: MAS estimates, DDRS

Chart 4F6 IR Derivatives by Currency: Monthly New Trades (excluding cross-currency swaps)



Source: MAS estimates, DDRS



Special Features on Financial Stability

Special Feature 1

Implications of USD Dominance on Capital Flows and Financial Stability in EMEs

Introduction

The USD is by far the most widely used currency in corporates' cross-border trade invoicing and banks' cross-border loans. As the "dominant currency", US monetary policy shocks and exchange rate fluctuations spill over across borders and create challenges for policymakers in EMEs, including Asian EMEs.

This feature investigates whether fluctuations in the broad value of the USD, measured against a broad basket of currencies, are associated with changes in capital flows to EMEs. The feature then explores factors that could potentially mitigate or amplify capital outflows following broad USD appreciation. Finally, the association between USD appreciation and financial crises (e.g. banking and currency crises) in EMEs is examined, focusing on the outcome of financial instability as opposed to the more intermediate indicators such as global risk appetite and business cycles typically covered in existing studies.

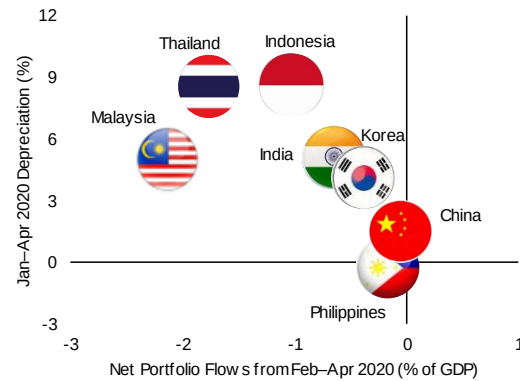
USD appreciation has been associated with capital outflows from Asian EMEs, and a reduction in cross-border loans.

US monetary policy shocks and exchange rate fluctuations could create challenges for Asian EMEs, which invoice a large proportion of their trade in USD and have large USD-denominated assets and liabilities. In the aftermath of the GFC, Asian EMEs experienced large foreign capital inflows, arguably driven by monetary policy decisions in major economies. More recently, USD appreciation against a broad range of currencies at the onset of COVID-19 in early 2020 also coincided with large net capital outflows from Asian EMEs (Chart S1.1).

Bruno and Shin (2015) and Avdjiev et al. (2018) show that broad USD appreciation reduces cross-border loans, working via both demand- and supply-side drivers. First, the bilateral appreciation of USD may worsen the net worth of unhedged local borrowers with currency mismatches and reduce their demand for credit. Second, broad USD appreciation increases the credit risk in an internationally-diversified portfolio of loans and the Value-at-Risk of global banks' portfolios, which in turn has the effect of reducing bank lending.

Chart S1.1 Between February and April 2020, significant broad USD appreciation was associated with large net capital outflows from Asian EMEs.

Exchange Rate Depreciation and Net Portfolio Flows in Selected Asian EMEs



Source: MAS estimates, BIS, IMF

USD appreciation has been associated with fluctuating global risk appetite and the business cycle.

Recent studies have identified fluctuations in the value of the USD as important factors driving the global financial cycle. Burcu et al. (2020) argue that the value of the USD has become a good measure of global risk appetite since the GFC, as it is tightly linked to the cost of funding global banks' lending activities. Shousha (2019) explores the link between USD strength and economic activity in EMEs using panel vector autoregression (VAR) methods. This feature finds that episodes of USD appreciation are important drivers of the business cycle in EMEs, leading to declines in GDP, investment and real credit to the private sector, as well as increases in sovereign spreads.

Empirical methodology

This feature first uses the baseline equation (1) below, to estimate the effect of broad USD appreciation on aggregate net capital flows, allowing for a decomposition into inflows and outflows⁴², as well as the type of flows (i.e. foreign direct investment, portfolio or other flows). The analysis is carried out for Asian EMEs and other EMEs. To the extent that broad fluctuations in USD are driven by shifts in risk appetite by global financial markets, a relationship between USD fluctuations and net capital flows would indicate an important channel through which shocks in risk appetite are transmitted to EMEs. Second, an extended equation (2) is used to estimate the factors that mitigate or amplify capital flows. That is, whether there are country-specific characteristics ("pull factors") that interact with USD appreciation ("push factor"). Finally, equation (3) is used to assess whether banking and currency crises in EMEs are more likely during periods of strong USD.

⁴² In balance of payments statistics, the distinction between inflows and outflows pertains to the residency of the decision maker. Inflows are purchases or sales of domestic assets by non-residents; outflows are purchases or sales of foreign assets by residents.

The three equations are estimated using unbalanced panel data regression for a sample of EMEs, based on availability of data.⁴³

Equation 1

$$\begin{aligned} & flows_{i,t+k} - flows_{i,t-1} \\ &= \beta_k usd_t + \gamma_k VIX_t + \sum_{j=1:4} \theta_j^{flows} flows_{t-j} + \sum_{j=1:4} \theta_j^{reer} REER_{t-j} \\ &+ \sum_{j=1:4} \theta_j^{int} Interest_{t-j} + \sum_{j=1:4} \theta_j^{USint} USinterest_{t-j} + \omega_{ik} + \varepsilon_{i,t+k} \end{aligned}$$

where *flows* refer to capital flows, that are scaled by GDP; *usd_t* is the change in the USD value against a broad basket of currencies (a positive value represents USD appreciation); *VIX* is the Chicago Board Options Exchange index of equity price volatility that is used as a proxy for market perception of risk and an indicator of the global financial cycle⁴⁴; *REER* is the EME real effective exchange rate change; *Interest* and *USinterest* are the EME and US policy interest rates respectively; and ω_{ik} is a country fixed effect for horizon *k*. The specification also controls for seasonal effects, while ε is a random error term with mean=0.

Equation 2

$$\begin{aligned} & flows_{i,t+k} - flows_{i,t-1} \\ &= \beta_{k0} usd_t + \beta_{k1} (usd_t \times X_t) + \gamma_k VIX_t + \sum_{j=1:4} \theta_j^{flows} flows_{t-j} + \sum_{j=1:4} \theta_j^{reer} REER_{t-j} \\ &+ \sum_{j=1:4} \theta_j^{int} Interest_{t-j} + \sum_{j=1:4} \theta_j^{USint} USinterest_{t-j} \end{aligned}$$

where X_t is a dummy variable to represent high/above the median (dummy=one) or low/below the median values (dummy=zero) for a few country-specific characteristics (listed in Table S1.2 of Appendix), β_{k0} measures the sensitivity of capital flows *k* periods ahead to USD appreciation when the dummy is equal to zero and $(\beta_{k0} + \beta_{k1})$ when the dummy is equal to one. Since there are several factors that may affect the vulnerability of capital flows to movements in the broad USD value, this equation introduces one factor at a time.

Equation 3

$$\begin{aligned} & Prob(crisis_{t+k} = 1 | crisis_t = 0) \\ &= \Phi(usd_t, VIX_t, \{REER_{t-j}\}_{j=1:4}, \{Interest_{t-j}\}_{j=1:4}, \{USinterest_{t-j}\}_{j=1:4}, \omega_{ik}) \end{aligned}$$

where $Prob(crisis_{t+k} = 1 | crisis_t = 0)$ refers to dummies for crisis episodes, based on the dating of banking and currency crisis in Laeven and Valencia (2013, 2020).

⁴³ The EMEs are Brazil, Bulgaria, Chile, China, Colombia, Costa Rica, Croatia, Dominican Republic, Ecuador, Hungary, India, Indonesia, Kazakhstan, Malaysia, Mauritius, Mexico, Mongolia, Morocco, Paraguay, Peru, Philippines, Poland, Romania, Russia, Saudi Arabia, South Africa, Sri Lanka, Thailand, Turkey, Ukraine and Uruguay, for the period Q1 1996 to Q4 2018.

⁴⁴ VIX is a real-time measure derived from option prices on the S&P 500 and informs us about risk pricing in the US equity market. VIX is highly correlated with broader measures of financial stress (such as the Financial Stress Index developed by the Federal Reserve Bank of St. Louis).

Findings

The findings of the empirical estimations are presented in the form of impulse response functions, and summarised below.

First, fluctuations in the broad value of the USD are associated with changes in net capital flows to EMEs, with Asian EMEs subject to larger shifts than other EMEs (Chart S1.2). From equation (1), the feature finds that Asian EMEs see larger reductions in net capital flows in the immediate quarter following a broad USD appreciation. A 1% broad USD appreciation is associated with a reduction in net capital flows of 0.8% of GDP for Asian EMEs, compared to 0.3% of GDP for other EMEs. The reduction in net capital flows also persists for three more quarters.

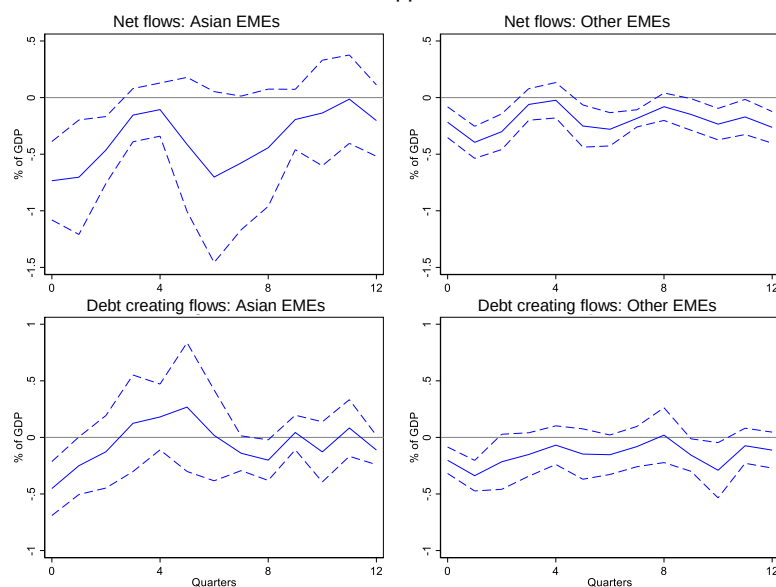
Second, fluctuations in the broad value of the USD are associated with larger reductions in debt-creating than in non-debt-creating flows to EMEs (Chart S1.3).⁴⁵ This finding is consistent with the widely held view that non debt-creating flows are more permanent and less sensitive to short-term market vagaries. The distinction is meaningful because debt-creating flows generate future liabilities for the recipient, even during times of volatility or instability when the borrowers' income streams are impacted. A reduction in debt-creating flows thus makes rolling over existing debt more difficult and increases financial stability risks. In contrast, non-debt creating flows are more akin to equity, and may or may not generate future cashflow liabilities for recipients.

Within debt-creating flows, EMEs experience reductions in portfolio debt and other inflows, but see larger and more persistent reductions in the "other capital flows" category, which includes flows intermediated by banks. This finding is in line with Burcu et al. (2020) who argue that reductions in inflows to EMEs in tandem with USD appreciation may be induced by risk management practices of global banks. A broad appreciation in USD would result in greater tail risk and tightening of value-at-risk constraints for global banks with an internationally diversified portfolio of loans, resulting in less cross-border lending.

⁴⁵ Debt-creating flows include bank loans and debt securities transactions; non-debt creating flows include foreign direct investment and equity securities transactions.

Chart S1.2 Capital flows to Asian EMEs are more “vulnerable” to USD appreciation than other EMEs

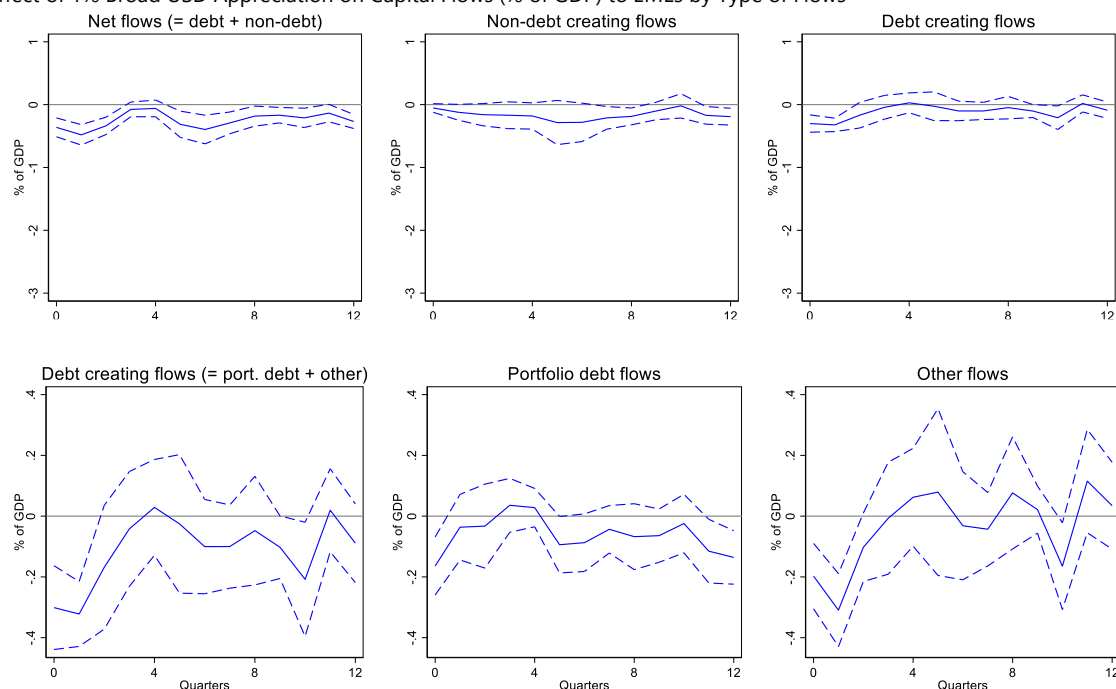
Effect of 1% Broad USD Appreciation on Capital Flows (% of GDP) to Asian EMEs and Other EMEs



Source: MAS estimates

Chart S1.3 USD appreciation is contemporaneously associated with larger reductions in net debt-creating flows, in comparison with non-debt-creating flows

Effect of 1% Broad USD Appreciation on Capital Flows (% of GDP) to EMEs by Type of Flows



Source: MAS estimates

Third, fluctuations in the broad value of the USD are associated with comparatively larger reductions in capital flows to high-growth and more financially open EMEs. The finding is aligned with the expectation that low-growth and financially closed EMEs are less likely to be

recipients of large capital flows in the first place. EMEs with higher reserves also see larger reductions in capital flows with USD appreciation. These findings suggest that the transitory experiences of capital flows in Asian EMEs associated with broad USD appreciation are possibly reflective of their larger FX reserves and net external positions.

Fourth, within-country domestic credit spreads rather than cross-country interest differentials are associated with a greater sensitivity of capital flows to broad USD fluctuations. This evidence is suggestive that capital flow volatility is more related to inefficiencies in domestic financial intermediation than to carry trades, which are driven by cross-country interest differentials.

Table S1.1 lists the detailed effects of various country-specific factors on capital flows, drawn from the estimation of equation (2).

Table S1.1 Effects of various country-specific factors on capital flows following broad USD appreciation

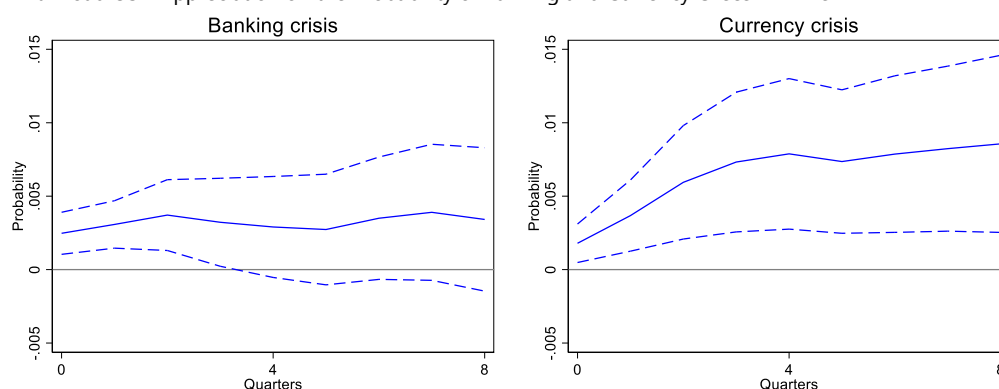
Country-specific factors	Effect on capital flows to EMEs	Potential explanations on transmission channels/ results
Higher expected future growth differentials	Larger outflows in short run	Countries with higher growth rates experience larger reversal in flows than countries with lower growth rates in the short run, possibly partly because they are structurally more open (financially). In the longer run, low-growth countries experience a protracted reversal in flows, likely reflecting their weaker fundamentals.
Higher foreign exchange reserves	Larger outflows	Countries may build up larger FX reserves as a precautionary response to existing vulnerabilities to capital flow fluctuations.
Higher gross external positions (or higher financial openness)	No significant differences	Countries that are more financially open (or integrated) tend to have larger inward and outward capital flows and this may be stabilising if they offset each other, but larger gross positions may also give rise to more significant currency mismatches.
Larger domestic interest rate spreads between lending and policy rates	Larger outflows	Higher domestic interest rate spreads may reflect inefficient financial intermediation and lower quality assets in the financial sector balance sheet, which may increase susceptibility to volatile capital flows.
Larger interest rate differential relative to USD rates	No significant difference	While countries with larger interest rate differentials with respect to USD interest rates are likely to be subject to carry trade strategies, many EME currencies are not used in such trades, because of higher transactions costs, capital flow management measures and lack of market depth.
Few capital flow management policies	No significant difference in short term, but larger outflows in long term	Countries with fewer capital controls experience more persistent reduction in net capital flows, suggesting the ease of a permanent shift in investments by foreign investors. However, since capital controls are a policy variable that is set in response to each country's perception of risks and vulnerabilities, there is the potential for endogeneity bias in these results.

Finally, broad USD appreciation has a significant association with banking and currency crises in EMEs (Chart S1.4). As broad USD appreciation is associated with larger reductions in net capital flows, including the ones intermediated by banks, this translates to reduced credit

for the private sector and consequent slowdown in economic activity. Together, the debt servicing abilities of corporates could eventually weaken, potentially triggering a systemic risk event if sufficiently widespread. From equation (3), the estimation reveals that an appreciation of 1% in the broad USD index in a quarter is associated with an increase in the probability of banking and currency crises of about 0.3 to 0.4 percentage points in the following quarter, equivalent to a 50% increase in the probability of each type of crisis.

Chart S1.4 USD appreciation is associated with increased occurrences of banking and currency crises

Effect of 1% Broad USD Appreciation on the Probability of Banking and Currency Crises in EMEs



Source: MAS estimates

Sum-up

EME policymakers should be watchful of episodes of generalised broad-based strength in the USD, as they are associated with increased occurrences of banking and currency crises that threaten financial stability in EMEs. High-growth EMEs, those with a high degree of financial openness, those with large domestic credit spreads, and those with large foreign exchange reserves are more exposed to fluctuations in capital flows. This raises the importance of pre-emptive macro- and micro-prudential measures that limit currency mismatches in these EMEs and reduce the risk that fluctuations in capital flows would trigger financial stresses.

Appendix

Table S1.2 Country-specific characteristics

List of country-specific factors	Sources
Expected future growth differentials	Consensus Forecasts
Size of foreign exchange reserves	HAVER
Size of gross external positions or financial openness	External Wealth of Nations database by Lane and Milesi-Ferretti, 2018
Size of domestic interest rate spreads	IMF International Financial Statistics (IFS)
Interest rate differential relative to USD rates	IMF IFS
Capital flow management policies	Fernández et al. (2015), based on the IMF's Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER)

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Special Feature 2**Transitioning to a New Interest Rate Benchmark Landscape in Singapore****Introduction**

LIBOR is expected to be discontinued after end-2021. Despite having undergone substantial reforms to enhance its robustness since 2012, declining liquidity in the underlying unsecured term funding markets has hampered the price discovery process and affected the credibility and reliability of LIBOR. LIBOR's existence is thus only guaranteed till end-2021. Financial markets globally have been ramping up efforts to transit away from LIBOR, towards the use of near risk-free rates (RFRs) such as the Secured Overnight Financing Rate (SOFR), Sterling Overnight Index Average (SONIA), Euro Short-Term Rates (ESTER), Swiss Average Rate Overnight (SARON) and Tokyo Overnight Average Rate (TONA).

LIBOR is referenced in about USD400 trillion of financial products globally (Schrimpf et al., 2019). Of this amount, an estimated 40-60% of assets and derivative exposures will mature after end-2021 (Financial Stability Board (FSB) et al., 2020). In Singapore, about SGD282 billion of LIBOR-linked cash products and a further SGD2.4 trillion of LIBOR-linked derivatives will mature after 2021.

Given the scale of LIBOR usage, it is imperative that market participants, both lenders and borrowers, as well as financial and non-financial corporates, prepare for the transition from LIBOR. The FSB has identified interest rate benchmark reform as a key priority, and reiterated that continued reliance of global financial markets on LIBOR poses risks to global financial stability.⁴⁶

Moving to a Singapore Overnight Rate Average-centred SGD interest rate landscape

Interest rates in Singapore are largely determined by global interest rate developments, given our exchange rate-centred monetary policy framework. As such, robust market-driven interest rate benchmarks play an important role in the domestic financial system and the broader economy. Similar to other economies, interest rate benchmarks are used widely by

⁴⁶ In October 2020, the FSB published a global transition roadmap for LIBOR, which sets out a timetable of actions that financial and non-financial sector firms should take, to ensure a smooth transition by end-2021 (FSB, 2020a). Key steps in the roadmap include:

- Firms should have already identified and assessed all existing LIBOR exposures, and agreed on a plan to transition in advance of end-2021.
- By end-2020, firms should be in a position of offer non-LIBOR linked loans to their customers.
- By mid-2021, firms should have established formalised plans to amend legacy contracts where possible, and have implemented the necessary system and process changes to enable the transition to alternative rates.
- By end-2021, firms should be prepared for LIBOR to cease.

Fls to price loans to individuals and businesses, financial market participants to price financial products such as bonds, floating rate notes (FRNs) and interest rate swaps, and corporates, as a discount rate for financial reporting of their assets.

The Singapore Interbank Offered Rate (SIBOR) and SGD Swap Offer Rate (SOR) are key interest rate benchmarks in SGD financial markets. They meet the needs of different user groups—SOR, referenced in SGD derivatives, has largely been used to price bonds and loans to large institutions with hedging requirements, while SIBOR is mainly referenced in banking products for smaller corporates and retail customers.

As SOR uses USD LIBOR in its computation, the impending end of LIBOR would mean that SOR would be discontinued after end-2021. In this regard, the financial industry in Singapore had, in August 2019, recommended a transition from SOR to the Singapore Overnight Rate Average (SORA) (Association Bank of Singapore (ABS) et al., 2019). SORA is a robust benchmark underpinned by actual transactions in a deep and liquid overnight unsecured SGD interbank funding market.

In July 2020, the financial industry issued a joint industry consultation⁴⁷ recommending the discontinuation of the SIBOR benchmark in three to four years to focus on a SORA-centred approach (ABS et al., 2020). This would bring significant long-term benefits such as enhanced market efficiency, increased price transparency and greater market competition, while meeting end-customer needs. The consultation closed on 30 September 2020, with the final recommendations expected by end-2020.

The shift towards a SORA-centred approach is expected to enhance the overall functionality and efficiency of SGD interest rate markets. For banks, this approach will reduce basis risk between assets and liabilities based on two different benchmarks and allow for greater pricing efficiencies, compared to managing both SIBOR and SOR exposures. For borrowers, the averaging effect from a compounded SORA will provide more stable rates compared to single-day SOR readings, which are exposed to idiosyncratic market factors, such as quarter- or year-end volatility. In addition, the adoption of a SORA-centred approach is aligned with the direction in key financial centres, and will support the continued participation of global institutions and investors in SGD markets.

Supporting the shift towards SORA

Completing the transition from SOR to SORA before end-2021 is important to mitigate risks to financial stability. After end-2021, the continued production of LIBOR and hence SOR cannot be guaranteed, and there is a material risk that these benchmarks could cease to be produced at short notice. Any delay in transition would inhibit the development of a sufficiently liquid market needed for price discovery, which in turn would impact pricing efficiencies in the market. Existing contracts referencing SOR after the discontinuation of LIBOR would also face contract frustration risk. In addition to the SGD2.7 trillion of LIBOR-linked contracts mentioned earlier that will be directly impacted by the discontinuation of LIBOR, close to SGD1.5 trillion of SOR-linked cash and derivative assets—comprising SGD90

⁴⁷ The joint industry consultation was issued by ABS, the Singapore Foreign Exchange Market Committee (SFEMC) and the Steering Committee for SOR Transition to SORA.

billion of SOR-linked cash assets (largely syndicated and bilateral, corporate loans) and SGD1.4 trillion of SOR-linked derivatives—will mature after 2021.

To reduce the risks of disorderly market outcomes and contract frustration for FIs and end users with SOR exposures, market participants need to take active steps to shift from SOR to SORA. To this end, MAS established the industry-led Steering Committee for SOR Transition to SORA (SC-STS) in August 2019, to provide strategic direction on developing new SORA-based products and markets, and to coordinate with relevant stakeholders. Good progress has been made through various initiatives by the SC-STS and MAS, including the following:

Establishing market conventions and infrastructure

- Key infrastructure: These initiatives formed the foundation to support trading of SORA derivatives, and included establishing central clearing of SORA derivatives by London Clearing House, making available SORA data on platforms such as Bloomberg and Refinitiv, and SORA quotes on broker screens.
- Standardised SORA references: MAS commenced daily publishing of 1-month, 3-month and 6-month compounded SORA and a SORA Index, to provide market participants with standardised references for use in SORA products (MAS, 2020).
- SORA market conventions: SC-STS published a comprehensive market compendium setting out SORA market conventions across SGD products, and considerations for contractual fallbacks for relevant SOR contracts, to facilitate widespread adoption and transparency (SC-STS, 2020).

Building liquidity in SORA markets

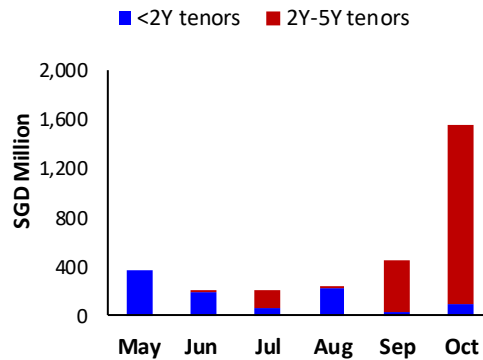
- MAS SORA FRNs: MAS established a SORA FRN programme, as part of its money market operations⁴⁸, to facilitate the adoption of SORA as a floating rate benchmark, and spur hedging activities in the SORA derivatives market.
- SORA loan products: The industry spearheaded a number of pilot SORA business and retail loans, paving the way for the launch of SORA loan products at scale in 2021. Some of these products include retail home loans and business property loans referencing MAS-published compounded SORA rates, dual currency SORA-SOFR and club or syndicated loans.⁴⁹
- SORA derivatives auctions: MAS established a daily auction process for SORA derivatives to facilitate price discovery in the growing SORA derivatives market. The daily auctions have contributed to an increase in derivatives volumes and extension in tenor (Chart S2.1), in tandem with the growth in usage of SORA in cash markets.

⁴⁸ See outstanding stock of MAS SORA FRN at <https://www.mas.gov.sg/bonds-and-bills/outstanding-mas-frn>.

⁴⁹ See SC-STS' announcements of notable SORA cash market issuances at <https://www.abs.org.sg/benchmark-rates/announcements>.

Chart S2.1 Increasing SORA derivatives transactions in Singapore

SORA Derivatives Turnover 2020



Source: MAS estimates, DRS

Facilitating legacy transition

- **Contractual fallbacks:** A number of market participants have signed up to the International Swaps and Derivatives Association (ISDA) Interbank Offered Rate (IBOR) Fallbacks Supplement and Protocol⁵⁰, which covers contractual fallbacks for SOR derivatives. SC-STS has also published guidance on contractual fallbacks for SOR loan contracts in the SORA market compendium. Incorporating contractual fallbacks into legacy contracts will provide a safety net to mitigate against risks of contractual frustration or settlement issues when SOR is discontinued.
- **Coordinated industry timelines:** Key timelines announced by the industry have enhanced transition momentum through coordinated action to reduce further usage of SOR in financial products, and would accelerate adoption of SORA:
 - By end-April 2021, all lenders and borrowers to cease issuance of SOR-linked loans and securities that mature after end-2021;
 - By end-February, all D-SIBs will offer a full-suite of SORA-based products to their customers. All non-DSIB banks to do so by end-April 2021;
 - By end-September 2021, all banks to have substantially reduced gross exposures to SOR derivatives.

⁵⁰ See ISDA's Press Release on the Launch of the IBOR Fallbacks Supplement and Protocol, October 2020 at <https://www.isda.org/2020/10/23/isda-launches-ibor-fallbacks-supplement-and-protocol/>

Looking to 2021: Next steps in the SOR endgame

In 2021, as we approach the SOR endgame, FIs and end-customers will need to make the necessary preparations to:

- Shift new business in loans and derivatives to SORA, and cease new SOR-linked originations, in line with the SC-STS recommended timelines mentioned above.
- Actively convert legacy SOR contracts to reference SORA.
- Ensure widespread adoption of appropriate contractual fallbacks in derivatives and cash market contracts, especially for market participants who are unable to complete the transition of all their SOR contracts to SORA by end-2021.

FIs would need to enhance their systems and processes, to adapt to the use of overnight rates like SORA. Failure to do so would expose FIs to additional market, liquidity, operational, technology and legal risks at end-2021. In this regard, MAS has conveyed supervisory expectations on FIs' preparations, and will continue to closely monitor the transition progress in 2021.

FIs would also need to engage customers in a clear, timely and transparent manner, when converting legacy SOR contracts to reference SORA. They need to provide clear explanations on: (a) the shift to SORA in the SGD interest rate market, (b) how this benchmark transition will impact them, and (c) how customer spreads over the loan reference rate may have to be adjusted to account for term and credit risk premia previously incorporated in SOR but not present in SORA. In this regard, FIs should also take reference from MAS' Fair Dealing Guidelines.

Customers should work with their FIs on a suitable pathway to transition from SOR towards alternative reference rates. The SC-STS has published several useful customer reference documents to support end-customers' awareness and understanding.⁵¹ To complement individual FIs' customer engagement efforts, MAS, ABS and SC-STS would be rolling out industry-wide public education programmes on SORA products and markets to raise customer awareness and understanding.

As these alternative benchmark products are increasingly used and market liquidity deepens, the bridging relationships across them will gradually be established, providing for anchorage in interest rate determination within the context of Singapore's exchange rate based monetary policy framework.

⁵¹ These included an end-user checklist (<https://abs.org.sg/docs/library/sc-sts-end-user-checklist-on-benchmark-transition.pdf>), to support internal preparatory work for non-financial corporates, investment firms, insurers and other buy-side participants, and a customer segmentation report (<https://abs.org.sg/docs/library/overview-on-the-usage-of-sora-in-loans-customer-segments-and-preferences.pdf>), which sets out how SORA can be used in loans for financial firms, corporates, and retail customers.

Conclusion

The shift away from LIBOR and SOR reflects an important change in the global interest rate landscape and will impact all financial market participants in Singapore—lenders, borrowers, investors, and savers.

The financial industry, together with MAS, has made good progress in its comprehensive and well-structured transition approach. However, further work remains in 2021. Industry timelines set out by the SC-STS is envisaged to steadily reduce exposures to SOR well before the end of next year, while ensuring that SORA-based products are available to meet the financing needs of customers. MAS will continue to work closely with industry stakeholders to ensure the adjustment process is smooth and our financial system is well prepared for this change.

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Special Feature 3

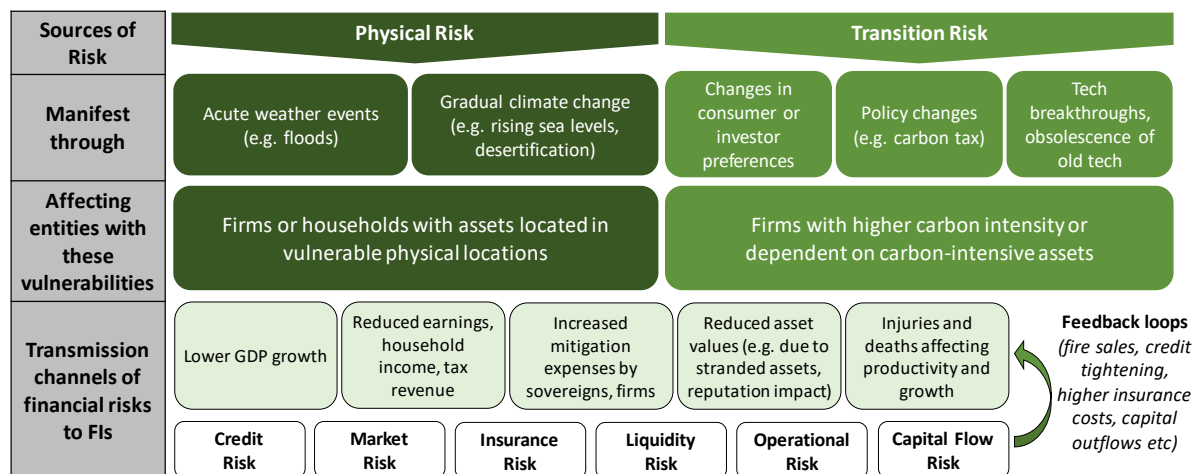
Climate Change and Financial Stability

Introduction

There is growing recognition among central banks and regulators globally that climate change presents an important source of risk to the financial system. Climate change could affect FIs either directly, but more so indirectly via their customers, through its environmental impact (“physical risk”), or through mitigation actions, new technologies, and changes in market preferences (“transition risk”).

This feature outlines the channels through which physical risk and transition risk can impact FIs (summarised in Figure S3.1), and details MAS’ planned approach in assessing these risks.

Figure S3.1 Implications of climate risks on the financial system



Source: MAS

Physical risk

Physical risk includes the economic costs and financial losses from the exposure of human and natural systems to climate-related events. These can arise from both acute weather events (such as flash floods) and more gradual, long-term changes in climate patterns (such as the gradual melting of polar ice sheets or desertification). In turn, such physical risk events could affect firms and households with assets located in vulnerable areas. Physical risk could also have broader macroeconomic implications, including through impairment of the productivity and stock of private-owned capital and public infrastructure in a jurisdiction, or through the labour force (such as due to migration).

Transition risk

Transition risk concerns the uncertain economic and financial costs of the adjustment to a low-emissions economy, including those induced by policy changes, technological breakthroughs, as well as shifts in investor preferences and social norms (Bolton et al., 2020). Transition risk can negatively affect asset values and income of firms and sovereigns, particularly those with a more intensive ongoing dependence on carbon-intensive assets. Conversely, firms with more environmentally-friendly processes (for instance, based on carbon capture and sequestration technologies) might stand to benefit. Ultimately, these changes in firms' balance sheets would also impact FIs and markets through their exposures to these firms.

Transition risk would be heightened with more rapid or disorderly change, whether in policy, social preferences or technology. For instance, the impact of the implementation of a carbon tax would be greater if unanticipated, and could lead to a sharp fall in the value of carbon-intensive assets, such as coal-fired power plants (Caldecott et al., 2015).

Transmission channels of climate risks

The impact of physical and transition risks on the financial system can materialise through the usual financial risk transmission channels. These include credit, market, insurance underwriting, liquidity, operational and capital flow risks. There are also potential feedback loops that could lead to second-order effects on the financial system.

Credit risk

Climate risks can impair borrowers' ability to repay their debts, be they sovereigns, corporates or households. The potential for a sharp devaluation of collateralised assets can also contribute to higher credit risk.

Physical risk can negatively affect a government's fiscal position by reducing the economy's growth prospects, and consequently tax income streams. Acute climate events can also cause significant injury, loss of life and infrastructure damage, affecting the overall labour force, physical capital and productivity. It may also lead to migration pressures. These could result in feedback loop effects whereby weaker fiscal health would also impair a sovereign's ability to invest in pro-growth strategies, with consequences for the economy's long-term potential path.

At the same time, governments may be forced to spend more on rebuilding or recovering destroyed physical assets, as well as on climate mitigation (Irwin, 2019). In regions where the insurance protection gap for natural catastrophes varies significantly, such as Emerging Asia, governments would bear more of the cost of rebuilding after a physical risk event, leaving the government with fewer resources to spend on other productive uses such as education, social services, or healthcare.

Governments also face transition risk directly through their holdings of fossil fuel assets. Public sector entities in some jurisdictions own a significant portion of global oil, gas and coal

resources, and collect taxes and royalties on the portions they do not own. The potential impact of stranded assets⁵² would be compounded in countries where both direct government holdings and economic activity are concentrated in carbon-intensive industries. Thus, stranded assets may affect the government's ability to meet its debt obligations.

Corporates with physical assets in locations susceptible to physical risk or with carbon-intensive processes could face higher operating expenses, for instance, due to increased spending on flood mitigation or taxes on carbon emissions. They also face obsolescence risk or declining revenue if their business models are affected by changes in technology or preferences. The corresponding impact on profits could reduce their ability to roll over their debts (Colas et al., 2019) or raise their funding costs, resulting in increased credit risk.

Meanwhile, climate risks could reduce household wealth and income through several channels. These include (i) the fall in home values either due to physical damage or the pricing-in of future physical risk, (ii) the devaluation of investment portfolios due to declining asset values (e.g. securities holdings of firms subject to climate risks), or (iii) reduced income due to increased unemployment or slower GDP growth as a result of climate risks. A reduction in household wealth and income could lead to a deterioration in households' debt servicing ability and increase credit risk.

Market risk

Climate change could lead to sharp changes in the market prices of securities and derivatives, particularly those related to carbon-intensive industries and commodities. For instance, a significant decline in existing asset values could occur under a transition scenario characterised by sudden changes in government policy (e.g. an unanticipated ban on the use of internal combustion engines (Unsworth et al., 2020)), or the adoption of new technology that either renders certain businesses obsolete or causes discontinuous shifts in consumer preferences. Meanwhile, under such a scenario, other "greener" industries might conversely see increases in market values. These changes in financial valuations pose market risk to investors, depending on their trading and investment positions.

Insurance underwriting risk

Insurance companies could face large underwriting losses in the event of sustained or cumulative catastrophic claims (Schulten et al., 2019; Bevere et al., 2018) due to physical risk (e.g. death, illness, destruction of property), especially if the occurrence and severity of such events become more unpredictable and correlated.⁵³ In the medium to longer term, the increased frequency and severity of physical damage from climate change could force insurers to increase premiums, or even render certain risks completely uninsurable. This could reduce or even impede risk transfers within the economy (FSB, 2020), making it more difficult for firms and households to mitigate climate risks, ultimately worsening economic and welfare effects.

⁵² Stranded assets generally refer to assets suffering from unanticipated or premature write-downs, devaluations or conversions to liabilities as a result of climate change.

⁵³ Based on data from the Swiss Re Institute, global insured losses in 2017 from disaster events hit a historical high of USD144 billion, with uninsured losses estimated at USD337 billion for the same period.

Liquidity risk

For insurers, the unpredictability of climate-related events could lead to larger-than-expected insurance claim payouts, especially where there has been sustained and repetitive physical damage over a short time period. That said, insurers typically hold short-term assets that match their liability profiles, and may have some lead time before paying out claims. Additionally, where direct insurers use proportional reinsurance contracts, these usually have a cash call provision that allows a direct insurer to receive immediate payments for a large loss, mitigating immediate liquidity strains.

For banks, a deterioration in the credit quality or an increase in volatility for certain assets might disqualify them from being considered as High Quality Liquid Assets (HQLA). Where banks hold similar HQLA (for instance, bonds issued by carbon-intensive firms), a transition risk event may drive these banks to sell and replace their HQLA at the same time.⁵⁴

Operational risk

FIs are also exposed to climate-related operational risk as their infrastructure (e.g. physical offices, ATM networks, network lines and servers) can be damaged by severe and persistent physical risk, impeding their business operations. They could also be subject to increasing regulatory requirements and legal risk from climate-related legislation and lawsuits.

Capital flow risk

As environmental, social and corporate governance (referred to in this special feature as ESG) criteria increasingly guide investors' investments (Organisation for Economic Co-operation and Development (OECD), 2017), the demand for financial assets that fulfil such criteria would correspondingly rise, shifting capital away from non ESG-compliant assets. EMEs, which tend to have fewer ESG-compliant investment opportunities, could face capital outflows as investments are reallocated. This could lead to currency depreciation and potential refinancing difficulties for sovereigns and corporates borrowing in foreign currencies. Where these capital flows are particularly large, balance of payments stresses could potentially emerge. Relatedly, if the supply of such ESG-compliant assets does not rise in tandem, there is the potential for "bubbles" in ESG-compliant assets, in a situation of "too much money chasing too few assets". This raises the potential for market dislocations (e.g. if a change in ESG assessment criteria causes shifts in the ESG classifications of these assets).

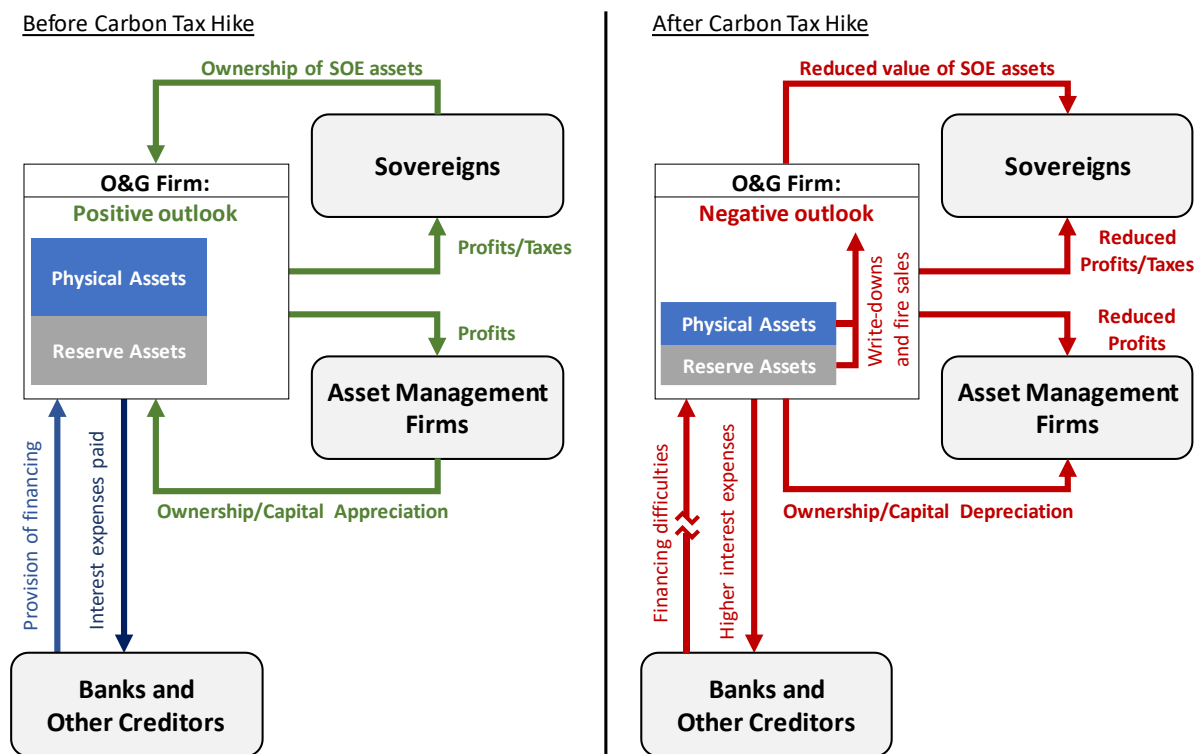
Feedback loops

The risks identified above may also give rise to negative feedback loops. For example, rising sea levels and more frequent extreme weather events could diminish property values and cause losses for homeowners, increasing the risks to FIs' mortgage portfolios. FIs may also choose not to lend to buyers in high-risk areas.

⁵⁴ Nevertheless, if an asset no longer qualifies as HQLA, the Basel LCR framework allows banks to consider the asset as HQLA for an additional 30-day period, which partially mitigates the risk of potential fire-sales of the asset.

A similar feedback loop could arise in the corporate sector. To illustrate, Figure S3.2 considers the impact of a hypothetical carbon tax hike on an O&G firm, with spillover effects on the wider financial system and government. Depending on the intensity of the carbon tax, the impact could range from simply higher costs, to more extreme scenarios where reserve assets are completely written off. The resulting losses may lead to a fire sale of the company's stock, further driving down its market capitalisation. FIs and markets will demand a higher interest rate to compensate for the increased risk, or may even cease providing financing. The increasing cost of running the firm with declining prospects would strain the firm's ability to operate as a going concern. In the event that the O&G industry is dominated by state-owned enterprises (SOEs), as is the case in many oil-producing jurisdictions, there is a potential for increased sovereign risk as the revenues that form the bulk of the government budget may be threatened (FSB, 2020).

Figure S3.2 Illustrative impact of a carbon tax hike on an O&G firm



Source: MAS

Approaches used to assess the financial stability impact of climate change

This section provides an overview of the various approaches adopted by regulators, academics and the industry for assessing climate-related impacts, along with the challenges of these assessment frameworks.

Current work on climate stress testing and related quantitative assessments

The quantitative techniques for assessing financial stability impact can be broadly classified by whether they adopt *top-down* or *bottom-up* estimation methods.

Top-down methods rely on aggregated data at a jurisdiction or sector basis, and usually model more gradual, longer-term effects. These methods often use Integrated Assessment Models (IAMs)⁵⁵ together with other satellite models to estimate macrofinancial parameters under alternative climate scenarios.

Bottom-up methods rely on granular entity-level data⁵⁶, and typically model the effect of sudden acute shocks, such as an unanticipated carbon tax or a natural catastrophe. Such entity-level assessments can then be aggregated to conduct broader sectoral analyses, and can serve as a robustness check to top-down results.

As an example of such quantitative approaches, De Nederlandsche Bank (DNB) conducted an exploratory energy transition risk stress test (Vermeulen et al., 2018) to assess the potential impact of policy, technological or confidence shocks on the Netherlands' financial system over a five-year period. The Banque de France (BdF) and Bank of England (BoE) have also released details regarding their ongoing and proposed climate stress tests (Allen et al., 2020; French Prudential Supervision and Resolution Authority (ACPR), 2020; BoE, 2019; Bailey, 2020), which will focus on assessing the longer-term impacts of climate change over a 30-year scenario horizon.

Current work on the surveillance of climate risks

Beyond stress testing, regulators have conducted preliminary analyses to estimate potential climate risks, such as characterising exposures by (i) geography, to capture physical risk, and (ii) carbon-intensive sectors, to capture transition risk. While exposures can provide some indication of vulnerabilities, they would be insufficient for an in-depth assessment of potential systemic implications.

Meanwhile, some international organisations have developed indices that provide a point-in-time proxy measure of climate risks, such as the World Economic Forum's annual Energy Transition Index (Alidi et al., 2020). While such indices could influence the design of stress tests and climate risk surveillance frameworks, more work is required to develop indices

⁵⁵ IAMs typically consist of interlinked modules featuring climate, economic, energy use and land use components, and are frequently used to assess possible outcomes under different scenarios.

⁵⁶ Examples include individual firms' carbon intensity and the geographical location of their assets.

specific to industry sectors and sub-national geographies that can be updated more frequently.

Challenges faced in assessing the financial stability impact of climate change

Assessments of the financial stability impact of climate change are still at a preliminary stage due to both data-related and methodological challenges. At the broad, conceptual level, climate risks have only begun to materialise and hence their manifestation in the data is difficult to identify.

In terms of data, there remain challenges with the quantity and quality of entity-level data relevant to estimating climate risks (e.g. individual firms' carbon intensity). This is partially attributed to the nascent state of climate-related disclosures and taxonomies. While it is possible to use aggregated data at a sectoral or national level instead, such analyses will miss out intra-group variance, including differences in production processes and technologies used by firms in the same sector—indeed, such details are often crucial in assessing actual financial stability impacts.

From a methodological standpoint, top-down approaches reliant on IAMs typically generate a limited range of macrofinancial parameters for broad geographical regions, necessitating substantial additional work to derive more granular estimates useful for detailed assessments. Such models may also be subject to the “Lucas Critique”⁵⁷ since climate change affects structural relationships and deep behavioural parameters. In comparison, while bottom-up approaches face less methodological challenges, they are subject to the data issues cited above, and are typically confined to sectoral level analysis.

MAS' work on assessing the financial stability impact of climate change

Central banks and regulators globally, including the MAS, have begun efforts to better assess the impact of climate change on financial stability. This section outlines MAS' ongoing and planned work in this area, including through climate stress testing as well as surveillance of climate risks.

Climate stress testing in MAS' Industry-Wide Stress Test

MAS has already started to incorporate scenarios with climate risks as part of the annual Industry-Wide Stress Test (IWST). In the 2018 IWST, insurers were subjected to a scenario featuring extreme flooding, and had to consider the impact of higher claims on their balance sheets arising from damage to insured properties. Within the next two years, MAS will incorporate a broader range of climate risks in thematic scenarios as part of the IWST.

⁵⁷ The Lucas Critique argues that one cannot predict the effects of a change in economic policy entirely on the basis of relationships observed in historical data, because changes in economic policy modify the environment within which consumers and investors make their decisions.

Table S3.1 provides a broad outline of the possible scenarios for the stress testing of climate risks. These scenarios are illustrative in nature, and form part of a wider range of possible pathways that incorporate the effects of climate change.⁵⁸ The inclusion of both short-term and long-term scenarios featuring climate risks will help regulators and the industry to jointly increase awareness of the economic and financial implications of such risks, and encourage the collaborative development of relevant capabilities in stress testing and related quantitative assessment techniques.

Table S3.1 Outline of potential scenarios for the stress testing of climate risks

	Short-term Scenario	Long-term Scenario
Climate risks	• Low physical risk • High transition risk	• Moderate physical risk • Moderate transition risk
Physical risk drivers	• Global warming and rising sea levels • No significant increase in natural catastrophe frequency or severity	• Global warming and rising sea levels • Flooding in some major cities
Transition risk drivers	• Sharp, abrupt carbon tax hike globally driven by increased recognition of the adverse effects of climate change⁵⁹	• Gradual carbon tax hike globally • New technologies drive shift from fossil fuels towards renewable energy
Macro economic-financial backdrop	• Disorderly transition leads to short-term decline in economic growth • Sharp rise in energy prices results in inflationary pressures • Yields and credit spreads widen for entities with low transition readiness	• Orderly transition leads to slightly below-trend growth in the interim • New technologies lead to write-offs on equipment dependent on fossil fuels • Increased awareness of flood risk leads to a repricing of assets

Source: MAS

Given the nascent stage of research in quantifying climate risks, MAS' work on climate stress testing and climate-related modelling will involve a multi-year iterative process. MAS intends to adopt a sequenced approach (Figure S3.3), building on the work done in this area by other regulators, academia and the financial industry, such as the pilot stress testing approaches adopted in other jurisdictions.

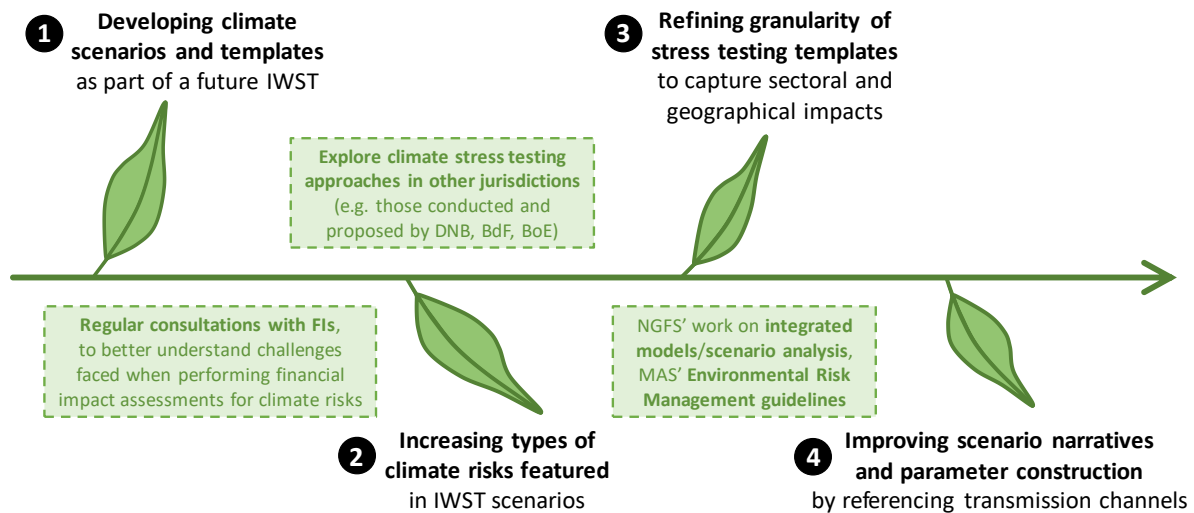
In addition, MAS has been engaging FIs to better understand the issues they face when performing financial impact assessments related to climate risks. The key challenges cited were data availability and quality, as well as a lack of widely accepted modelling approaches for the robust analysis of climate-related impacts. Over time, such gaps will be gradually addressed, through the increased adoption of climate-related disclosures globally as well as the development of climate risk assessment methodologies by regulators, academia and the

⁵⁸ Examples of such scenario pathways include the Network of Central Banks and Supervisors for Greening the Financial System (NGFS) scenarios (NGFS, 2020) that feature varying levels of physical and transition risk, as well as the Shared Socioeconomic Pathways (van Vuuren et al., 2017) that feature different assumptions regarding global mitigation and adaptation efforts regarding climate change.

⁵⁹ For instance, the Principles for Responsible Investment has developed the concept of an "Inevitable Policy Response" (Principles of Responsible Investment, 2019) that outlines a possible strengthening of global climate policy in 2023–2025 due to increased pressure for policy action over environmental, social and economic concerns.

financial industry. This will also be supported by MAS' upcoming Guidelines on Environmental Risk Management, that will set out the supervisory expectations for banks, insurers and asset managers on the assessment and management of such risks.

Figure S3.3 MAS' planned work on climate stress testing and climate-related modelling



Source: MAS

Surveillance of climate risks

In addition to stress tests, it would also be important to conduct surveillance of climate risks and assess their financial stability implications for our financial system on an ongoing basis. Various indicators and data would be required for such surveillance and assessment work, such as the geospatial information of firms' assets (for physical risk), the carbon intensity of entities' operations (for transition risk), and FIs' exposures to entities vulnerable to climate risks. MAS is exploring how new and existing datasets can be used to augment current surveillance approaches and better capture the impact of climate risks.

Conclusion

Climate change has become the defining challenge facing both present and future generations. From shifting weather patterns to rising sea levels, the risks posed by climate change are both multi-faceted and pervasive in their manifestation. Research has suggested that there are substantial transition risks associated with abrupt policy action, and eventually, even higher physical risks associated with policy inaction. As such, climate change requires an interdisciplinary approach and multi-dimensional global response involving the close collaboration of governments, academics and the private sector.

The financial system plays a key role in catalysing this global response. Financial markets are pivotal in the allocation of capital and resources throughout the wider economy. As such, the financial system will be instrumental in enabling economies to make a successful transition to a more sustainable trajectory. At the same time, FIs are exposed to climate risks because they provide financing and insurance services to entities which could be significantly impacted

by physical or transition risk. Many of these risks are complex, and assessing them will require the necessary data, expertise, and models. It is thus imperative for the financial industry to build up these resources in a timely manner, in order to better assess, manage and mitigate climate risks.

MAS will continue to work in partnership with industry, academia, other regulators as well as international organisations to strengthen the resilience of the financial system to climate risks. This includes, but is not limited to, initiatives in improving data, refining methodologies on climate risk assessment, and developing green finance solutions. In developing a deeper understanding of the transmission channels and the possible severity of climate risks, stakeholders will be better equipped to recognise their potential impact on financial stability.

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Special Feature 4

S-REITs' Performance During the COVID-19 Pandemic

The COVID-19 pandemic has severely disrupted businesses, with spillovers to commercial real estate markets globally...

To stem the spread of COVID-19, governments have implemented measures such as increasing social distancing and, in some cases, by imposing temporary lockdowns. Such interventions have had a wide-ranging impact on business, with resulting spillovers to commercial real estate (CRE) markets. The hospitality segment has experienced a precipitous drop in demand due to travel restrictions. At the same time, retail tenants have had to close or limit operations, while the office segment has had to grapple with companies transitioning to work-from-home arrangements.

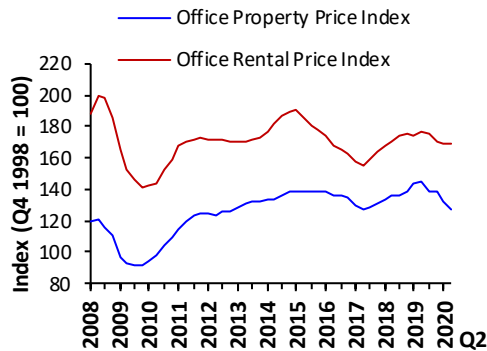
This Special Feature provides an update on how Singapore-listed Real Estate Investment Trusts (S-REITs) fared during the COVID-19 pandemic. It also assesses the resilience of S-REITs under alternative adverse conditions when the CRE rental environment remains challenging for an extended period.

...and in Singapore.

In Singapore, the economic downturn has reduced business demand for CRE property space. Prices have declined across the commercial and industrial property market segments in 2020 (Charts S4.1, S4.2, S4.3 and S4.4), while vacancy rates have risen, weighed down by the weakening economic and business outlook. On the supply side, the muted pipeline supply coming on-stream could mitigate the risk of sharp price corrections.

Chart S4.1 Property prices in the office segment have been falling since H2 2019, while rental prices remained more stable in 2020

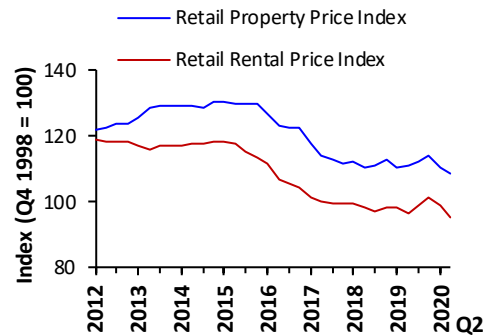
Office Property Price and Rental Price Indices (Q4 1998=100)



Source: URA

Chart S4.2 Retail property and rental prices have both declined in 2020 after increases in 2019

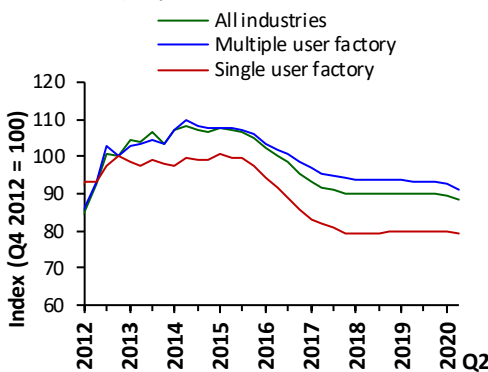
Retail Property Price and Rental Price Indices (Q4 1998=100)



Source: URA

Chart S4.3 Property prices in the industrial segment have remained fairly stable...

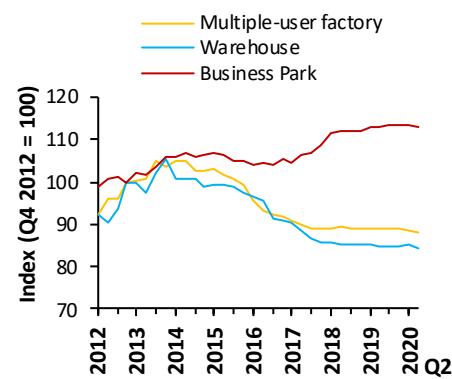
Industrial Property Price Index (Q4 2012=100)



Source: URA

Chart S4.4 ...as have rental prices

Industrial Rental Price Index (Q4 2012=100)



Source: URA

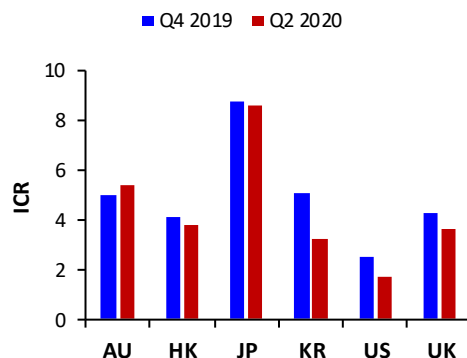
In addition to the market-driven fall in rental prices, measures to offer rental relief for tenants have also affected landlords' rental revenue. Under the COVID-19 (Temporary Measures) Act 2020, qualifying SME tenants receive additional rental relief from landlords and are allowed to take up a repayment scheme where tenants can pay their arrears in instalments over an extended period of time. They are also allowed to terminate their leases early if they are unable to renegotiate the terms of their agreement with their landlords. Further, some landlords have proactively provided relief to tenants over and above government stipulation. These extensions have impacted REITs' effective earnings and cashflows negatively.

REITs are an important barometer of health in the CRE sector.

At the critical nexus between the CRE market and the financial sector, global REITs saw their entire valuation gains over the past five years erased in H1 2020. The BIS also found that REITs lagged the strong recovery in equity markets after Q1 2020, faring worse than comparable stocks in cyclical industries as well as smaller-sized firms (BIS, 2020). The median ICRs of REITs have also deteriorated slightly in most major REIT markets globally (Chart S4.5).

Chart S4.5 REITs in most major REIT markets have seen median ICRs deteriorate in 2020

Median ICRs of REITs in Major REIT Markets Globally

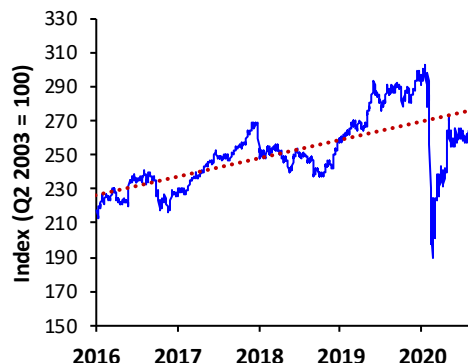


Source: MAS estimates, Thomson Reuters

In Singapore, S-REIT valuations grew at an average of 2.7% per annum over the period from 2016 to 2020, and S-REITs now account for some 9% of total market capitalisation of the Singapore Exchange (SGX), compared to 8% in 2016. S-REITs experienced a sharp decline of about 40% in February and March 2020 but have since recovered about 85% of their pre-pandemic valuations (Chart S4.6).

Chart S4.6 S-REITs have recovered most of their pre-pandemic valuations

S&P Singapore REIT Index (Q2 2003=100)



Source: S&P Global

Due to the size of S-REITs and the procyclicality of the markets in which they invest, there are potential financial stability concerns in Singapore through their influence on rental and property prices in the CRE segment.

Any distress in S-REITs could lead to contagion in the banking system, as they have strong exposures to the loan and bond markets via their borrowings. S-REITs obtain most of their funding in the form of loans from banks (80%). In aggregate, S-REITs were approximately SGD59 billion in debt as of Q2 2020, representing 3.5% of Singapore's banking sector's non-bank loans, with a third of these borrowings secured. Nevertheless, S-REITs have generally been prudent with leverage, with a median leverage ratio of 35%. The leverage ratio is well within the allowable limit of 50% as set out under the Code on Collective Investment Schemes.

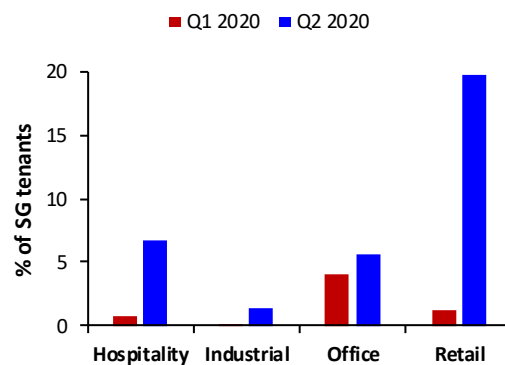
There are also indirect linkages between S-REITs and the banking system. In times of distress, S-REITs might be forced to offload distressed assets, which could have an adverse effect on property prices, at a time when sentiments are weak. Lower property prices could cause a knock-on effect on the asset quality of bank exposures to the commercial property sector as collateral values fall. The impact on asset quality would in turn trigger negative confidence effects that could lead to reduced banks' willingness to lend to the property sector, and further affect demand for and prices of commercial properties.

S-REITs have remained resilient to income shocks during COVID-19...

S-REITs have experienced a decline in income as market rentals have eased. In addition, an estimated 8% of their Singapore-based tenants took up some form of rental relief due to COVID-19 as of Q2 2020 (Chart S4.7). The retail segment saw the highest proportion of tenants seeking assistance with rentals, likely due to the confluence of the circuit-breaker impact as well as the higher proportion of SMEs in that segment.

Chart S4.7 Retail segment saw the highest proportion of tenants taking up rental relief

Average Proportion of Singapore-based Tenants Taking Up Rental Relief

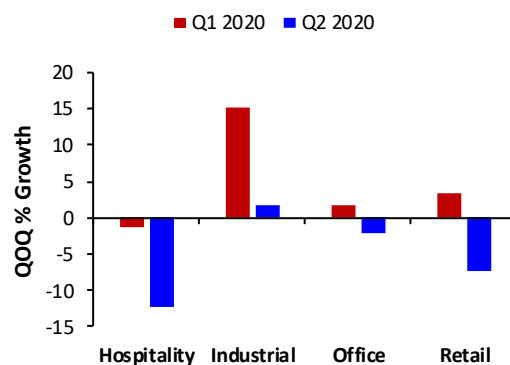


Source: MAS estimates, REIT Association of Singapore (REITAS)

The fall in market rents, combined with the rental relief measures, resulted in reduced revenues for S-REITs with significant Singapore-based assets (Chart S4.8). Hospitality REITs had already experienced revenue decline of –1.3% in Q1 2020. Revenues fell even more sharply by –12.4% in Q2 2020 as travel restrictions started to gather pace globally. Retail REITs' revenues deteriorated significantly by –7.3% in Q2 2020 as Singapore's circuit-breaker measures took effect, whereas industrial REITs have remained relatively unscathed.

Chart S4.8 The hospitality and retail segments have been hardest hit

Q-o-q Changes in Annual Gross Revenues of S-REITs with Significant Singapore-based Assets

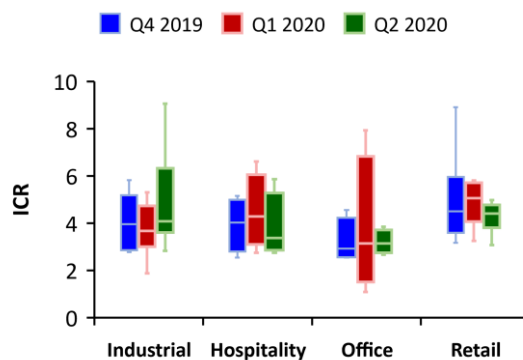


Source: MAS estimates

While the drop in revenue has weakened S-REITs' short-term performance and cash buffers, their underlying financial positions remain generally resilient at this stage. Looking at S-REITs' ongoing ability to service their debt obligations, the vast majority of S-REITs have median ICRs above 2.0 as of Q2 2020, indicating that earnings remain sufficient to make interest payments. From Q4 2019 to Q2 2020, median ICRs have declined for the hospitality (4.0 to 3.4) and retail (4.5 to 4.4) segments but have improved for the industrial (4.0 to 4.1) and office (2.9 to 3.1) segments (Chart S4.9).

Chart S4.9 ICRs have remained resilient, even after declines in Q2 2020

ICRs of S-REITs with Significant Singapore-based Assets



Source: MAS estimates

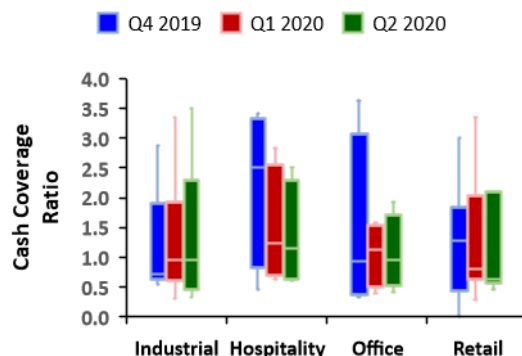
Note: The light-coloured line within the box denotes the median while the top and bottom of the boxes denote the 75th and 25th percentile values respectively.

Conversely, S-REITs' cash coverage ratios—which indicates ability to meet debt obligations coming due based solely on cash buffers—showed some signs of weakness, likely reflecting their business model of distributing most of the income as dividends to shareholders. With low retained earnings, S-REITs have had less opportunity to build cash buffers to withstand prolonged periods of liquidity and earnings stress. As shown in Chart S4.10, median cash coverage ratios of S-REITs with significant Singapore-based assets had declined since Q4 2019, with the falls most pronounced in the hospitality (2.5 to 1.2) and retail

(1.4 to 0.6) segments. Median cash coverage ratios in the office and retail segments were less than one, implying that cashflows and cash reserves might be insufficient to cover current liabilities, and that REITs would likely need to refinance or borrow to meet these obligations.

Chart S4.10 Cash buffers have weakened, especially for hospitality and retail segments

Cash Coverage Ratios of S-REITs With Significant Singapore-based Assets



Source: MAS estimates

Note: The light-coloured line within the box denotes the median while the top and bottom of the boxes denote the 75th and 25th percentile values respectively.

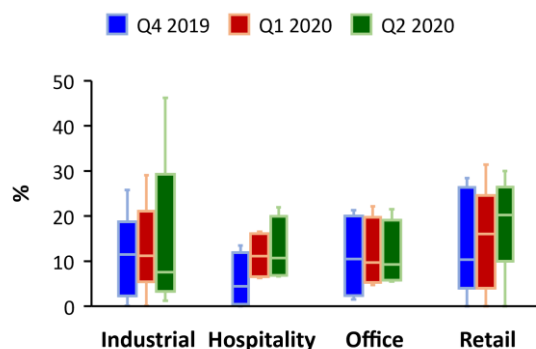
The decline in cash coverage has been mitigated by the proactive measures that S-REITs have taken to preserve cash buffers, including borrowing from banks, issuing new debt securities and not redeeming perpetual debt securities on the first call date. Bond issuance has been supported by a resilient Singapore corporate bond market, which saw a recovery between May and August 2020 (see Box A for a more detailed analysis of the Singapore corporate bond market). It should be noted, however, that attempts to raise cash buffers have been accompanied in part by an increase in short-term debt and correspondingly an increase in refinancing risk. Overall, short-term debt as a proportion of total debt increased from 9.5% at end-2019 to 14.6% in Q2 2020. In particular, the hospitality and retail segments increased short-term debt significantly in 2020, raising their median short-term debt ratios by 4.5 and 6.9 percentage points respectively (Chart S4.11).

Nevertheless, recently introduced measures⁶⁰ by the relevant authorities have provided more flexibility on the timing of distributions and ensured S-REITs' continued access to different funding channels (including borrowing from banks, issuing bonds, and raising equity). In Q2 2020, 40% of S-REITs with significant Singapore-based assets retained a median of 30% of distributable income. The increased flexibility will allow S-REITs to better manage their cash buffers in a sustainable manner.

⁶⁰ On 16 April 2020, the Ministry of Finance, Inland Revenue Authority of Singapore, and MAS announced an extension of the deadline for distribution of taxable income, as well as an increase in the leverage limit from 45% to 50%. To support issuers amid the challenging business and economic climate due to COVID-19 and accelerate fund raising efforts, SGX Regulation (SGX RegCo) allowed Mainboard issuers to seek a general mandate for an issue of pro-rata shares and convertible securities of up to 100% of its share capital (compared to 50% previously).

Chart S4.11 The proportion of short-term debt has increased for hospitality and retail segments

Proportion of Short-term Debt to Total Debt of S-REITs With Significant Singapore-based Assets



Source: MAS estimates

Note: The light-coloured line within the box denotes the median while the top and bottom of the boxes denote the 75th and 25th percentile values respectively.

...but there remains a need to be watchful if the macroeconomic environment worsens.

At present, the baseline scenario assumes that S-REITs' performance stabilise, with both debt servicing and liquidity ratios (represented by ICRs and cash coverage ratios) remaining broadly similar to those in Q2 2020. Under the baseline, it is assumed that the COVID-19 pandemic will remain contained in the domestic economy for the rest of the year. This assumption implies some continued easing of social distancing restrictions, resulting in a gradual resumption of footfall to commercial properties. Government initiatives to promote local tourism and the opening of travel bubbles would also provide a further boost to the retail and hospitality segments. With the recovery in economic activity, S-REITs would be able to resume, to some extent, the collection of rental payments that were deferred earlier in the year.

Nevertheless, to assess S-REITs' resilience against further income shocks from a potentially protracted and severe COVID-19 crisis, MAS simulated an alternative adverse scenario, where SME tenants are assumed to experience continued declines in business revenues. Under this downside scenario, recurrent outbreaks of COVID-19 infections lead to prolonged episodes of tighter social distancing restrictions. Footfall to commercial properties would be further reduced as consumer confidence worsens. Some SME tenants would therefore find it increasingly difficult to continue servicing rental payments out of diminishing revenues, especially as government relief measures taper off and existing cash buffers are run down.

As a result, SME tenants could terminate their leases prematurely. Based on feedback from S-REIT managers, the proportion of tenants that are SMEs that could terminate their leases early due to a significant decline in their sales revenues is estimated to be about 16%. Vacancy rates would be expected to rise, increasing search costs for replacement tenants and resulting in a loss of rental income for S-REITs. The increase in termination of leases is

expected to lead to higher vacancy rates of between 5% (industrial and office) to 15% (retail) based on the proportion of SME tenants typical to each property segment.

Under this adverse scenario, S-REITs would face declines in annual revenue of between 5–10% relative to the baseline, due to foregone rentals and increased search costs for replacement tenants. The drop in revenues translates to a 25% fall in earnings before interest, taxes, depreciation and amortisation on average from baseline.

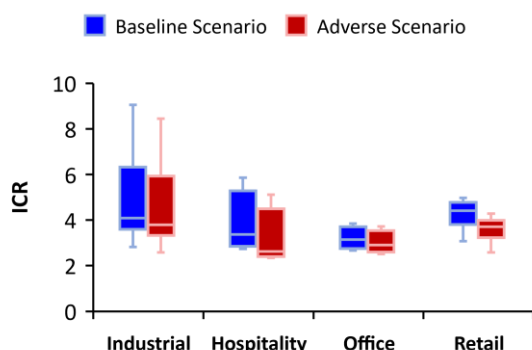
The results of the impact analysis show that even under the adverse scenario, debt servicing ability of S-REITs remains adequate with median ICRs healthy, keeping above 2.0 (Chart S4.12). However, once again, the hospitality and retail segments are expected to be impacted the most, with median ICRs declining by 0.8 and 0.7 percentage points respectively relative to the baseline scenario.

That said, shorter-term refinancing pressures could persist, with cash coverage ratios decreasing under the adverse scenario (Chart S4.13), reflecting the drop in cash flows from operating activities. The median cash coverage ratio for the industrial segment deteriorates to 0.9, compared to 1.0 under the baseline scenario. The office and retail segments, which already had cash coverage ratios below 1.0 in the baseline scenario, would face similar pressure. Of note, there are many retail REITs with low cash coverage ratios under both the baseline and adverse scenarios, highlighting that this is a segment that bears watching for any potential liquidity stress.

Since REITs generally have buffers to borrow before their leverage ratios hit regulatory limits, they could borrow to tide over any short-term cashflow shortfalls. However, their success in doing so would depend on banks' and markets' risk tolerance and prognosis of the longer-term health of these REITs, once COVID-19 is more permanently under control.

Chart S4.12 ICRs remain healthy under adverse scenario

ICRs of S-REITs With Significant Singapore-based Assets

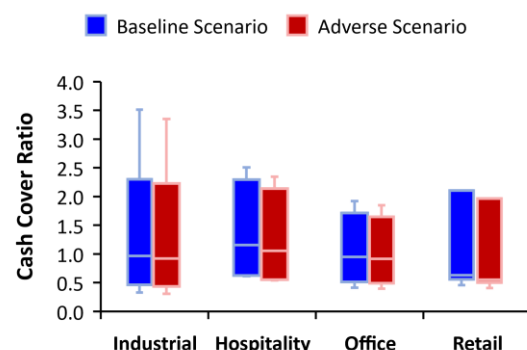


Source: MAS estimates

Note: The light-coloured line within the box denotes the median while the top and bottom of the boxes denote the 75th and 25th percentile values respectively.

Chart S4.13 Cash coverage ratios for industrial segment falls below one under adverse scenario

Cash Coverage Ratios of S-REITs With Significant Singapore-based Assets



Source: MAS estimates

Note: The light-coloured line within the box denotes the median while the top and bottom of the boxes denote the 75th and 25th percentile values respectively.

Conclusion

S-REITs have weathered the earnings shock brought about by the COVID-19 storm relatively well. However, cash buffers have been stressed by lower rental revenue amid COVID-19 containment measures and increased operating costs to adhere to safe distancing measures, resulting in weak cash coverage ratios. Current macroeconomic headwinds and continued uncertainty surrounding business prospects as well as the operating environment could pose some risks to maintaining dividend pay-outs while still meeting debt service obligations. Simulations under both baseline and adverse scenarios on S-REITs with significant exposures in Singapore suggest that they remain well-placed to meet their longer-term ability to service interest payments, although near-term refinancing pressures could persist. S-REITs have nevertheless taken proactive measures to preserve liquidity, supported by regulatory efforts to provide greater flexibility. The anticipated recovery in economic activity in 2021 should also bolster S-REITs' financial positions, with cash flows improving as domestic demand picks up and tenants' business prospects improve. S-REIT managers should continue to exercise prudence in managing their cash buffers and monitor their debt servicing obligations closely.

References

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